UNITED STATES OF AMERICA
BEFORE THE
FEDERAL ENERGY REGULATORY COMMISSION

Interconnection of Large Loads to the Interstate Transmission System	)))	Docket No. RM26-4-000

SUPPLEMENTAL COMMENTS OF EOLIAN L.P. ON THE ADVANCE NOTICE OF PROPOSED RULEMAKING

Eolian L.P. (Eolian) respectfully submits comments in response to the Advance Notice of Proposed Rulemaking (ANOPR) on Ensuring the Timely and Orderly Interconnection of Large Loads.[1] On April 14, 2026, the Commission issued an Order Regarding Intent to Act in this proceeding.[2] We agree with the Commission that, "'[i]n light of the unprecedented current and expected growth of large loads' such as data centers, there is an urgent need for thoughtful reforms to ensure that large loads are 'able to connect to the transmission system in a timely, orderly, and non-discriminatory manner.'"[3] The Commission's Order Regarding Intent to Act states that the Commission will "address the problems discussed in the ANOPR in a manner that is quick, efficient, and legally durable," potentially suggesting a faster procedural path than a rulemaking, such as action through a set of Orders to Show Cause.

Eolian is an independent developer of both load and generation, with a broad portfolio of projects spanning many transmission provider territories. For the past decade, Eolian has been working to develop supply resources and load in a coordinated and efficient manner, including through co-location and at adjacent (that is, electrically proximate) sites. Drawing on that experience, Part I.B of these supplemental comments provide the Commission with specific

[1] *Ensuring the Timely and Orderly Interconnection of Large Loads*, Advanced Notice of Proposed Rulemaking, Docket No. RM26-4-000 (Oct. 23, 2025) (ANOPR).
[2] Order Regarding Intent to Act, 195 FERC ¶ 61,045 (Apr. 16, 2026).
[3] *Id.* at P 2 (citations omitted).

factual examples concerning the scope of deficiencies the company has seen in transmission provider tariffs, so as to ensure that any solution is broad enough to address the challenge at hand.

To achieve its goals of ensuring that large loads are able to connect in a timely, orderly, and efficient manner, the Commission must reform transmission provider tariffs across the country. In our experience, we have not encountered any transmission provider that currently has a process which (i) adequately integrates or coordinates the interconnection of supply resources and load, (ii) provides developers of supply resources such as Eolian with adequate incentive to locate generation and storage assets in a manner that opens additional system headroom for load interconnection, or (iii) offers the full range of transmission service options necessary to allow supply and load to coordinate to optimally use the existing system and reduce transmission infrastructure needs. The lack of tariff provisions to afford all three elements results in unjust and unreasonable or unduly discriminatory rates by exacerbating the reliability and affordability impacts of large load growth, while impeding speed-to-power for new large loads. The lack of coordination between load and generation interconnection processes, as well as the lack of transmission services to meet the needs and capabilities of configurations where generation is adjacent or located at the same point of interconnection as load, prevent new customers from accessing the electricity system and leads to an over-built transmission system by forcing the construction of unnecessary network upgrades.

I. COMMENTS

A. The Commission must require nationwide reforms to enable coordinated interconnection of generation and load and provide transmission services that match the needs of load using a broad range of technologies and supply arrangements

Eolian has focused on developing supply resources adjacent to large load customers to facilitate more timely service to the large load customer while reducing infrastructure needs (and their associated costs). In addition, Eolian has identified sites where the addition of generation or storage opens system transmission capacity to add new load without the need for additional network infrastructure. In Eolian's experience, locating supply at these strategic locations—near the site of an incoming large load customer or at places where a new large load customer could connect—holds the potential to significantly reduce the amount of network upgrades required to integrate new load. Because long network upgrade construction timelines and lack of adequate supply are increasingly causing substantial delays for new load interconnection requests, developing generation or storage at these locations could allow large customers to connect to the system much faster. But existing rules do not adequately encourage generation and storage developers to pursue these strategies. Any action in this proceeding must fix this design failure, providing an incentive to locate efficiently.

Eolian's unique experience as a developer of both supply and load has revealed the lack of coordination in load and generation interconnection processes across the country, as well as a lack of consistent standards for how load is reflected in generation interconnection studies. Because generation and load are studied independently, load is often studied in a manner that ignores the presence of generation that could minimize system impacts, and vice versa. Transmission providers have refused Eolian's requests to model supply and load together, even where the relevant assets are both Eolian's own projects and their development is specifically

coordinated. The lack of integration in load and supply interconnection processes may result in both types of projects triggering significant network upgrades that may not be needed if the two projects were studied together, or studied in a manner that accounted for one another. As described below in Section I.B, Eolian has seen this lack of integration cause transmission providers to identify and assign the costs of network upgrades that would not be needed under coordinated load and generation interconnection studies, and cause cancellation or reduced commercial scale of both supply and load projects that would have been viable had transmission providers coordinated interconnection studies.

While a single integrated study process would be the most efficient path for considering interconnection of load and generation together,[4] the Commission can require load and generation study coordination even if it decides against asserting jurisdiction over load interconnection by requiring transmission providers to coordinate in generation interconnection study processes with the relevant transmission owners or electric distribution companies conducting load interconnection studies. Such coordination should include the opportunity for generation developers to nominate load(s) to be studied together with the generation. Further, to prevent undue discrimination and guard against inefficient system buildout, the Commission should require consistent standards for when and how load is included in generator interconnection studies.

In addition, transmission providers do not allow load to avoid the need for network upgrades by offering non-firm network transmission services to loads willing and able to limit withdrawals from the broader transmission system.

[4] *See* Eolian, Comments, Docket No. RM26-4-000, at Attachment A (filed Nov. 21, 2025) (describing a process through which transmission providers could study interconnection requests by hybrid facilities that include generation and load elements).

As an increasing number of large load customers have sought to interconnect to the system, the challenges stemming from lack of generation and load study integration and the lack of differentiated transmission services matching the needs of co-located or adjacent customers have become more acute. However, these grid integration challenges pre-date the growth of demand from large load customers. In Eolian's experience, inefficient grid planning outcomes are occurring even in areas with comparatively fewer large load customers. Every single area in which Eolian develops projects would benefit from reform.

Eolian's prior Comments and Reply Comments in this proceeding outline the key elements of reform needed to facilitate timely interconnection of new large load customers,[5] including:

- The Commission must require transmission providers to provide for a generator interconnection process that allows generator and load interconnection requests to be studied together. While a single joint process would be the ideal, at minimum there must be an *option* for a generator to identify load to be included in its study.
- The Commission should require transmission providers to adopt transparent and consistent standards for when and how new load additions are reflected in generation interconnection studies.
- The relevant interconnection process must provide a path for permanent interconnection service. Temporary frameworks that hinge on subsequent processing through a standard queue create commercial risks that are likely to inhibit widespread use of these interconnection pathways.

[5] *See Id.* Comments at 25; Eolian, Reply Comments, Docket No. RM26-4-000, at 15-16 (filed Dec. 5, 2025).

- The Commission must require transmission providers to offer transmission services that match the ability of loads to use the capabilities of adjacent generation, including non-firm and reduced service options. Such service must allow the load to take service when the grid is unconstrained and the supply resource is not available (including when a battery resource is charging) while electing to curtail when the grid is constrained.
- The Commission must ensure that these service options can be linked to load interconnection. To the extent the Commission takes jurisdiction over load interconnection processes, it should require transmission providers to expedite interconnection of large loads that are able to limit withdrawals from the system during grid constraints, including by relying on co-located or adjacent supply resources. Regardless of whether the Commission asserts jurisdiction over large load interconnections, the Commission must act to ensure that transmission service options and associated transmission study processes (at the local or regional level) enable and are not a barrier to states' efforts to manage the interconnection of large loads. Specifically, transmission service options and associated operations and planning frameworks must recognize and account for the commitments of flexible loads to limit their withdrawals from the transmission system (including loads that achieve such flexibility through reliance on co-located or adjacent supply resources).

These reform recommendations stand regardless of the procedural path the Commission takes, whether the Commission acts via rulemaking, through a series of show cause orders, or via another path.

Significantly, however, the Commission must require additional reforms beyond recently approved initiatives that some regions have enacted to ensure timely and affordable integration of large loads. While recent reforms in Southwest Power Pool (SPP) and PJM Interconnection, LLC (PJM) are positive steps, these reforms fall short of what is needed to fully address the large load interconnection challenges identified in the ANOPR. In SPP, while we support development of Conditional High Impact Large Load Service (CHILLS) and the High Impact Large Load Generation Assessment (HILLGA) process, **these reforms are not adequate to facilitate interconnection of most new large load customers** because of the operational risks and time-limited nature of the service provided for the load and the generation.[6] During the service terms for load under CHILLS and for generation under HILLGA, both are subject to unquantifiable curtailment risk. Furthermore, because HILLGA and CHILLS are not permanent service options, interconnection of both load and generation via this process remains subject to network upgrades that may be subsequently identified in the load and generation's permanent interconnection study processes. And such permanent interconnection study process, which occurs under SPP's Consolidated Planning Process, may be years after the facilities are operational. This creates potentially unworkable commercial risks for both load and generation that will inhibit use of these processes, as compared to what would be enabled via tariff mechanisms that allow for non-time limited (i.e., permanent) non-firm transmission service, and non-time limited (i.e., permanent) interconnection rights for generation directly supporting that load.

Likewise in PJM, the Commission must build on the Co-Location Order[7]—which, conditional on effective PJM compliance and implementation, is an important and significant step toward enabling faster and more affordable service—to maximize impact. First, the

[6] *See* Eolian, Reply Comments, Docket No. RM26-4-000, at 46-48 (filed Dec. 5, 2025).
[7] *See PJM Interconnection, LLC*, 193 FERC ¶ 61,217 (2025) (Co-Location Order).

Commission should explicitly require transmission providers to adopt modeling assumptions in interconnection and transmission study processes to account for the ability of coordinated load and generation arrangements to limit withdrawals from the broader system, where the relevant load and generation arrangements commit to limit withdrawals. Second, the Commission should explicitly direct transmission providers to make the non-firm services available to all loads willing and able to limit withdrawals from the transmission system rather than limiting availability only to co-located loads. And third, the Commission should make clear that whatever non-firm transmission services that the Commission directs transmission providers to offer in this proceeding must be designed to allow for broad availability of non-firm service during non-emergency conditions and not only during associated generator outages.

B. Eolian's experience developing projects demonstrates the need for these reforms

Eolian develops energy generation, storage, and load assets throughout the United States. It has originated more than 30 GW of operating generation capacity, and more than 3.5 GW of commercialized battery energy storage capacity across 10 different utilities and power markets. Eolian also has more than 2 GW of data center and large industrial load applications in process in locations across the country. The company's experiences illustrate lessons regarding the need for reform, including the scope of changes necessary to meet the speed-to-power challenge.

To further support the Commission's development of a record on the need for reform, Eolian provides three project development examples that illustrate the benefits of a more integrated and coordinated process for considering energy supply and load additions to the system. Each of these examples demonstrates how the current framework of uncoordinated study and undifferentiated transmission service options leads to study results that trigger infrastructure that could be avoided through a more sophisticated approach, which in turn delays speed to

power, increases costs for customers, and can result in reliability impacts where supply additions lag behind load growth. They also demonstrate how the bluntness of the current framework removes generators' incentives to site facilities where they could avoid the need for otherwise unnecessary grid infrastructure, let alone to maximize the creation of additional headroom, thereby driving costs higher than needed to maintain reliable service and slowing down load interconnection processes that require significant network upgrades. Where studies are not coordinated with one another, a load study may show a lack of adequate headroom, triggering network upgrades that may delay or imperil the proposed project, while a generation developer, unable to capture benefits of the headroom it creates, may not be commercially viable or have a path to an offtake agreement. The Commission in this proceeding must cut through this misalignment, incenting and recognizing the benefits of efficient and coordinated load and generation siting.

1. Lack of coordination between load and generation interconnection studies caused cancellation of the Matta Grid battery energy storage project in Virginia

Eolian's proposed Matta Grid battery energy storage project, a 200 MW / 800 MWh battery energy storage project, entered the PJM queue in 2020 and was assigned to PJM's Transition Cycle 1. The project was directly adjacent to Amazon's proposed 770 MW Mattameade Data Center and proposed to interconnect to the same 230 kV line. Although the data center and related substation were long-planned, the large load and associated infrastructure were not in the PJM base case for the Phase II study of Transition Cycle 1. In Eolian's experience developing the project, PJM's rules provide individual utilities with discretion regarding what infrastructure needed for load interconnection is input into PJM's study models that form the basis of the generator interconnection studies. In this case, Dominion's discretion to

omit infrastructure and load from its inputs into PJM's models caused the data center substation to be entirely omitted from PJM's generator interconnection studies for Transition Cycle 1.

This omission proved fatal to the Matta Grid project. PJM's Phase II study results assigned $158 million in network upgrade costs to Matta Grid, largely driven by a grid constraint 25 miles from both the Matta Grid project and the proposed Amazon data center. A Timmons Group study that Eolian commissioned[8] demonstrated that, had PJM included even a very conservative 150 MW of the 770 MW of Amazon load in its Phase II study for Transition Cycle 1, the study results would have shown over $100 million less in system reinforcements assigned to the Matta Grid project.[9] The failure of the generator interconnection study process to reflect reasonably expected conditions on the grid caused the cancellation of supply that is greatly needed in PJM and that could have reliably served adjacent load growth without extensive network upgrades.

Eolian's efforts to develop other projects have revealed that this is not an isolated experience. For example, in the American Electric Power (AEP) service territory, Eolian has observed that AEP appears not to submit transmission lines or substations into PJM models until the infrastructure is fully energized. Given the lead time inherent to developing generation and storage projects, this makes little sense. It provides for supply resources to be studied using models that will be years out of date when considering the proposed commercial operations date of the relevant infrastructure. Infrastructure can be omitted even when it is under construction, and sure to be operating by the time the relevant generation or storage projects come online. In many cases, such as with Matta, the omission of relevant load and related infrastructure will cause proposed generation to trigger more network upgrades. Thus, through their decisions

[8] The Timmons Group study is attached hereto as Exhibit A.
[9] *See* Exhibit A at 4.

regarding whether and when to include load and related infrastructure in generator interconnection studies, utilities may provide for outcomes that maximize their own return on equity while triggering the construction of infrastructure that would not be needed were the process to more accurately reflect planned system additions.

2. An uncoordinated approach to load and generation interconnection has triggered significant network upgrades at projects in the Eagle Mountain technology corridor in Utah

Eolian has experienced similarly uncoordinated study processes outside of Regional Transmission Organization (RTO) regions. In Utah, Eolian is part of a joint venture developing a multi-technology approach to serving large load customers in the Eagle Mountain technology corridor that includes over 1000 MW of gas, 1200 MW of solar, and 1200 MW of solar. The projects are located adjacent to several operating and proposed large loads —including from hyperscalers, independent data center developers, and Eolian itself via its joints venture—in an area surrounding the Mercer substation that could become the next "Data Center Alley" of the West.

Yet despite the fact that requests for a number of the proposed loads were submitted years before the generation cluster was studied, the proposed loads were not reflected in PacifiCorp's Cluster Study Report.[10] PacifiCorp omitted even the proposed load that had been requested by Eolian's Joint Venture, despite direct requests from to model the load and generation together. The report assigned the cluster significantly more than $100 million in costs, including for remotely located upgrades.[11] PacifiCorp's failure to include the proposed load has persisted even as one data center developer signed a load service agreement and began construction. PacifiCorp

[10] PacifiCorp's Cluster Study Report is attached hereto as Exhibit B.
[11] Exhibit B at 59.

has represented its position now is that it will model load within the generator interconnection study process only when the load is fully operational.

In sum, PacifiCorp's study approach has provided Eolian with no benefit for efficiently locating at the site of several proposed loads, even as one of them has moved into the construction phase. An alternate approach that recognized that the projects would primarily serve local planned loads could avoid significant network upgrade requirements, allowing for faster, cheaper, and more commercially viable supply to come online to serve booming load growth. The current process of ignoring known, in-process load interconnections makes developing such new supply much more commercially difficult, as growing resource adequacy shortfalls in many regions illustrate.

3. Eolian's Flint storage facility demonstrates that strategic location of energy supply assets can make additional transmission system capacity available to large loads, but current planning frameworks do not provide developers with adequate incentive to locate efficiently

Eolian's Flint storage facility in New Albany, Ohio demonstrates the ability of strategic location of energy assets to facilitate speed to power for large loads. Eolian's Flint storage facility is a 200 MW battery energy storage system located in a manner such that discharge of the battery opens transmission capacity by relieving thermal overloads seen during N-1-1 conditions that would limit large load interconnection absent the battery. As shown in Exhibits C and D, attached, Eolian commissioned a study by PowerGEM,[12] which identified thermal overloads under N-1-1 conditions without the benefit of the storage facility.[13] PowerGEM found that discharging Eolian's Flint storage facility would open 174 MW of additional system headroom to

[12] A summary of the PowerGEM study conducted by Aurora Energy Research is attached hereto as Exhibit C. The PowerGEM study is attached as Exhibit D.

[13] Exhibit D at 13; *see also* Exhibit C at 2 ("Data center development in New Albany, OH is constrained by the local transmission availability under the N-1 contingency study.").

host load at the Badger bus, 167 MW at the Jug St bus, and 110 MW at the Innovation Lab bus.[14] This represents a significant system benefit that could avoid the need for transmission system upgrades to enable new load to interconnect.

Under PJM's current rules, however, there is little incentive for developers such as Eolian to locate assets to provide these system benefits. Developers do not get any credit or earn any additional revenues for opening additional system capacity. Further, the lack of coordination between generation and load interconnection studies makes it challenging for developers like Eolian to bilaterally contract with loads to enable their development. This discourages coordination between load and generation development, which may result in a situation where (1) unnecessary network upgrades are constructed; or (2) neither the load nor the generation is built because each appears commercially non-viable when considered in isolation, though the load and generation could be viable if developed in coordination/awareness of the other.

An even more effective incentive would be provided if the generation interconnection process afforded generators the ability to generate "tags" representing the transmission capacity that the proposed generation opens. Such a transmission capacity tag could then be used in the load interconnection process, allowing load to contract with generation to support development that enables development to move forward that is currently on hold. This mechanism would encourage innovative or creative generation operators to finance and build projects in locations that maximize interconnection opportunities for new loads by allowing them to monetize some of the value that such development can provide. Even where load interconnection remains a matter of state jurisdiction, the lack of information on the impacts of generation interconnection

[14] Exhibit D at 6. The Flint storage facility would create even more system headroom during peak load conditions: 221 MW of additional system capacity to host load at the Innovation Lab bus, 168 MW at the Badger bus, and 163 MW at the Jug St bus. *See id.* at 4.

on transmission capacity is a barrier within FERC-jurisdictional transmission provisions that inhibits action by retail authorities to realize the benefits of siting generation so as to enable faster, more affordable and reliable development of loads.

II. CONCLUSION

Eolian urges the Commission to take bold, nationwide action to correct the inefficiencies illustrated above in both RTO and non-RTO regions that threaten affordability and resource adequacy. Inability to efficiently connect large load customers and adequate supply to support them threatens the economy of United States, and its leadership on artificial intelligence. The Commission can make significant strides in meeting this challenge by requiring transmission providers to integrate load and supply interconnection processes, incentivize efficient generation and storage assets siting to open additional system headroom for load interconnection, and offer the full range of transmission service options necessary to allow supply and load coordination to optimally use the existing system and reduce transmission infrastructure needs.

Respectfully submitted,

/s/ Miles Farmer

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Exhibit A

Timmons Group Memorandum Re:
Mattameade Data Center Sensitivity Analysis

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MEMORANDUM

TO: Eolian Energy

FROM: Timmons Group

DATE: July 7th, 2025

RE: Mattameade Data Center Sensitivity Analysis

Project Overview

The Mattameade Data Center is a prospective data center up to 770 MW expected to be online by 2027. The location of the data center is adjacent on the south side of the AG1-412 queued project. At Decision Point II (DPII), AG1-412 was allocated over $120M for an underground upgrade from Garrisonville to Aquia Harbor on the 230 kV system. However, the Mattameade Data Center did not appear to be considered within the Transition Cycle 1 (TC1) Phase II (PhII) analysis. Herein, Timmons Group explores the sensitivity of load growth in relation to the PJM TC1 study process and how it may have adjusted the outcome for AG1-412. The Mattameade Data Center is just one example within the extreme load growth in Dominion Energy Virginia's (DEV's) territory within PJM.

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Interconnection Information

The TC1 PhII Summer Peak model and TC2 PhI Summer Peak models were compared to identify load demand changes between the 2027 and 2028 models. Image 1 and Image 2 below detail the topology of the Ladysmith CT to Fredericksburg 230 kV line where the Mattameade Data Center is expected to connect in addition to the previous Point of Interconnection (POI) for AG1-412.

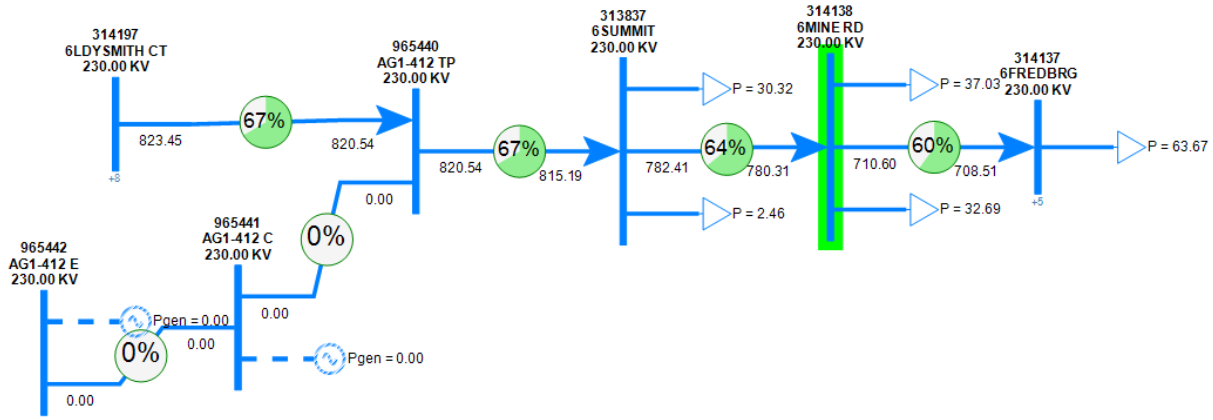

Image 1. Oneline diagram of the TC1 PhII Summer Peak 2027 model. A total load of 166.17 MW is shown within.

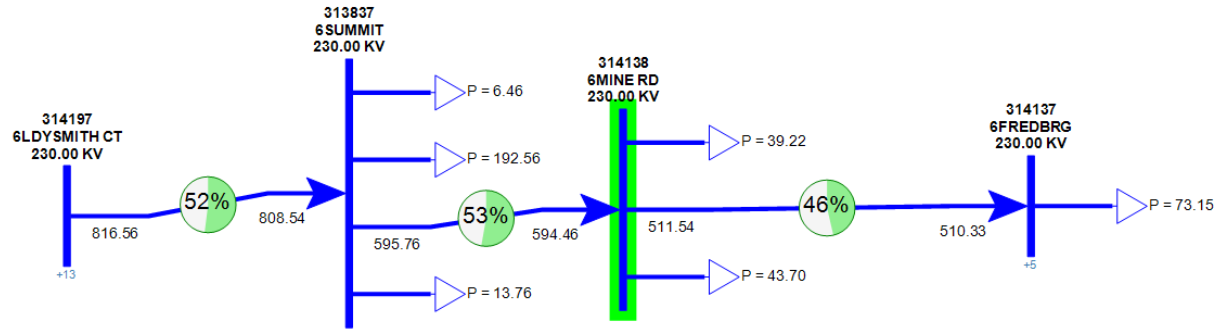

Image 2. Oneline diagram of the TC2 PhI Summer Peak 2028 model. A total load of 368.85 MW is shown within.

The difference in load shown between the two cases totals just over 200 MW.

Study Methodology and Assumptions

Although the AG1-412 project is withdrawn from the PJM queue, it was utilized for the analysis herein as the project is a prime example of co-located prospective generation to prospective load growth.

AG1-412 was sited between Ladysmith CT and Summit, with the Mattameade data center expected to interconnect on the south side of the generating project between Ladysmith CT and AG1-412. Although some load increase is shown at the Summit and Mine Road substations, the Mattameade data center itself does not appear to be considered in the model.

When building the case, PJM will include approved upgrades expected in-service by June 1st of the study year. Typically, data centers will increase their load through the years until they achieve their maximum withdrawal from the system. It is understandable to see load growth between the two models; however, given that the data center is expected to have a 770 MW withdrawal by 2027[1], it appears that Mattameade was not included in the TC1 or TC2 models as no load is shown at its expected location.

[1] https://www.datacenterdynamics.com/en/news/amazon-requests-permit-for-11-building-data-center-campus-in-caroline-county-virginia/

The primary cause of the cost increase for AG1-412 is a facility approximately 25 miles north of the project site between Garrisonville and Aquia Harbor. This facility was loaded at 102.69% with AG1-412 allocated over $120M in network upgrade cost[2]. Timmons Group included an arbitrary 150 MW load to mimic Mattameade slowly increasing their load demand as it comes online. Again, this data center is expected to withdraw up to 770 MW by 2027, so the 150 MW is a very conservative estimate. This additional load added to the model will show whether the Aquia Harbor to Garrisonville 230 kV line would NOT have been overloaded within the TC1 PhII analysis if Mattameade was considered in the study. If so, the $120M allocated cost for this upgrade falsely inflated the network upgrade costs for AG1-412. Image 3 below shows the adjustment to the TC1 PhII model to mimic 150 MW of load at Mattameade.

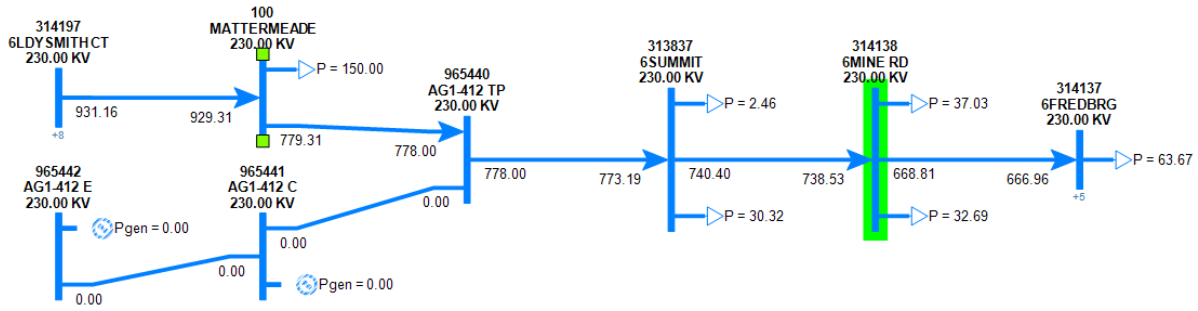

Image 3. Oneline diagram of the TC1 PhII Summer Peak 2027 model with an additional 150 MW load for the Mattameade Data Center.

Results

The original TC1 PhII study identified the risk of overloading the Aquia Harbor B – Garrisonville 230 kV circuit under the "DVP_P1-2: LN 2157_SRT-A" (Cranes to Fredericksburg) contingency. The loading showed at 102.69% of the applicable 898.64 MVA rating. When adding the 150 MW load as shown in Image 3, the same contingency is identified as the worst-case loading; however, it is only at 99.9%. Although this is very near the 100% threshold identified by PJM, it is not yet over.

Table 1. Final loadings of the Aquia Harbor B – Garrisonville 230 kV line with added 150 MW of load.

Bus #	Bus Name	Dfax	MW Impact	Pre-Queue AC %LD	Post-Queue AC %LD	Monitored Facility	Rating	Cont Name
965441	AG1-412 C	0.2058	16.46	94.55	99.9	314145 6AQUI_HARB_B 314140 6GARSVL 1 230	898.6	DVP_P1-2: LN 2157_SRT-A
965441	AG1-412 C	0.2058	16.46	90.36	95.7	314145 6AQUI_HARB_B 314140 6GARSVL 1 230	898.6	DVP_P1-2: LN 2104_SRT-A

When only considering 150 MW of load, the facility causing over a $120M cost to the AG1-412 project is no longer overloaded and therefore eliminating the need for the upgrade identified. If the full output of 770 MW was included in the study; then, it is expected that the loading would further decrease along Aquia Harbor B – Garrisonville.

[2] https://www.pjm.com/pjmfiles/pub/planning/project-queues/TC1/PHASE_2/AG1-412/AG1-412_imp_PHASE_2.htm

Conclusions

When developers are siting new generation interconnection projects, many are strategically located near data centers to directly serve increasing load demands. However, if PJM is not considering the increased load within the interconnection process, new generators are at risk of extremely high upgrade costs that could be avoided when the data centers come online. Unfortunately, projects like AG1-412 were strategically located, but received overstated yet avoidable network upgrade costs.

The PJM interconnection process has shifted from a serial process to a cycle process in order to efficiently study a larger number of requests simultaneously. Timmons Group commends the effort that PJM and the stakeholders went through to finalize the cycle process; however, there may be developmental challenges with the process as it is implemented in full for the first time with Transition Cycle 1. It is widely known that Northern Virginia is experiencing some of the highest data center growth in the world, requiring additional capacity to serve the load. If generation interconnection projects continue to receive unprecedented costs for upgrades that could be alleviated through appropriate load modeling, it will be nearly impossible to bring new projects in service in time to serve the increased load.

Exhibit B

PacifiCorp Generation Interconnection Cluster 1 Study Report Cluster Area 5

PACIFICORP

Generation Interconnection

Cluster 1 Study Report

Cluster Area 5

November 10, 2021

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1.0 SCOPE OF THE STUDY

Cluster Area 5 (CA5) is generally described as the Clover area and includes the following Interconnection Requests: C1-08, C1-09, C1-16, C1-34, C1-50, C1-51, C1-52, and C1-54.

Consistent with Attachment W, Section 3.4.2 and Section 51.4 of PacifiCorp's ("Transmission Provider") Open Access Transmission Tariff ("OATT"), this interconnection Transition Cluster Study ("Cluster Study") evaluated the impact of the proposed interconnections on the reliability of the Transmission System. The Cluster Study considered the Base Case as well as all generating facilities (and with respect to (iii) below, any identified Network Upgrades associated with such higher queued interconnection) that, on the date the Cluster Request Window closes:

(i) are existing and directly interconnected to the Transmission System;
(ii) are existing and interconnected to Affected Systems and may have an impact on the Interconnection Request;
(iii) have a pending higher queued or higher clustered interconnection request to interconnect to the transmission system; and
(iv) have executed an LGIA or requested that an unexecuted LGIA be filed with FERC.

The Cluster Study consisted of power flow, stability, and short circuit analyses.

This Cluster Study report provides the following information:

- identification of any circuit breaker short circuit capability limits exceeded as a result of the interconnection;
- identification of any thermal overload or voltage limit violations resulting from the interconnection;
- identification of any instability or inadequately damped response to system disturbances resulting from the interconnection and
- description and non-binding, good faith estimated cost of facilities required to interconnect the Generating Facilities to the Transmission System and to address the identified short circuit, instability, and power flow issues.

2.0 STUDY ASSUMPTIONS

- All active higher priority transmission service and/or generator interconnection requests that were considered in this study are listed in Appendix 2. If any of these requests are withdrawn, the Transmission Provider reserves the right to restudy this request, and the results and conclusions could significantly change.
- For study purposes there are two separate queues:
    - Transmission Service Queue: to the extent practical, all network upgrades that are required to accommodate active transmission service requests were modeled in this study.
    - Generation Interconnection Queue: Interconnection Facilities and network upgrades associated with higher queued or higher clustered interconnection requests were modeled in this study.
- The Interconnection Customers' request for energy or network resource interconnection

service in and of itself does not request or convey transmission service. Only a Network Customer may make a request to designate a generating resource as a Network Resource. Because the queue of higher priority transmission service requests may be different when a Network Customer requests network resource designation for this Generating Facility, the available capacity or transmission modifications, if any, necessary to provide Network Integration Transmission Service may be significantly different. Therefore, Interconnection Customers should regard the results of this study as informational rather than final.

- Under normal conditions, the Transmission Provider does not dispatch or otherwise directly control or regulate the output of generating facilities. Therefore, the need for transmission modifications, if any, that may be required to provide Network Resource Interconnection Service will be evaluated on the basis of 100 percent deliverability (i.e., this study did not model displacement of other resources in the same area).
- This study assumed the Projects will be integrated into the Transmission Provider's system at agreed upon and/or proposed Points of Interconnection ("POI" or "POIs").
- If Interconnection Customers proceed through the interconnection process, they will be required to construct and own any facilities required between the Point of Change of Ownership and the Project unless specifically identified by the Transmission Provider.
- Line reconductor or fiber underbuild required on existing poles were assumed to follow the most direct path on the Transmission Provider's system. If during detailed design the path must be modified it may result in additional cost and timing delays for the Interconnection Customer's Project.
- Generator tripping may be required for certain outages.
- All facilities will meet or exceed the minimum Western Electricity Coordinating Council ("WECC"), North American Electric Reliability Corporation ("NERC"), and the Transmission Provider's performance and design standards.
- Power flow analysis requires WECC base cases to reliably balance under peak load conditions the aggregate of generation in the local area, with the Generating Facility at full output, to the aggregate of the load in the Transmission Provider's Transmission System. As the ("PACE") balancing authority area ("BAA") has more existing and proposed generation than load, it is necessary to assume some portion of other resources are displaced by this Project's output in order to assess the impact of interconnecting this Project's generation to transmission system operations. For the purposes of this study, generation in the Transmission Provider's Wyoming, Idaho, Northern Utah, and Southeast Utah areas were assumed to be displaced.
- The following Transmission Provider planned system improvements were assumed in service:
    - Energy Gateway South Planned project (Q4 2024)
    - Lakeside I Remedial Action Scheme (RAS) modification (Q2 2022)
    - Magna Cap Bank (Q4 2023)
    - Camp Williams bus improvements (Q4 2024)
    - Cottonwood - Snyderville Reconductor (Q4 2024)
- The following system improvements assigned to higher priority Interconnection and/or Transmission Service Requests were assumed in service:
    - Milford 138 kV three breaker ring bus and Blundell 138 kV breaker (Q0820)
    - New South Milford – Milford 46 kV line (Q0820)

  - Rebuild of Cameron – Tushar - Sevier Tap – Sigurd 138 kV line (Q0820)
  - Upgrade of the Emery 345-138 kV transformers (Q0823)
  - Lakeside II RAS modifications (TSR Q2867)
  - Construction of a new 50-mile Spanish Fork -Mercer 345 kV line. The new line will be terminated in an existing bay in the Spanish Fork substation. A new circuit breaker will be installed in Mercer substation. (Transition Cluster)
  - Replace jumpers on the Huntington end of the Emery-Huntington 345 kV transmission line. (Transition Cluster)
  - Upgrade 75 MVA 138-46 kV LTC transformer at Milford substation to 125 MVA transformer. (Transition Cluster)
  - Rebuild both the existing and the new (identified for Q0820) 46 kV Milford – South Milford transmission lines. (Transition Cluster)
- Due to the relatively close locations of C1-50 and C1-54 to higher priority Interconnection Request TCS-09 the Transmission Provider assumes that both C1-50 and C1-54 will need to interconnect in the new POI substation proposed to be constructed as part of TCS-09
- This report is based on information available at the time of the study. It is the Interconnection Customer's responsibility to check the Transmission Provider's web site regularly for Transmission System updates at https://www.oasis.oati.com/ppw

3.0 GENERATING FACILITY REQUIREMENTS

The following requirements are applicable to all Interconnection Requests. The Transmission Provider will identify any site-specific generating facility requirements in addition to the following in this report and in facilities studies. Certain Interconnection Requests requesting service at a voltage level traditionally defined as distribution may be subject to the transmission interconnection request requirements listed below should the Transmission Provider make that determination.

3.1 Transmission Voltage Interconnection Requests

All interconnecting synchronous and non-synchronous generators are required to design their Generating Facilities with reactive power capabilities necessary to operate within the full power factor range of 0.95 leading to 0.95 lagging. This power factor range shall be dynamic and can be met using a combination of the inherent dynamic reactive power capability of the generator or inverter, dynamic reactive power devices and static reactive power devices to make up for losses.

For synchronous generators, the power factor requirement is to be measured at the POI. For non-synchronous generators, the power factor requirement is to be measured at the high-side of the generator substation.

The Generating Facility must provide dynamic reactive power to the system in support of both voltage scheduling and contingency events that require transient voltage support, and must be able to provide reactive capability over the full range of real power output.

If the Generating Facility is not capable of providing positive reactive support (i.e., supplying reactive power to the system) immediately following the removal of a fault or other transient low voltage perturbations, the facility must be required to add dynamic voltage support equipment.

These additional dynamic reactive devices shall have correct protection settings such that the devices will remain on line and active during and immediately following a fault event.

Generators shall be equipped with automatic voltage-control equipment and normally operated with the voltage regulation control mode enabled unless written authorization (or directive) from the Transmission Provider is given to operate in another control mode (e.g. constant power factor control). The control mode of generating units shall be accurately represented in operating studies. The generators shall be capable of operating continuously at their maximum power output at its rated field current within +/- 5% of its rated terminal voltage.

All generators are required to ensure the primary frequency capability of their Facility by installing, maintaining, and operating a functioning governor or equivalent controls as indicated in FERC Order 842.

As required by NERC standard VAR-001-4.2, the Transmission Provider will provide a voltage schedule for the POI. In general, Generating Facilities should be operated so as to maintain the voltage at the POI, typically between 1.00 per unit to 1.04 per unit, or other designated point as deemed appropriated by Transmission Provider. The Transmission Provider may also specify a voltage and/or reactive power bandwidth as needed to coordinate with upstream voltage control devices such as on-load tap changers. At the Transmission Provider's discretion, these values might be adjusted depending on operating conditions.

Generating Facilities capable of operating with a voltage droop are required to do so. Voltage droop control enables proportionate reactive power sharing among Generation Facilities. Studies will be required to coordinate voltage droop settings if there are other facilities in the area. It will be the Interconnection Customer's responsibility to ensure that a voltage coordination study is performed, in coordination with Transmission Provider, and implemented with appropriate coordination settings prior to unit testing.

For areas with multiple generating facilities additional studies may be required to determine whether or not critical interactions, including but not limited to control systems, exist. These studies, to be coordinated with Transmission Provider, will be the responsibility of the Interconnection Customer. If the need for a master controller is identified, the cost and all related installation requirements will be the responsibility of the Interconnection Customer. Participation by the Generation Facility in subsequent interaction/coordination studies will be required pre- and post-commercial operation in order ensure system reliability.

Interconnection Requests that are 75 MVA or larger may be required to facilitate collection and validation of accurate modeling data to meet NERC modeling standards. The Transmission Provider, in its roles as the Planning Coordinator, requires Phasor Measurement Units (PMUs) at all new Generating Facilities with an individual or aggregate nameplate capacity of 75 MVA or greater. In addition to owning and maintaining the PMU, the Generating Facility will be responsible for collecting, storing (for a minimum of 90 days) and retrieving data as requested by the Planning Coordinator. Data must be stored for a minimum of 90 days. Data must be collected and be able to stream to Planning Coordinator for each of the Generating Facility's step-up transformers measured on the low side of the GSU at a sample rate of at least 60 samples per

second and synchronized within +/- 2 milliseconds of the Coordinated Universal Time (UTC). Initially, the following data must be collected:

- Three phase voltage and voltage angle (analog)
- Three phase current (analog)

Data requirements are subject to change as deemed necessary to comply with local and federal regulations.

All generators must meet the Federal Energy Regulatory Commission (FERC), North American Electric Reliability Corporation (NERC) and WECC low voltage ride-through requirements as specified in the interconnection agreement. Inverters must be designed to stay connected to the grid in the case of severe faults and may not momentarily cease output within the no-trip area of the voltage curves. Figure 1 illustrates the voltage ride-through capability as per NERC PRC-024. Importantly, inverters should be designed such that a trip outside of the curves is a "may-trip" area (if needed to protect equipment) not a "must-trip" area. Inverters that momentarily cease active power output for these voltage excursions should be configured to restore output to pre-disturbance levels in no greater than five seconds, provided the inverter is capable of these changes. Generators must provide test results to the Transmission Provider verifying that the inverters for this Project have been programmed to meet all PRC-024 requirements rather than manufacturer IEEE distribution standards.

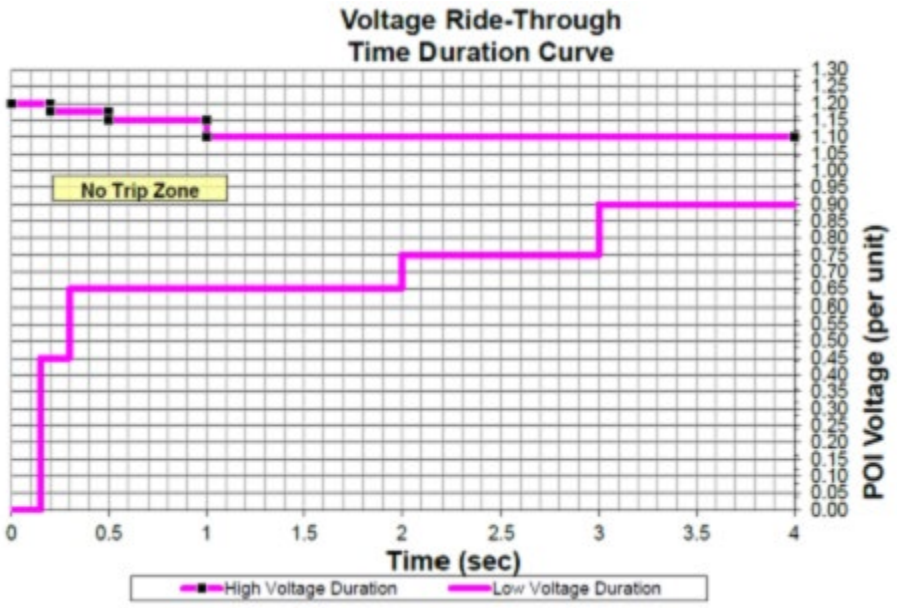

Figure 1 – Voltage Ride-Through Curve

As the Transmission Provider cannot submit a user written model to WECC for inclusion in base cases, a standard model from the WECC Approved Dynamic Model Library is required 180 days prior to trial operation. The list of approved generator models is continually updated and is available on the http://www.WECC.biz website.

Interconnection Customer with an Interconnection Request for a Generating Facility that is both 75 MVA or larger as well as being interconnected at a voltage higher than 100 kV shall register

with NERC as the Generator Owner ("GO") and Generator Operator ("GOP") for the Large Generating Facility and provide the Transmission Provider documentation demonstrating registration in order to be approved for Commercial Operation. This registration must be maintained throughout the lifetime of the Interconnection Agreement.

Interconnection Customers are responsible for the protection of transmission lines between the Generating Facility and the POI substation. For Interconnection Requests that are smaller than 75 MVA or are interconnected at a voltage less than 100 kV which have a tie line that is longer than 1,000 feet the Interconnection Customer shall construct and own a tie-line substation to be located at the change of ownership (separate fenced facility adjacent to the Transmission Provider's POI substation). The tie line substation shall include an Interconnection Customer owned protective device and associated transmission line relaying/communications. The ground grids of the Transmission Provider's POI substation and the Interconnection Customer's tie-line substation will be connected to support the use of a bus differential protection scheme which will protect the overhead bus connection between the two facilities.

3.2 Distribution Voltage Interconnection Requests

The Generating Facility and interconnection equipment owned by the Interconnection Customers are required to operate under constant power factor mode with a unity power factor setting unless specifically requested otherwise by the Transmission Provider. The Generating Facilities are expressly forbidden from actively participating in voltage regulation of the Transmission Provider's system without written request or authorization from the Transmission Provider. The Generating Facilities shall have sufficient reactive capacity to enable the delivery of 100 percent of the plant output to the applicable POI at unity power factor measured at 1.0 per unit voltage under steady state conditions.

Generators capable of operating under voltage control with voltage droop are required to do so. Studies will be required to coordinate the voltage droop setting with other facilities in the area. In general, the Generating Facility and Interconnection Equipment should be operated so as to maintain the voltage at the POI between 1.01 pu to 1.04 pu. At the Public Utility's discretion, these values might be adjusted depending on the operating conditions. Within this voltage range, the Generating Facility should operate so as to minimize the reactive interchange between the Generating Facility and the Public Utility's system (delivery of power at the POI at approximately unity power factor). The voltage control settings of the Generating Facility must be coordinated with the Public Utility prior to energization (or interconnection). The reactive compensation must be designed such that the discreet switching of the reactive device (if required by the Interconnection Customer) does not cause step voltage changes greater than +/-3% on the Public Utility's system.

All generators must meet applicable WECC low voltage ride-through requirements as specified in the interconnection agreement.

As per NERC standard VAR-001-1, the Public Utility is required to specify voltage or reactive power schedule at the POI. Under normal conditions, the Public Utility's system should not supply reactive power to the Generating Facility.

4.0 CLUSTER AREA DEFINITIONS

The Transmission Provider performed the Transition Cluster Study based on geographically and/or electrically relevant areas on the Transmission Provider's Transmission System known as Cluster Areas. The Transmission Provider has determined that the Interconnection Requests discussed in Section 5.0 are located in a geographically and/or electrically relevant area on Transmission Provider's Transmission System, and thus, were assigned Cluster Area 5 in the Transition Cluster Study process.

5.0 CLUSTER AREA 5

Cluster Area 5 (CA5) is generally described as the Clover area. This Cluster Area consists of the following Interconnection Requests.

5.1 Description of Interconnection Request – C1-08

The Interconnection Customer has proposed to interconnect 199 megawatts ("MW") of new solar and battery storage generation to PacifiCorp's ("Transmission Provider") 345 kV Clover substation located in Juab County, Utah. The Interconnection Request is proposed to consist of fifty-eight (58) TMEIC Ninja 4200 PV-PCS solar inverters for a total output of 199 MW at the POI. The Interconnection Request also consists of fifty-eight (58) TMEIC Ninja 4200 ESS-PCS battery storage inverters. The requested commercial operation date is May 1, 2024. Figure 2 below, is a one-line diagram that illustrates the interconnection of the proposed Generating Facility to the Transmission Provider's system.

Interconnection Customer will NOT operate this generator as a Qualified Facility as defined by the Public Utility Regulatory Policies Act of 1978 (PURPA).

The Interconnection Request will be studied for both Network Resource Interconnection Service ("NRIS") and Energy Resource Interconnection Service ("ERIS").

The Transmission Provider has assigned the Project Cluster Number "C1-08"

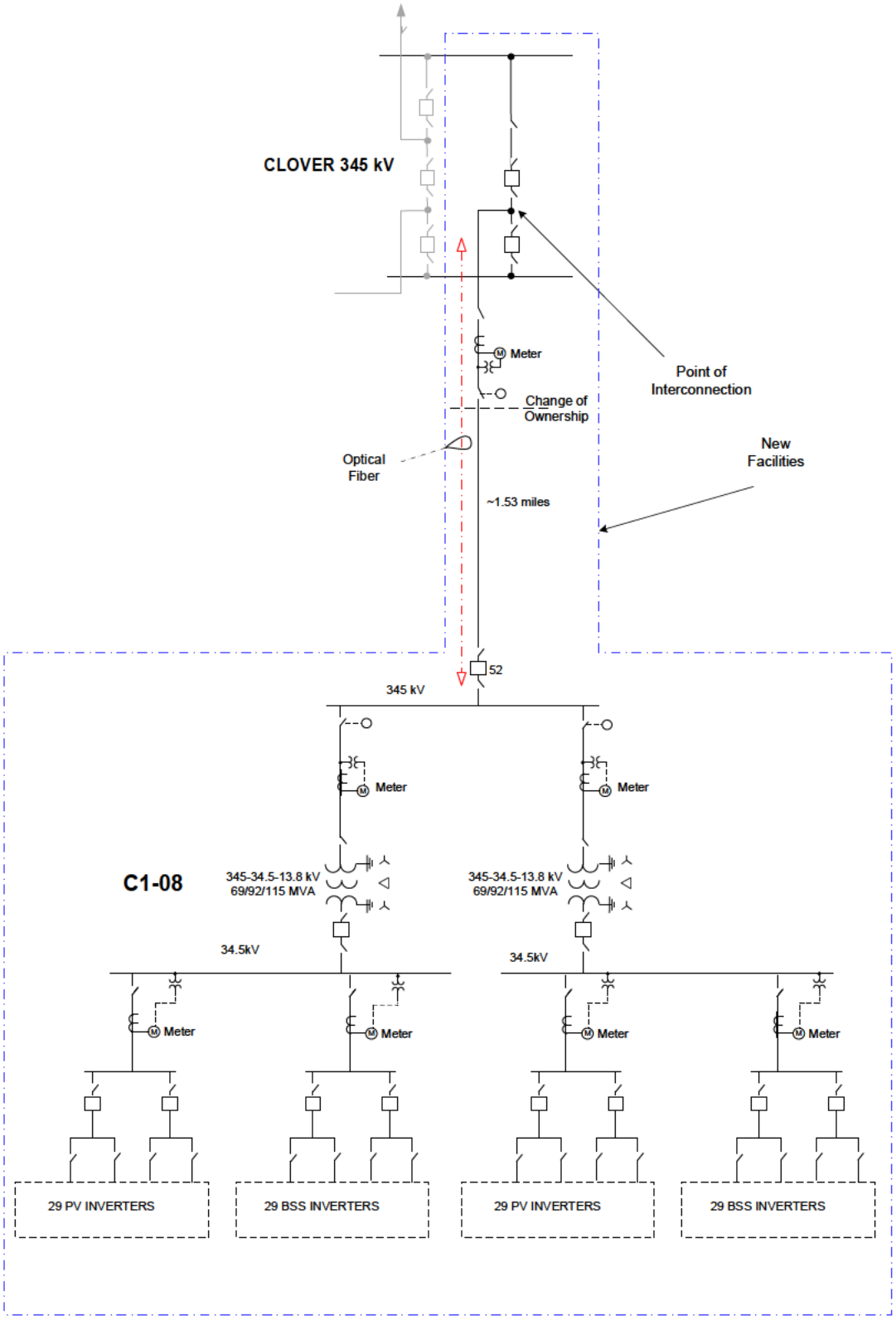

Figure 2: Simplified System One Line Diagram C1-08

5.2 Description of Interconnection Request – C1-09

The Interconnection Customer has proposed to interconnect 199 megawatts ("MW") of new solar and battery storage generation to PacifiCorp's ("Transmission Provider") 345 kV Clover substation located in Juab County, Utah. The Interconnection Request is proposed to consist of fifty-eight (58) TMEIC Ninja 4200 PV-PCS solar inverters for a total output of 199 MW at the POI. The Interconnection Request also consists of fifty-eight (58) TMEIC Ninja 4200 ESS-PCS battery storage inverters. The requested commercial operation date is May 1, 2024. Figure 3 below, is a one-line diagram that illustrates the interconnection of the proposed Generating Facility to the Transmission Provider's system.

Interconnection Customer will NOT operate this generator as a Qualified Facility as defined by the Public Utility Regulatory Policies Act of 1978 (PURPA).

The Interconnection Request will be studied for both Network Resource Interconnection Service ("NRIS") and Energy Resource Interconnection Service ("ERIS").

The Transmission Provider has assigned the Project Cluster Number "C1-09"

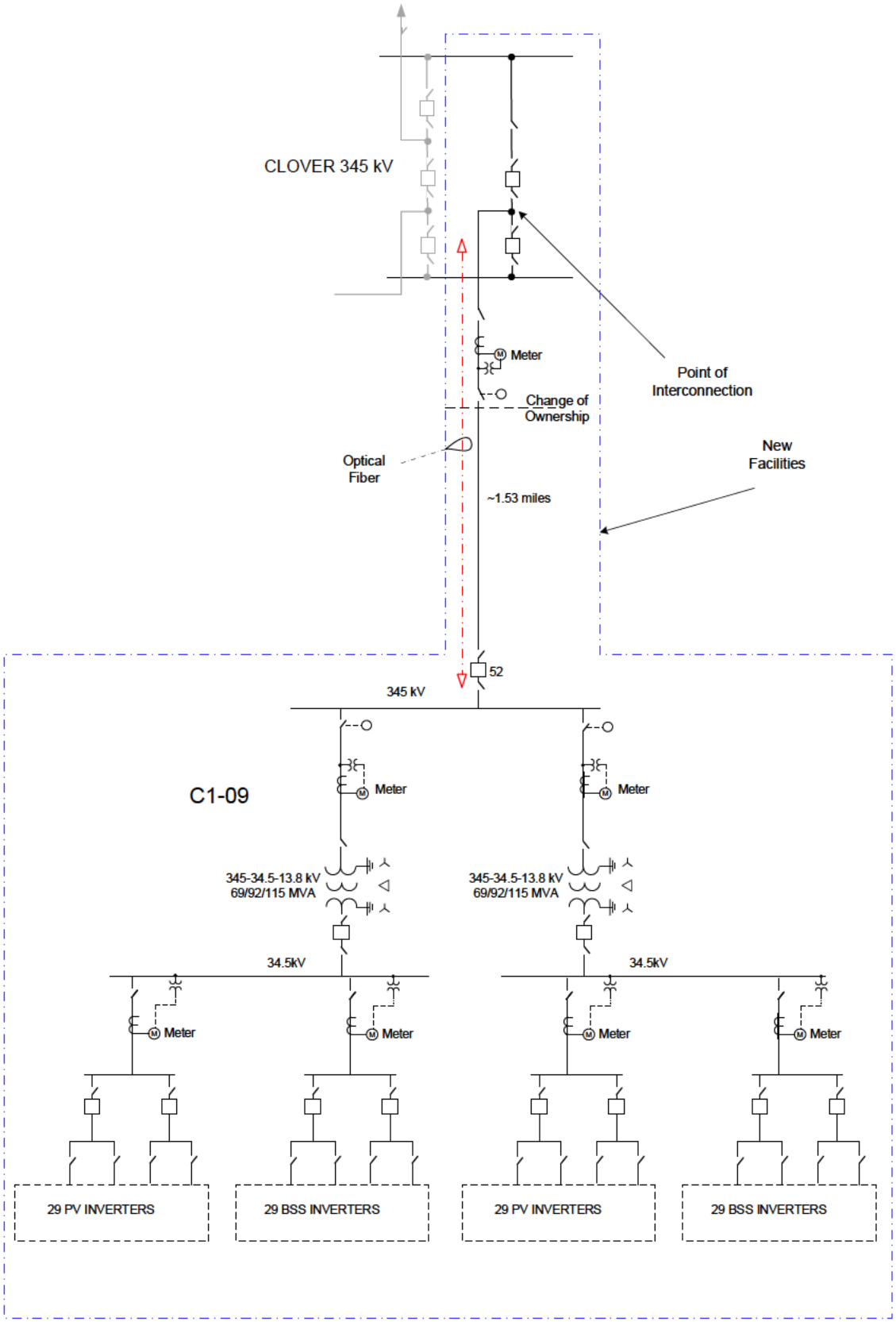

Figure 3: Simplified System One Line Diagram C1-09

5.3 Description of Interconnection Request – C1-16

The Interconnection Customer has proposed to interconnect 199 megawatts ("MW") of new solar and battery storage generation to PacifiCorp's ("Transmission Provider") 230 kV Blackrock substation located in Millard County, Utah. The Interconnection Request is proposed to consist of sixty-two (62) Sungrow SG3600UD-MV solar inverters for a total output of 199 MW at the POI. The Interconnection Request will also consist of sixty-five (65) Sungrow SC3450US-MV-US battery storage inverters. The requested commercial operation date is December 31, 2025. Figure 4 below, is a one-line diagram that illustrates the interconnection of the proposed Generating Facility to the Transmission Provider's system.

Interconnection Customer will NOT operate this generator as a Qualified Facility as defined by the Public Utility Regulatory Policies Act of 1978 (PURPA).

The Interconnection Request will be studied for Network Resource Interconnection Service ("NRIS").

The Transmission Provider has assigned the Project Cluster Number "C1-16"

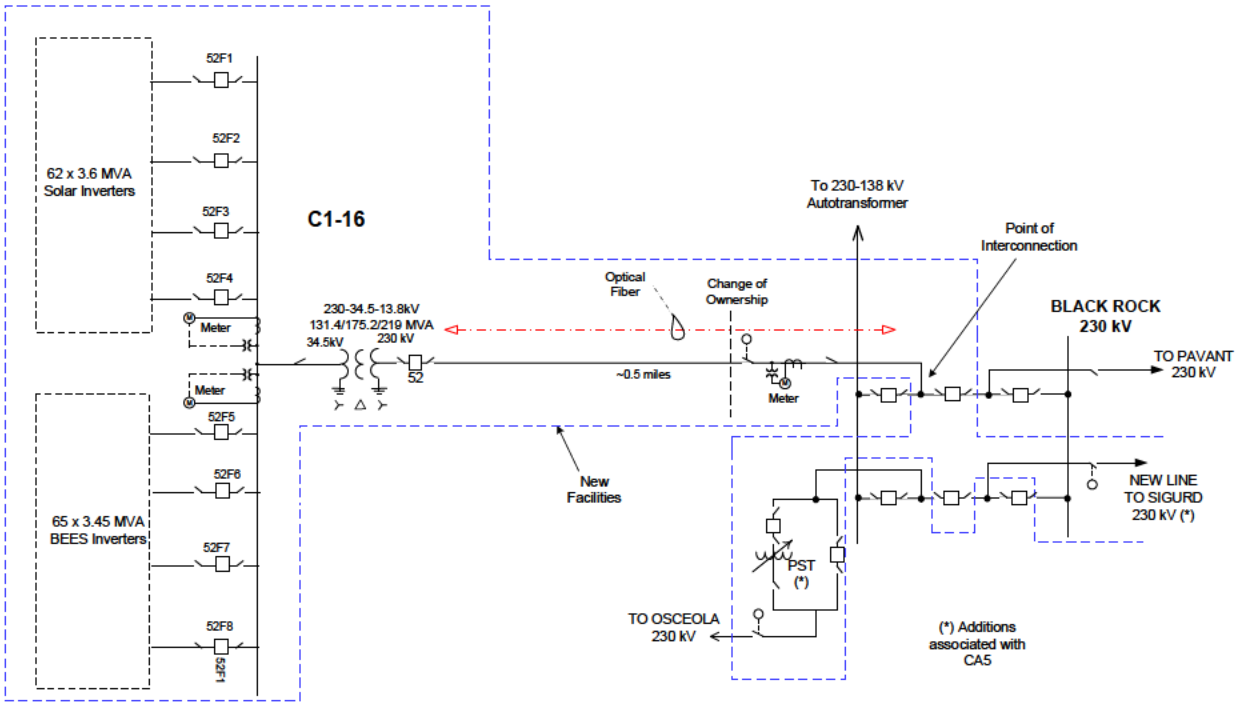

Figure 4: Simplified System One Line Diagram C1-16

5.4 Description of Interconnection Request – C1-34

The Interconnection Customer has proposed to interconnect 200 megawatts ("MW") of new solar generation to PacifiCorp's ("Transmission Provider") 138 kV Mercer substation located in Utah County, Utah. The Interconnection Request is proposed to consist of seventy-two (72) Power Electronics FS3190M solar inverters for a total output of 200 MW at the POI. The requested

commercial operation date is December 31, 2024. Figure 5 below, is a one-line diagram that illustrates the interconnection of the proposed Generating Facility to the Transmission Provider's system.

Interconnection Customer will NOT operate this generator as a Qualified Facility as defined by the Public Utility Regulatory Policies Act of 1978 (PURPA).

The Interconnection Request will be studied for both Network Resource Interconnection Service ("NRIS") and Energy Resource Interconnection Service ("ERIS").

The Transmission Provider has assigned the Project Cluster Number "C1-34"

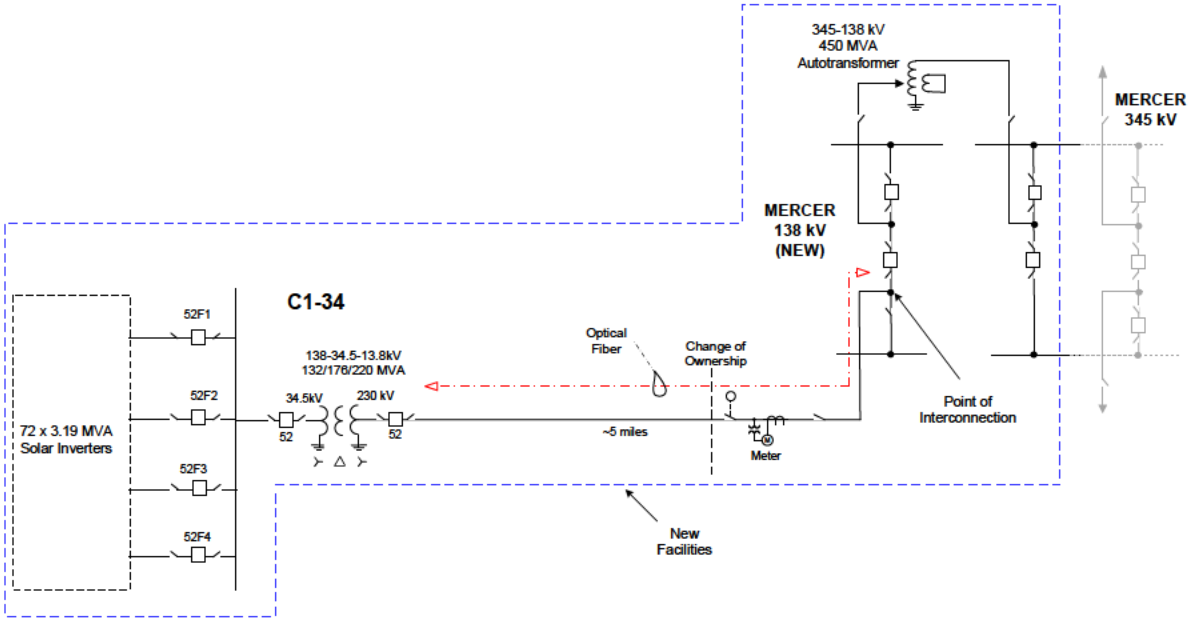

Figure 5: Simplified System One Line Diagram C1-34

5.5 Description of Interconnection Request – C1-50

The Interconnection Customer has proposed to interconnect 400 megawatts ("MW") of new solar and battery storage generation to PacifiCorp's ("Transmission Provider") Camp Williams-Mona #1 345 kV transmission line located in Utah County, Utah. The Interconnection Request is proposed to consist of one hundred forty (140) Sungrow SG3150U solar inverters for a total output of 400 MW at the POI. The Interconnection Request also consists of thirty-six (36) Sungrow SG3150U battery storage inverters. The requested commercial operation date is October 1, 2024. Figure 6 below, is a one-line diagram that illustrates the interconnection of the proposed Generating Facility to the Transmission Provider's system.

Interconnection Customer will NOT operate this generator as a Qualified Facility as defined by the Public Utility Regulatory Policies Act of 1978 (PURPA).

The Interconnection Request will be studied for both Network Resource Interconnection Service ("NRIS") and Energy Resource Interconnection Service ("ERIS").

The Transmission Provider has assigned the Project Cluster Number "C1-50"

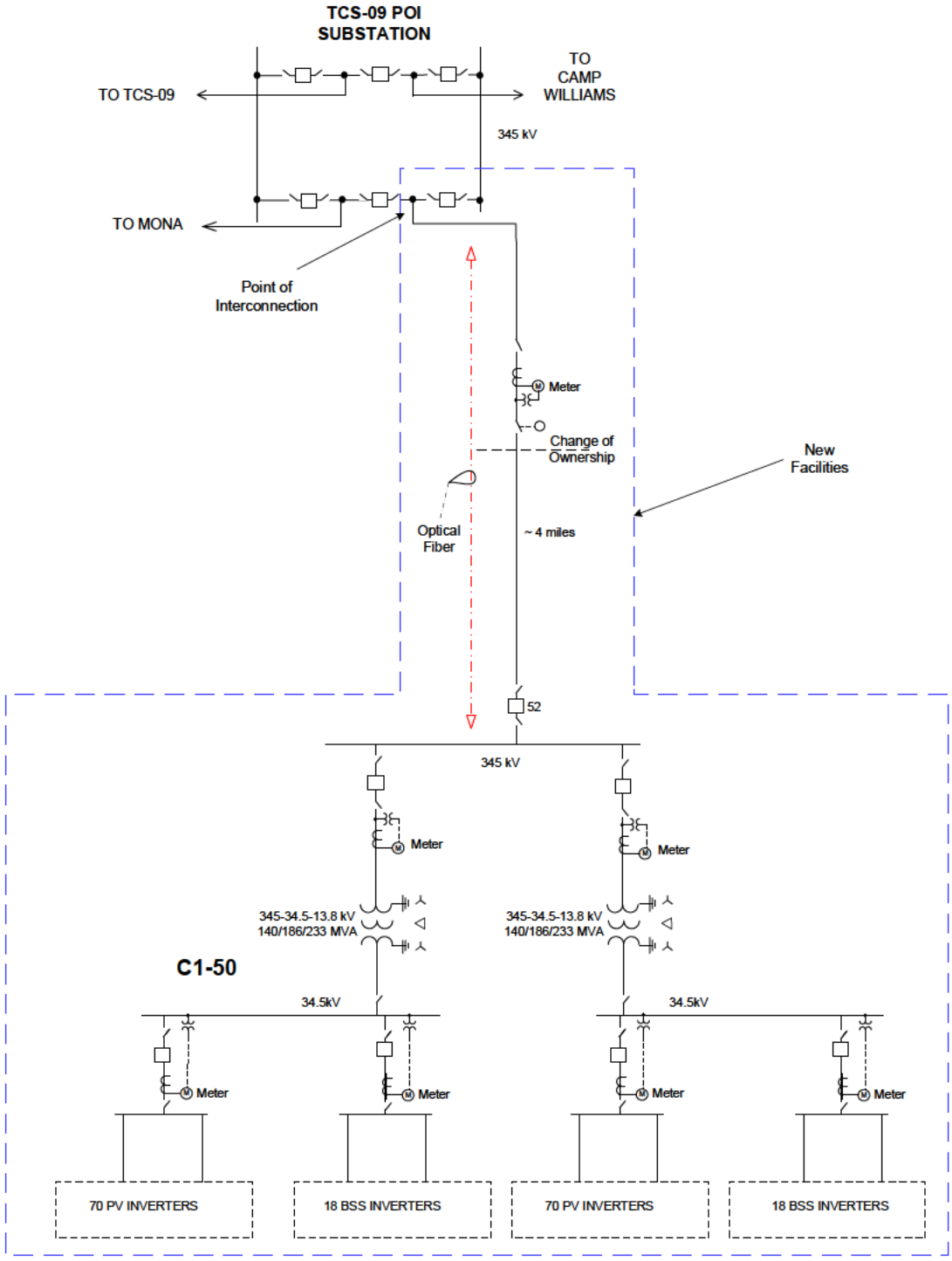

Figure 6: Simplified System One Line Diagram C1-50

5.6 Description of Interconnection Request – C1-51

The Interconnection Customer has proposed to interconnect 300 megawatts ("MW") of new solar and battery storage generation to PacifiCorp's ("Transmission Provider") 345 kV Clover substation located in Juab County, Utah. The Interconnection Request is proposed to consist of eighty-four (84) TMEIC PVU-L0840GR, 4.20 MVA solar inverters for a total output of 300 MW at the POI. The Interconnection Request will also consist of twenty-eight (28) TMEIC BSU=L0640GR, 3.20 MVA battery storage inverters. The requested commercial operation date is October 1, 2024. Figure 7 below, is a one-line diagram that illustrates the interconnection of the proposed Generating Facility to the Transmission Provider's system.

Interconnection Customer will NOT operate this generator as a Qualified Facility as defined by the Public Utility Regulatory Policies Act of 1978 (PURPA).

The Interconnection Request will be studied for both Network Resource Interconnection Service ("NRIS") and Energy Resource Interconnection Service ("ERIS").

The Transmission Provider has assigned the Project Cluster Number "C1-51"

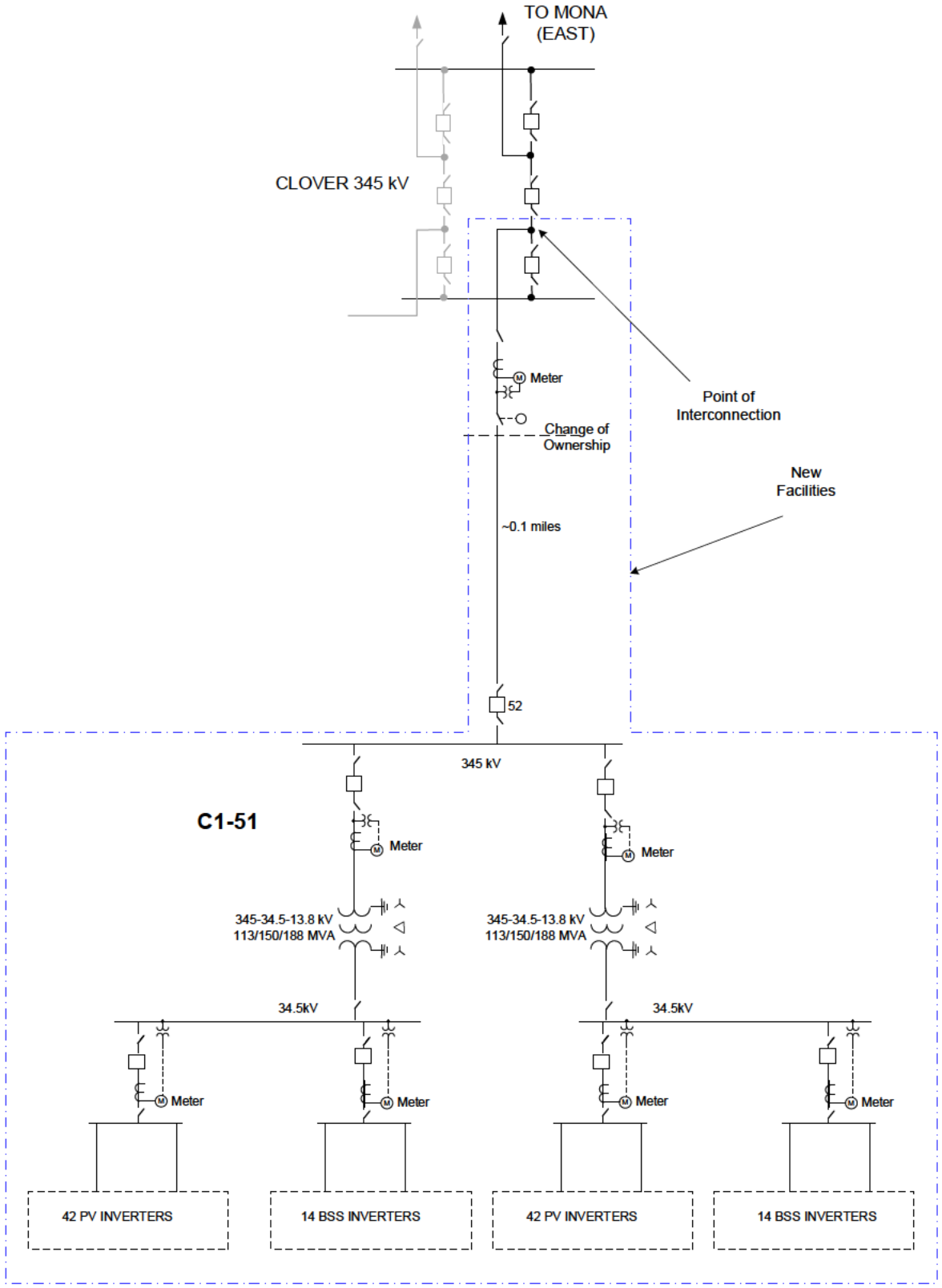

Figure 7: Simplified System One Line Diagram C1-51

5.7 Description of Interconnection Request – C1-52

The Interconnection Customer has proposed to interconnect 250 megawatts ("MW") of new solar and battery storage generation to PacifiCorp's ("Transmission Provider") planned 345 kV Limber substation located in Tooele County, Utah. The Interconnection Request is proposed to consist of seventy (70) TMEIC PVU-L0840GR, 4.20 MVA solar inverters for a total output of 250 MW at the POI. The Interconnection Request will also consist of twenty-four (24) TMEIC BSU-L0640GR, 3.20 MVA battery storage inverters. The requested commercial operation date is October 1, 2024. Figure 2 below, is a one-line diagram that illustrates the interconnection of the proposed Generating Facility to the Transmission Provider's system.

Interconnection Customer will NOT operate this generator as a Qualified Facility as defined by the Public Utility Regulatory Policies Act of 1978 (PURPA).

The Interconnection Request will be studied for both Network Resource Interconnection Service ("NRIS") and Energy Resource Interconnection Service ("ERIS").

The Transmission Provider has assigned the Project Cluster Number "C1-52"

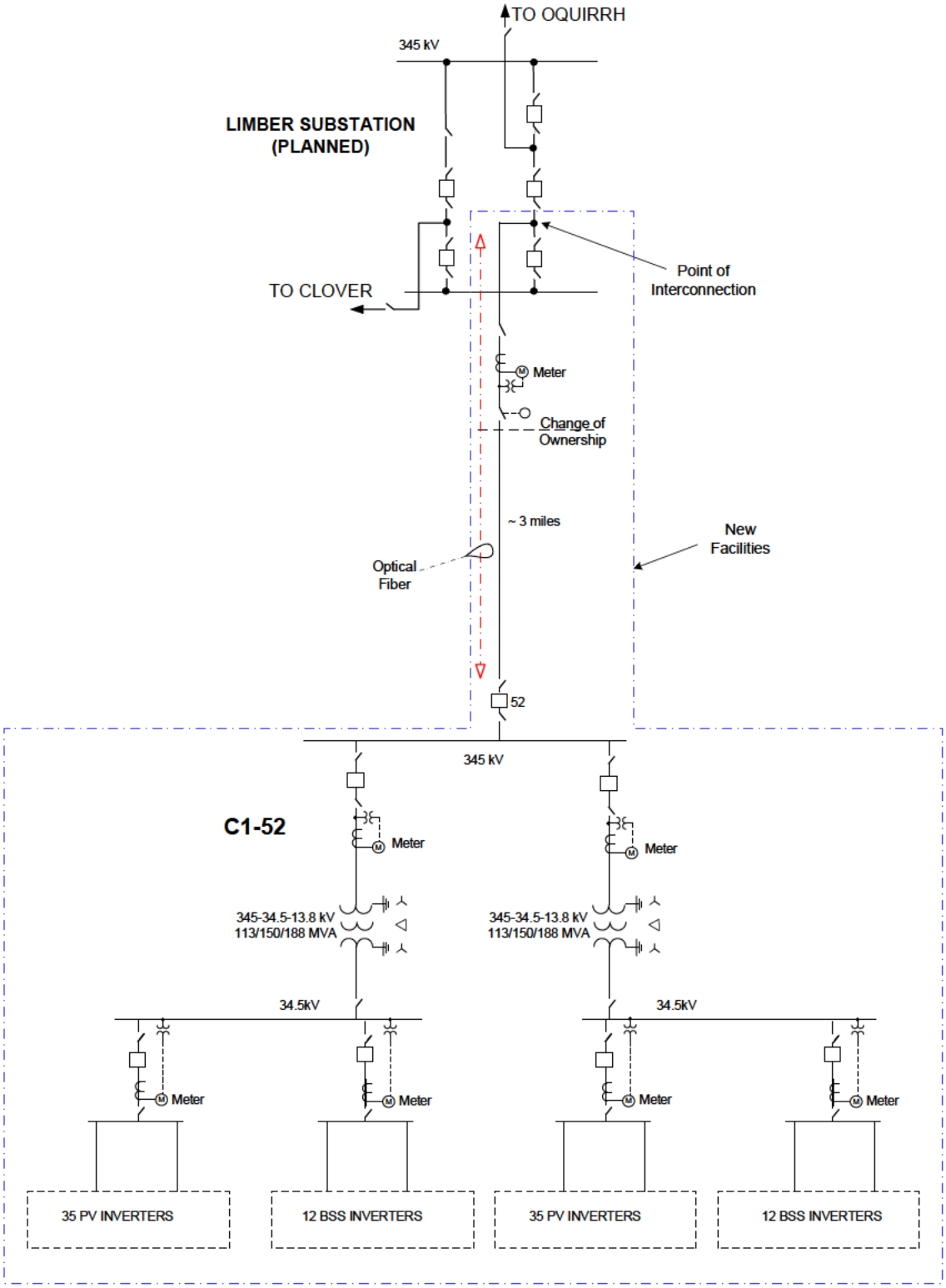

Figure 7: Simplified System One Line Diagram C1-52

5.8 Description of Interconnection Request – C1-54

The Interconnection Customer has proposed to interconnect 199 megawatts ("MW") of new solar and battery storage generation to PacifiCorp's ("Transmission Provider") Camp Williams-Mona #1 345 kV transmission line located in Utah County, Utah. The Interconnection Request is proposed to consist of seventy (70) Sungrow SG3150U, 3.15 MVA solar inverters for a total output of 199 MW at the POI. The Interconnection Request will also consist of twenty (20) Sungrow SC2500U, 2.75 MVA battery storage inverters. The requested commercial operation date is October 31, 2024. Figure 8 below, is a one-line diagram that illustrates the interconnection of the proposed Generating Facility to the Transmission Provider's system.

Interconnection Customer will NOT operate this generator as a Qualified Facility as defined by the Public Utility Regulatory Policies Act of 1978 (PURPA).

The Interconnection Request will be studied for Energy Resource Interconnection Service ("ERIS").

The Transmission Provider has assigned the Project Cluster Number "C1-54"

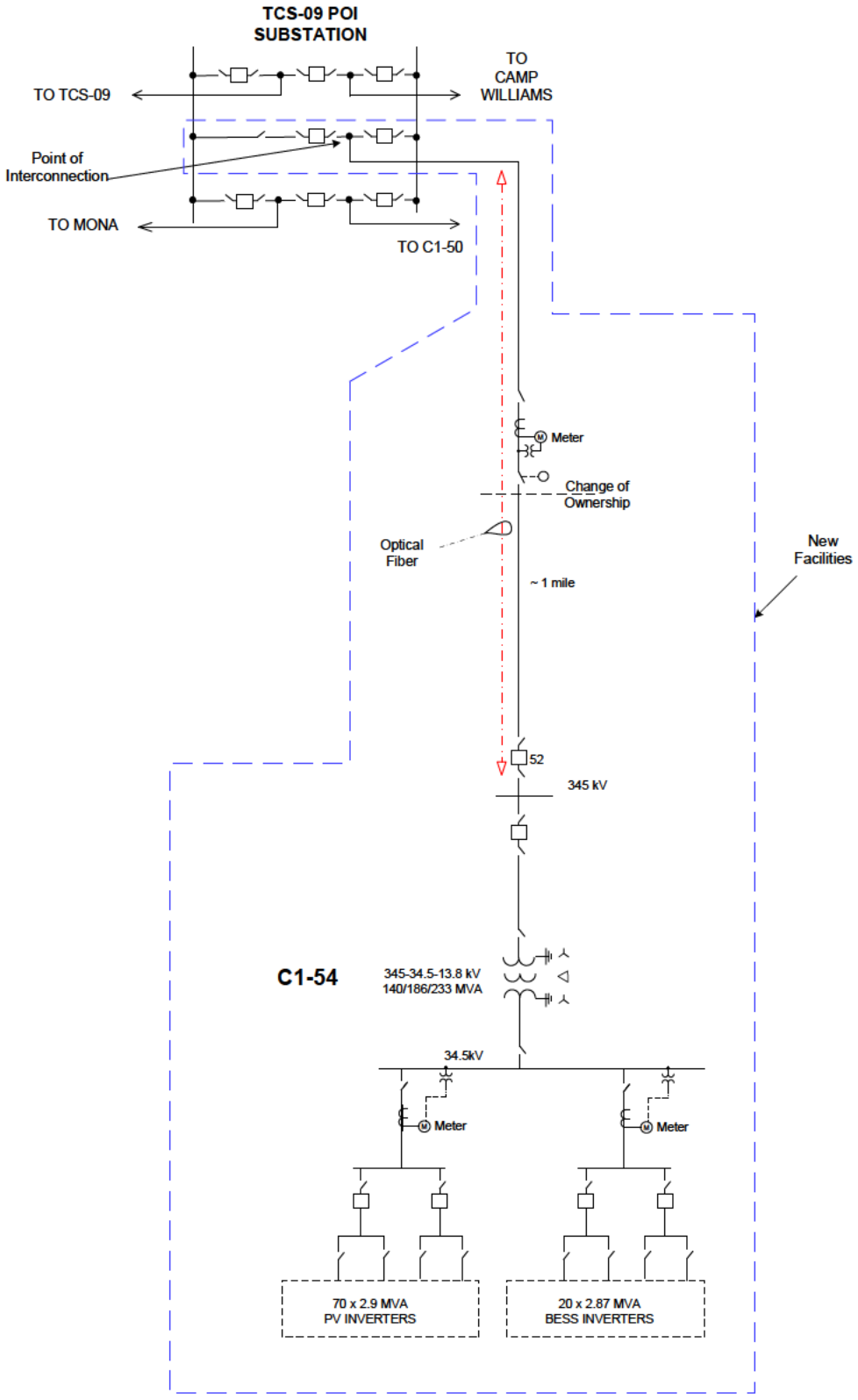

Figure 8: Simplified System One Line Diagram C1-54

6.0 SITE SPECIFIC GENERATING FACILITY REQUIREMENTS

In addition to the requirements described above the following Generating Facility are required for the specific Interconnection Requests listed below.

None.

7.0 TRANSMISSION PROVIDER SYSTEM REQUIREMENTS - ERIS

7.1 Transmission System Requirements

The following transmission system improvements are required to accommodate the Interconnection Requests in this Cluster Area:

- Construct a new 63.5-mile Black Rock–Sigurd 230 kV line with 1272 ACSR. Build a new 230 kV line position at Black Rock and one at Sigurd
- Install a 230 kV phase shifting transformer on the Black Rock–Osceola line 1272 ACSR
- Install a new 345/138 kV transformer at Oquirrh substation
- Rebuild McClelland–Midvalley 138 kV line to 1272 ACSR (6.5 miles)
- Loop the 90th South-Terminal 345 kV line into the Midvalley substation (Energy Gateway planned project)
- Fix clearance issue on 5 spans of the Emery–Spanish Fork 345 kV line
- Fix clearance issue on 20 spans of the Camp Williams–Mona #1 345 kV line
- Construct a new 14-mile Oquirrh–Terminal #3 345 kV line (Energy Gateway planned project)
- Rebuild 5.6 miles of the Clover–Nebo 138 kV line to 1272 ACSR
- Replace 345/230 kV transformers #5 and #6 at Sigurd with 700 MVA units each

Refer to Appendix 1 for more details regarding the necessity for these required upgrades.

The following are station upgrades required for each of the Interconnection Requests within this Cluster Area.

C1-08/C1-09
Construct a new 345 kV bay at Clover substation with two breakers and associated switches.

C1-16
Install a new breaker and associated switches into one of the existing 230 kV bays at Black Rock substation.

C1-34
Construct the following upgrades at Mercer Substation

- Construct a new 345 kV bay with 2 breakers and associated switches
- Install a new 345/138 kV autotransformer with an initial capacity of 450 MVA
- Construct a new partially developed 138 kV breaker-and-a-half substation, with two breakers and associated switches.

C1-50
Add a new breaker and associated switches to the TCS-09 POI substation. A voltage coordination study will be required for the Interconnection Requests at this location.

C1-51
Install a new breaker into existing 345 kV bay in Clover substation.

C1-52
Construct the Transmission Provider's planned Limber substation. The substation will initially be constructed three breaker 345 kV ring bus expandable to breaker and a half configuration.

C1-54
Install a new 345 kV bay with four circuit breakers and associated switches to the TCS-09 POI substation. A voltage coordination study will be required for the Interconnection Requests at this location.

7.2 Distribution System Requirements

None.

7.3 Transmission Line Requirements

The following new transmission line is required to be constructed. The Transmission Provider assumes new rights-of-way will be required and that portions of the line will need to be constructed on federal land requiring environmental permitting processes.

- A new ~63.5-mile 230 kV line has been identified to be required between Black Rock and Sigurd substations.

The following existing transmission lines are required to be rebuilt. The Transmission Provider assumes all structures will need to be replaced.

- A 6.5-mile line segment of the existing McClelland - Midvalley 138kV line has been identified to be rebuilt to accommodate a larger conductor.
- A 5.6-mile line segment of the existing Clover – Nebo 138kV line has been identified to be rebuilt with a larger conductor.

The following existing transmission lines will need to be modified to meet higher clearance standards. The Transmission Provider assumes all structures will need to be replaced to meet clearance.

- The Emery - Spanish Fork 345kV line has 5-spans limiting the line rating due to NESC required clearances.
- The Camp Williams – Mona #1 345kV line has 20-spans limiting the line rating due to NESC required clearances.

C1-52

- This interconnection request requires connection of the existing Clover – Oquirrh 345 kV line to Limber substation. The exact location of Limber substation and the associated line routes are currently unknown. Cost estimates provided for this work assume that Limber substation is located directly adjacent to the existing Clover – Oquirrh 345kV line.

Each of the Interconnection Requests in this Cluster Area shall construct its last structure and span/bus connection into the POI substation to Transmission Provider standards. The Transmission Provider will review the design of the Interconnection Customer line for the last span into the POI substations. The Interconnection Customers shall coil enough fiber and conductor on the last dead-end structure to make the span into the POI substations. The Transmission Provider shall construct the final terminations into the POI substations.

If the Interconnection Customer's tie line is required to cross a Transmission Provider line, the Interconnection Customer shall make application with the Transmission Provider to do so. The Customer's line shall cross below the Transmission Provider's line in all cases unless the Customer's line is of a greater voltage.

7.4 Existing Circuit Breaker Upgrades – Short Circuit

The following Table summarizes the circuit breakers whose interrupting capacity must be increased from 40 kA to 63 kA (minimum), due to the combined effect of all Interconnection Requests of the cluster.

Substation	Circuit breaker to replace
SPANISH FORK 138.kV	CBC146
CLOVER 345.kV	CB 10B48

C1-08

With the Interconnection Customer's system information, including the transformers' impedances, the calculated short-circuit currents considering this plant alone do not violate the interrupting capacity of the Transmission Provider to values higher than the presently existing ratings. The table presented in the common requirements shows the circuit breaker interrupting capacity violation due to the combined requirements considering all the plants of the cluster.

C1-09

With the Interconnection Customer's system information, including the transformers' impedances, the calculated short-circuit currents considering this plant alone do not violate the interrupting capacity of the Transmission Provider to values higher than the presently existing ratings. The table presented in the common requirements shows the circuit breaker interrupting capacity violation due to the combined requirements considering all the plants of the cluster.

C1-16

With the Interconnection Customer's system information, including the transformers' impedances, the calculated short-circuit currents considering this plant alone do not violate the interrupting capacity of the Transmission Provider to values higher than the presently existing ratings. The table presented in the common requirements shows the circuit breaker interrupting capacity violation due to the combined requirements considering all the plants of the cluster.

C1-34

With the Interconnection Customer's system information, including the transformers' impedances, the calculated short-circuit currents considering this plant alone do not violate the interrupting capacity of the Transmission Provider to values higher than the presently existing ratings. The table presented in the common requirements shows the circuit breaker interrupting capacity violation due to the combined requirements considering all the plants of the cluster.

C1-50
With the Interconnection Customer's system information, including the transformers' impedances, the calculated short-circuit currents considering this plant alone do not violate the interrupting capacity of the Transmission Provider to values higher than the presently existing ratings. The table presented in the common requirements shows the circuit breaker interrupting capacity violation due to the combined requirements considering all the plants of the cluster.

C1-51
With the Interconnection Customer's system information, including the transformers' impedances, the calculated short-circuit currents considering this plant alone do not violate the interrupting capacity of the Transmission Provider to values higher than the presently existing ratings. The table presented in the common requirements shows the circuit breaker interrupting capacity violation due to the combined requirements considering all the plants of the cluster.

C1-52
With the Interconnection Customer's system information, including the transformers' impedances, the calculated short-circuit currents considering this plant alone do not violate the interrupting capacity of the Transmission Provider to values higher than the presently existing ratings. The table presented in the common requirements shows the circuit breaker interrupting capacity violation due to the combined requirements considering all the plants of the cluster.

C1-54
With the Interconnection Customer's system information, including the transformers' impedances, the calculated short-circuit currents considering this plant alone do not violate the interrupting capacity of the Transmission Provider to values higher than the presently existing ratings. The table presented in the common requirements shows the circuit breaker interrupting capacity violation due to the combined requirements considering all the plants of the cluster.

7.5 Protection Requirements

Use standard optical-fiber-based line current differential for the new 230 kV line between Black Rock and Sigurd. The multifunction line relays will also perform the functions for breaker failure and synchro-check. This implies the installation of new line relay panels at Black Rock and Sigurd as well as optical fiber communications. Adapt the existing bus differential systems to the new substation configurations for Black Rock and Sigurd.

Use standard Transmission Provider protection for the new 230 kV phase shifting transformer at Black Rock (to Osceola).

Use modern redundant Transmission Provider protection standard panels for the new 700 MVA 345-230 kV transformers at Sigurd. Replace all electromechanical relays that apply.

Use standard redundant transmission autotransformer protection for the new 345-138kV transformer at Oquirrh.

Review the line relay settings for the 138 kV line from McClelland to Midvalley.

Review the line relay settings for the 345 kV line from Emery to Spanish Fork.

Review the line relay settings for the 345 kV #1 line from Camp Williams to Mona.

Review relay settings for the 138 kV line from Clover to Nebo.

C1-08/C1-09
Extend the Clover 345 kV bus differential to include the new bay (both buses). Use standard optical-fiber-based line current differential for the 345 kV interconnection line. The multifunction line relays will also perform the functions for breaker failure and synchro-check as well as the under/over voltage under/over frequency to disconnect the plant in case of voltage and/or frequency fall out of the Transmission Provider tolerance zone.

C1-16
Use standard optical-fiber-based line current differential for the 230 kV interconnection line to Black Rock. The multifunction line relays will also perform the functions for breaker failure and synchro-check as well as the under/over voltage under/over frequency to disconnect the plant in case of voltage and/or frequency fall out of the Transmission Provider tolerance zone.

C1-34
Use standard optical-fiber-based line current differential for the 138 kV interconnection line to the new Mercer 138 kV yard. The multifunction line relays will also perform the functions for breaker failure and synchro-check as well as the under/over voltage under/over frequency to disconnect the plant in case of voltage and/or frequency fall out of the Transmission Provider tolerance zone. Use standard transmission autotransformer protection for the new 345-138 kV auto transformer. Add the two breakers of the new bay to the 345 kV Mercer bus differentials.

C1-50
Use standard optical-fiber-based line current differential for the 345 kV interconnection line to the planned TCS-09 POI 345 kV substation. The multifunction line relays will also perform the functions for breaker failure and synchro-check as well as the under/over voltage under/over frequency to disconnect the plant in case of voltage and/or frequency fall out of the Transmission Provider tolerance zone. New bus differentials will be needed as the substation is being converted from a three-breaker ring to a breaker-and a-half substation.

C1-51
Use standard dual bus differential system for the 345 kV interconnection line segment to Clover 345 kV substation. The grounding grids of Clover and the Interconnection Customer's substation will have to be connected. The bushing current transformers full-winding ratio at the Customer's substation must match the maximum ratio at of the CTs at Clover substation. Add a

multifunction directional overcurrent relay as back up and to serve as the under/over voltage under/over frequency to disconnect the plant in case of voltage and/or frequency fall out of the Transmission Provider tolerance zone. Adapt the existing bus differential system to the new substation configuration. Add synchro-check and breaker failure relays.

C1-52
Use standard optical-fiber-based line current differential for the 345 kV interconnection line to the planned Limber 345 kV substation. The multifunction line relays will also perform the functions for breaker failure and synchro-check as well as the under/over voltage under/over frequency to disconnect the plant in case of voltage and/or frequency fall out of the Transmission Provider tolerance zone. Adapt the existing bus differential system to the new substation configuration.

C1-54
Use standard optical-fiber-based line current differential for the 345 kV interconnection line to the planned TCS-09 POI 345 kV substation. The multifunction line relays will also perform the functions for breaker failure and synchro-check as well as the under/over voltage under/over frequency to disconnect the plant in case of voltage and/or frequency fall out of the Transmission Provider tolerance zone. New bus differentials will be needed as the substation is being converted from a three-breaker ring to a breaker-and a-half substation.

7.6 Data (RTU) Requirements

Black Rock Substation
Provide a controls interface for one new 230 kV breaker, a circuit switcher and a new phase shifting transformer.

Mona Substation
Provide a controls interface for one new 345 kV breaker.

Oquirrh Substation
Provide a SCADA interface for one new 345-138 kV autotransformer. Provide a controls interface for two new 345 kV breakers and one new 138 kV breaker.

Sigurd Substation
Provide a SCADA interface for two new 345-230 kV autotransformers. Provide a controls interface for two new 230 kV breakers.

C1-08
Hard-wired signals to the Primary RTU at the collector substation:

Analogs:

- Real power flowing through the #1 345 – 34.5 kV transformer
- Reactive power flowing through the #1 345 – 34.5 kV transformer
- 34.5 kV Real power PV Breaker 1
- 34.5 kV Reactive power PV Breaker 1
- 34.5 kV Real power PV Breaker 2
- 34.5 kV Reactive power PV Breaker 2

- 34.5 kV Real power BSS Breaker 1
- 34.5 kV Reactive power BSS Breaker 1
- 34.5 kV Real power BSS Breaker 2
- 34.5 kV Reactive power BSS Breaker 2
- Real power flowing through the #2 345 – 34.5 kV transformer
- Reactive power flowing through the #2 345 – 34.5 kV transformer
- 34.5 kV Real power PV Breaker 3
- 34.5 kV Reactive power PV Breaker 3
- 34.5 kV Real power PV Breaker 4
- 34.5 kV Reactive power PV Breaker 4
- 34.5 kV Real power BSS Breaker 3
- 34.5 kV Reactive power BSS Breaker 3
- 34.5 kV Real power BSS Breaker 4
- 34.5 kV Reactive power BSS Breaker 4
- Global Horizontal Irradiance (GHI)
- Average Plant Atmospheric Pressure (Bar)
- Average Plant Temperature (Celsius)
- BESS current energy capacity (MWh)
- BESS current energy capacity (%)
- BESS cycles or health (cycle count or % health)
- Max Gen Limit MW Set Point Feed Back
- Potential Power MW
- 345 kV A phase voltage
- 345 kV B phase voltage
- 345 kV C phase voltage

Analog Written to the RTU:

- Max Gen Limit MW Set Point

Status:

- 34.5 kV transformer 1 low side breaker
- 34.5 kV PV breaker 1
- 34.5 kV PV breaker 2
- 34.5 kV BSS breaker 1
- 34.5 kV BSS breaker 2
- 34.5 kV transformer 2 low side breaker
- 34.5 kV PV breaker 3
- 34.5 kV PV breaker 4
- 34.5 kV BSS breaker 3
- 34.5 kV BSS breaker 4

Primary RTU acquiring serial data from the six primary meters at the collector substation:

Analogs:

- Generation MW

- Generator MVAr
- Energy Register kWh

Backup RTU acquiring serial data from the six backup meters at the collector substation:

Analogs:

- Generation MW
- Generator MVAr
- Energy Register kWh

From Clover Substation:

Net Metering Analogs:

- Net Generation MW
- Net Generator MVAr
- Energy Register kWh

Provide a controls interface for two new 345 kV breakers.

C1-09

Hard-wired signals to the Primary RTU at the collector substation:

Analogs:

- Real power flowing through the #1 345 – 34.5 kV transformer
- Reactive power flowing through the #1 345 – 34.5 kV transformer
- 34.5 kV Real power PV Breaker 1
- 34.5 kV Reactive power PV Breaker 1
- 34.5 kV Real power PV Breaker 2
- 34.5 kV Reactive power PV Breaker 2
- 34.5 kV Real power BSS Breaker 1
- 34.5 kV Reactive power BSS Breaker 1
- 34.5 kV Real power BSS Breaker 2
- 34.5 kV Reactive power BSS Breaker 2
- Real power flowing through the #2 345 – 34.5 kV transformer
- Reactive power flowing through the #2 345 – 34.5 kV transformer
- 34.5 kV Real power PV Breaker 3
- 34.5 kV Reactive power PV Breaker 3
- 34.5 kV Real power PV Breaker 4
- 34.5 kV Reactive power PV Breaker 4
- 34.5 kV Real power BSS Breaker 3
- 34.5 kV Reactive power BSS Breaker 3
- 34.5 kV Real power BSS Breaker 4
- 34.5 kV Reactive power BSS Breaker 4
- Global Horizontal Irradiance (GHI)
- Average Plant Atmospheric Pressure (Bar)
- Average Plant Temperature (Celsius)
- BESS current energy capacity (MWh)

- BESS current energy capacity (%)
- BESS cycles or health (cycle count or % health)
- Max Gen Limit MW Set Point Feed Back
- Potential Power MW
- 345 kV A phase voltage
- 345 kV B phase voltage
- 345 kV C phase voltage

Analog Written to the RTU:

- Max Gen Limit MW Set Point

Status:

- 34.5 kV transformer 1 low side breaker
- 34.5 kV PV breaker 1
- 34.5 kV PV breaker 2
- 34.5 kV BSS breaker 1
- 34.5 kV BSS breaker 2
- 34.5 kV transformer 2 low side breaker
- 34.5 kV PV breaker 3
- 34.5 kV PV breaker 4
- 34.5 kV BSS breaker 3
- 34.5 kV BSS breaker 4

Primary RTU acquiring serial data from the six primary meters at the collector substation:

Analogs:

- Generation MW
- Generator MVAr
- Energy Register kWh

Backup RTU acquiring serial data from the six backup meters at the collector substation:

Analogs:

- Generation MW
- Generator MVAr
- Energy Register kWh

From Clover Substation:

Analogs:

- Net Generation MW
- Net Generator MVAr
- Energy Register kWh

Provide a controls interface for two new 345 kV breakers.

C1-16

Hard-wired signals to the Primary RTU at the collector substation:

Analogs:

- Real power flowing through the 230 – 34.5 kV transformer
- Reactive power flowing through the 230 – 34.5 kV transformer
- 34.5 kV Real power PV Breaker 52F1
- 34.5 kV Reactive power PV Breaker 52F1
- 34.5 kV Real power PV Breaker 52F2
- 34.5 kV Reactive power PV Breaker 52F2
- 34.5 kV Real power PV Breaker 52F3
- 34.5 kV Reactive power PV Breaker 52F3
- 34.5 kV Real power PV Breaker 52F4
- 34.5 kV Reactive power PV Breaker 52F4
- 34.5 kV Real power BSS Breaker 52F5
- 34.5 kV Reactive power BSS Breaker 52F5
- 34.5 kV Real power BSS Breaker 52F6
- 34.5 kV Reactive power BSS Breaker 52F6
- 34.5 kV Real power BSS Breaker 52F7
- 34.5 kV Reactive power BSS Breaker 52F7
- 34.5 kV Real power BSS Breaker 52F8
- 34.5 kV Reactive power BSS Breaker 52F8
- Global Horizontal Irradiance (GHI)
- Average Plant Atmospheric Pressure (Bar)
- Average Plant Temperature (Celsius)
- BESS current energy capacity (MWh)
- BESS current energy capacity (%)
- BESS cycles or health (cycle count or % health)
- Max Gen Limit MW Set Point Feed Back
- Potential Power MW
- 230 kV A phase voltage
- 230 kV B phase voltage
- 230 kV C phase voltage

Analog Written to the RTU:

- Max Gen Limit MW Set Point

Status:

- 230 kV transformer high side breaker 52
- 34.5 kV PV breaker 52F1
- 34.5 kV PV breaker 52F2
- 34.5 kV PV breaker 52F3
- 34.5 kV PV breaker 52F4
- 34.5 kV BSS breaker 52F5
- 34.5 kV BSS breaker 52F6
- 34.5 kV BSS breaker 52F7

- 34.5 kV BSS breaker 52F8

Primary RTU acquiring serial data from the two primary meters at the collector substation:

Analogs:

- Generation MW
- Generator MVAr
- Energy Register kWh

Backup RTU acquiring serial data from the two backup meters at the collector substation:

Analogs:

- Generation MW
- Generator MVAr
- Energy Register kWh

From Black Rock Substation:

Net Metering Analogs:

- Net Generation MW
- Net Generator MVAr
- Energy Register kWh

Provide a controls interface for two new 230 kV breakers and a new Phase shifting transformer.

C1-34

Hard-wired signals to the RTU at the collector substation:

Analogs:

- Real power flowing through the 138 – 34.5 kV transformer
- Reactive power flowing through the 138 – 34.5 kV transformer
- 34.5 kV Real power PV Breaker 52F1
- 34.5 kV Reactive power PV Breaker 52F1
- 34.5 kV Real power PV Breaker 52F2
- 34.5 kV Reactive power PV Breaker 52F2
- 34.5 kV Real power PV Breaker 52F3
- 34.5 kV Reactive power PV Breaker 52F3
- 34.5 kV Real power PV Breaker 52F4
- 34.5 kV Reactive power PV Breaker 52F4
- Global Horizontal Irradiance (GHI)
- Average Plant Atmospheric Pressure (Bar)
- Average Plant Temperature (Celsius)
- Max Gen Limit MW Set Point Feed Back
- Potential Power MW
- 138 kV A phase voltage
- 138 kV B phase voltage
- 138 kV C phase voltage

Analog Written to the RTU:

- Max Gen Limit MW Set Point

Status:

- 138 kV transformer high side breaker 52
- 138 kV transformer low side breaker 52
- 34.5 kV PV breaker 52F1
- 34.5 kV PV breaker 52F2
- 34.5 kV PV breaker 52F3
- 34.5 kV PV breaker 52F4

From new Mercer 138 Substation:

Net Generation Meter Analogs:

- Net Generation MW
- Net Generator MVAr
- Energy Register kWh

Install an RTU in the new control house to support the controls interface for two 345 kV breakers, two 138 kV breakers and provide a SCADA interface for one new 345-138 kV autotransformer.

C1-50

Hard-wired signals to the Primary RTU at the collector substation:

Analogs:

- Real power flowing through the #1 345 – 34.5 kV transformer
- Reactive power flowing through the #1 345 – 34.5 kV transformer
- 34.5 kV Real power PV Breaker 1
- 34.5 kV Reactive power PV Breaker 1
- 34.5 kV Real power BSS Breaker 1
- 34.5 kV Reactive power BSS Breaker 1
- Real power flowing through the #2 345 – 34.5 kV transformer
- Reactive power flowing through the #2 345 – 34.5 kV transformer
- 34.5 kV Real power PV Breaker 2
- 34.5 kV Reactive power PV Breaker 2
- 34.5 kV Real power BSS Breaker 2
- 34.5 kV Reactive power BSS Breaker 2
- Global Horizontal Irradiance (GHI)
- Average Plant Atmospheric Pressure (Bar)
- Average Plant Temperature (Celsius)
- BESS current energy capacity (MWh)
- BESS current energy capacity (%)
- BESS cycles or health (cycle count or % health)
- Max Gen Limit MW Set Point Feed Back
- Potential Power MW

- 345 kV A phase voltage
- 345 kV B phase voltage
- 345 kV C phase voltage

Analog Written to the RTU:

- Max Gen Limit MW Set Point

Status:

- 345 kV breaker 52
- 345 kV transformer 1 high side breaker
- 34.5 kV PV breaker 1
- 34.5 kV BSS breaker 1
- 34.5 kV transformer 2 high side breaker
- 34.5 kV PV breaker 2
- 34.5 kV BSS breaker 2

Primary RTU acquiring serial data from the three primary meters at the collector substation:

Analogs:

- Generation MW
- Generator MVAr
- Energy Register kWh

Backup RTU acquiring serial data from the three backup meters at the collector substation:

Analogs:

- Generation MW
- Generator MVAr
- Energy Register kWh

From the TCS-09 POI Substation:

Net Metering Analogs:

- Net Generation MW
- Net Generator MVAr
- Energy Register kWh

Install an RTU in the control house to support the controls interface for one 345 kV breaker.

C1-51

Hard-wired signals to the Primary RTU at the collector substation:

Analogs:

- Real power flowing through the #1 345 – 34.5 kV transformer
- Reactive power flowing through the #1 345 – 34.5 kV transformer
- 34.5 kV Real power PV Breaker 1
- 34.5 kV Reactive power PV Breaker 1
- 34.5 kV Real power BSS Breaker 1

- 34.5 kV Reactive power BSS Breaker 1
- Real power flowing through the #2 345 – 34.5 kV transformer
- Reactive power flowing through the #2 345 – 34.5 kV transformer
- 34.5 kV Real power PV Breaker 2
- 34.5 kV Reactive power PV Breaker 2
- 34.5 kV Real power BSS Breaker 2
- 34.5 kV Reactive power BSS Breaker 2
- Global Horizontal Irradiance (GHI)
- Average Plant Atmospheric Pressure (Bar)
- Average Plant Temperature (Celsius)
- BESS current energy capacity (MWh)
- BESS current energy capacity (%)
- BESS cycles or health (cycle count or % health)
- Max Gen Limit MW Set Point Feed Back
- Potential Power MW
- 345 kV A phase voltage
- 345 kV B phase voltage
- 345 kV C phase voltage

Analog Written to the RTU:

- Max Gen Limit MW Set Point

Status:

- 345 kV breaker 52
- 345 kV transformer 1 high side breaker
- 34.5 kV PV breaker 1
- 34.5 kV BSS breaker 1
- 34.5 kV transformer 2 high side breaker
- 34.5 kV PV breaker 2
- 34.5 kV BSS breaker 2

Primary RTU acquiring serial data from the three primary meters at the collector substation:

Analogs:

- Generation MW
- Generator MVAr
- Energy Register kWh

Backup RTU acquiring serial data from the three backup meters at the collector substation:

Analogs:

- Generation MW
- Generator MVAr
- Energy Register kWh

From Clover Substation:

Net Metering Analogs:

- Net Generation MW
- Net Generator MVAr
- Energy Register kWh

Provide a controls interface for one 345 kV breaker.

C1-52

Hard-wired signals to the Primary RTU at the collector substation:

Analogs:

- Real power flowing through the #1 345 – 34.5 kV transformer
- Reactive power flowing through the #1 345 – 34.5 kV transformer
- 34.5 kV Real power PV Breaker 1
- 34.5 kV Reactive power PV Breaker 1
- 34.5 kV Real power BSS Breaker 1
- 34.5 kV Reactive power BSS Breaker 1
- Real power flowing through the #2 345 – 34.5 kV transformer
- Reactive power flowing through the #2 345 – 34.5 kV transformer
- 34.5 kV Real power PV Breaker 2
- 34.5 kV Reactive power PV Breaker 2
- 34.5 kV Real power BSS Breaker 2
- 34.5 kV Reactive power BSS Breaker 2
- Global Horizontal Irradiance (GHI)
- Average Plant Atmospheric Pressure (Bar)
- Average Plant Temperature (Celsius)
- BESS current energy capacity (MWh)
- BESS current energy capacity (%)
- BESS cycles or health (cycle count or % health)
- Max Gen Limit MW Set Point Feed Back
- Potential Power MW
- 345 kV A phase voltage
- 345 kV B phase voltage
- 345 kV C phase voltage

Analog Written to the RTU:

- Max Gen Limit MW Set Point

Status:

- 345 kV breaker 52
- 345 kV transformer 1 high side breaker
- 34.5 kV PV breaker 1
- 34.5 kV BSS breaker 1
- 34.5 kV transformer 2 high side breaker
- 34.5 kV PV breaker 2

- 34.5 kV BSS breaker 2

Primary RTU acquiring serial data from the three primary meters at the collector substation:

Analogs:

- Generation MW
- Generator MVAr
- Energy Register kWh

Backup RTU acquiring serial data from the three backup meters at the collector substation:

Analogs:

- Generation MW
- Generator MVAr
- Energy Register kWh

From Limber Substation:

Net Metering Analogs:

- Net Generation MW
- Net Generator MVAr
- Energy Register kWh

Provide a controls interface for one new 345 kV breaker.

C1-54

Hard-wired signals to the RTU at the collector substation:

Analogs:

- Real power flowing through the 345 – 34.5 kV transformer
- Reactive power flowing through the 345 – 34.5 kV transformer
- 34.5 kV Real power PV Breaker 1
- 34.5 kV Reactive power PV Breaker 1
- 34.5 kV Real power PV Breaker 2
- 34.5 kV Reactive power PV Breaker 2
- 34.5 kV Real power BSS Breaker 1
- 34.5 kV Reactive power BSS Breaker 1
- 34.5 kV Real power BSS Breaker 2
- 34.5 kV Reactive power BSS Breaker 2
- Global Horizontal Irradiance (GHI)
- Average Plant Atmospheric Pressure (Bar)
- Average Plant Temperature (Celsius)
- BESS current energy capacity (MWh)
- BESS current energy capacity (%)
- BESS cycles or health (cycle count or % health)
- Max Gen Limit MW Set Point Feed Back
- Potential Power MW
- 345 kV A phase voltage

- 345 kV B phase voltage
- 345 kV C phase voltage

Analog Written to the RTU:

- Max Gen Limit MW Set Point

Status:

- 345 kV breaker 52
- 345 kV transformer 1 high side breaker
- 34.5 kV PV breaker 1
- 34.5 kV BSS breaker 1
- 34.5 kV transformer 2 high side breaker
- 34.5 kV PV breaker 2
- 34.5 kV BSS breaker 2

RTU acquiring serial data from the primary meter at the collector substation:

Analogs:

- Generation MW
- Generator MVAr
- Energy Register kWh

Fiber connection to channel bank at TCS-09 POI Substation for acquiring serial data from the backup meter at the collector substation:

Analogs:

- Generation MW
- Generator MVAr
- Energy Register kWh

From the TCS-09 POI Substation:

Net Metering Analogs:

- Net Generation MW
- Net Generator MVAr
- Energy Register kWh

Install an RTU in the control house to support the controls interface for four 345 kV breakers.

7.7 Substation Requirements

Black Rock Substation

The substation will be expanded to support the installation of a 230 kV phase shifting transformer yard. A new 230 kV transmission line to Sigurd substation will be terminated in the southeast position of the existing 230 kV ring bus. A CDEGS grounding analysis is required. The following equipment has been identified as being required and may change during detailed design.

- 1 – 230 kV, breaker

- 2 – 230 kV, switch, breaker disconnect
- 2 – 230 kV, switch, line disconnect
- 1 – 230 kV, switch, metering disconnect
- 1 – 230 kV, transformer, phase shifting
- 1 – 230 kV, breaker
- 1 – 230 kV, circuit switcher
- 4 – 230 kV, switch

McClelland Substation
Conductor inside the substation will be replaced to support the new line ratings.

Mona Substation
One (1), 345 kV circuit breaker will be replaced with a breaker that has greater fault interrupting capability. If the ground fault duty increases above the growth factor, a CDEGS grounding analysis will be required.

Nebo Substation
Conductor inside the substation will be replaced to support the new line ratings.

Oquirrh Substation
A new 345-138 kV, autotransformer will be installed. A new 345 kV bay will be constructed. The low side of the transformer will be terminated in an existing 138 kV bay. If the ground fault duty increases above the growth factor, a CDEGS grounding analysis will be required. The following equipment has been identified as being required and may change during detailed design.

- 2 – 345 kV, breaker
- 5 – 345 kV, switch, breaker disconnect
- 1 – 345 kV, switch, transformer disconnect
- 3 – 345 kV, CCVT
- 1 – 345-138 kV, autotransformer
- 1 – 138 kV, breaker
- 2 – 138 kV, switch, breaker disconnect
- 1 – 138 kV, switch, transformer disconnect
- 1 – 138 kV, CCVT

Sigurd Substation
The substation will be expanded to support the installation of the new 230 kV transmission line to Black Rock Substation. The two, 345-230 kV autotransformers will be replaced with larger units. A CDEGS grounding analysis is required. The following equipment has been identified as being required and may change during detailed design.

- 2 – 230 kV, breaker
- 5 – 230 kV, switch, breaker disconnect
- 1 – 230 kV, switch, line disconnect

- 3 – 230 kV, CCVT
- 3 – 230 kV, arrester
- 2 – 345-230 kV, 700 MVA, autotransformer

C1-08/C1-09

C1-08 and C1-09 Collector Substation

The Interconnection Customer will provide a separate graded, grounded, and fenced area along the perimeter of the Interconnection Customer's collector substation for the Transmission Provider to install a control house for any required metering, protection and/or communication equipment. This area will share a fence and ground grid with the collector substation and have separate, unencumbered access for the Transmission Provider. The Interconnection Customer shall perform and provide a CDEGS grounding analysis. AC station service will be supplied by the Interconnection Customer. DC power for the control house will be supplied by the Transmission Provider. Six (6), 345 kV and twelve (12), 34.5 kV combined CT/VT metering instrument transformers will be installed. A disconnect switch shall be installed on each side of each metering instrument transformer such that a visible open can be achieved.

Clover Substation

A new 345 kV bay will be installed to support the new line position. If the ground fault duty increases above the growth factor, a CDEGS grounding analysis will be required. The following equipment has been identified as being required and may change during detailed design.

- 2 – 345 kV, breaker
- 5 – 345 kV, switch, breaker disconnect
- 1 – 345 kV, switch, line disconnect
- 3 – 345 kV, arrester
- 3 – 345 kV, combined CT/VT metering instrument transformer

C1-16

C1-16 Collector Substation

The Interconnection Customer will provide a separate graded, grounded, and fenced area along the perimeter of the Interconnection Customer's collector substation for the Transmission Provider to install a control house for any required metering, protection and/or communication equipment. This area will share a fence and ground grid with the collector substation and have separate, unencumbered access for the Transmission Provider. The Interconnection Customer shall perform and provide a CDEGS grounding analysis. AC station service will be supplied by the Interconnection Customer. DC power for the control house will be supplied by the Transmission Provider. Six (6), 34.5 kV combined CT/VT metering instrument transformers will be installed. A disconnect switch shall be installed on each side of each metering instrument transformer such that a visible open can be achieved.

Black Rock Substation

The new 230 kV line will terminate in the northwest position of the existing 230 kV ring bus. The following equipment has been identified as being required and may change during detailed design.

- 1 – 230 kV, breaker

- 2 – 230 kV, switch, breaker disconnect
- 1 – 230 kV, 2000 A, line disconnect
- 1 – 230 kV, 2000 A, meter disconnect
- 3 – 230 kV, arrester
- 3 – 230 kV, combined CT/VT metering instrument transformer

C1-34

C1-34 Collector Substation

The Interconnection Customer will provide a separate graded, grounded, and fenced area along the perimeter of the Interconnection Customer's collector substation for the Transmission Provider to install a control house for any required metering, protection and/or communication equipment. This area will share a fence and ground grid with the collector substation and have separate, unencumbered access for the Transmission Provider. The Interconnection Customer shall perform and provide a CDEGS grounding analysis. AC station service will be supplied by the Interconnection Customer. DC power for the control house will be supplied by the Transmission Provider.

Mercer Substation

The substation will be expanded to support the installation of a new 345 kV bay, a 345-138 kV transformer, a 138 kV bay, control house, and associated equipment. A CDEGS grounding analysis is required. The following equipment has been identified as being required and may change during detailed design.

- 2 – 345 kV, breaker
- 5 – 345 kV, switch, breaker disconnect
- 1 – 345 kV, switch, transformer disconnect
- 1 – 345 kV, CCVT
- 1 – 345-138 kV, autotransformer
- 2 – 138 kV, breaker
- 5 – 138 kV, switch, breaker disconnect
- 1 – 138 kV, switch, transformer disconnect
- 1 – 138 kV, switch, line disconnect
- 1 – 138 kV, switch, meter disconnect
- 3 – 138 kV, combined CT/VT metering instrument transformer
- 3 – 138 kV, CCVT
- 3 – 138 kV, arrester
- 1 – Control house

C1-50

C1-50 Collector Substation

The Interconnection Customer will provide a separate graded, grounded, and fenced area along the perimeter of the Interconnection Customer's collector substation for the Transmission Provider to install a control house for any required metering, protection and/or communication equipment. This area will share a fence and ground grid with the collector substation and have separate, unencumbered access for the Transmission Provider. The Interconnection Customer shall perform and provide a CDEGS grounding analysis. AC station service will be supplied by

the Interconnection Customer. DC power for the control house will be supplied by the Transmission Provider. Six (6), 345 kV and twelve (12), 34.5 kV combined CT/VT metering instrument transformers will be installed. A disconnect switch shall be installed on each side of each metering instrument transformer such that a visible open can be achieved.

TCS-09 POI Substation

The new 345 kV line position will be terminated in an existing 345 kV bay. If the ground fault duty increases above the growth factor, a CDEGS grounding analysis will be required. The following equipment has been identified as being required and may change during detailed design.

- 1 – 345 kV, breaker
- 2 – 345 kV, breaker disconnect
- 1 – 345 kV, switch, line disconnect
- 3 – 345 kV, combined CT/VT metering instrument transformer
- 3 – 345 kV, arrester

C1-51

C1-51 Collector Substation

The C1-51 collector substation and Clover substation ground grids will be tied together. Six (6), 345 kV and twelve (12), 34.5 kV combined CT/VT metering instrument transformers will be installed. A disconnect switch shall be installed on each side of each metering instrument transformer such that a visible open can be achieved. The Transmission Provider assumes that control cable can be installed from its instrument transformers back into Clover substation. If the distance is determined to be too far during design the Transmission Provider will require a control building adjacent to the Interconnection Customer collector substation.

Clover Substation

The new 345 kV line will be installed in an existing bay, opposite the east 345 kV line to Mona Substation. The Clover Substation and C1-51 Collector station ground grids will be tied together. A CDEGS grounding analysis will be required. A marshalling cabinet will be installed to facilitate the installation of the control cables between the two yards. The following equipment has been identified as being required and may change during detailed design.

- 1 – 345 kV, breaker
- 2 – 345 kV, switch, breaker disconnect
- 1 – 345 kV, switch, line disconnect
- 3 – 345 kV, arrester
- 3 – 345 kV, combined CT/VT metering instrument transformer

C1-52

C1-52 Collector Substation

The Interconnection Customer will provide a separate graded, grounded, and fenced area along the perimeter of the Interconnection Customer's collector substation for the Transmission Provider to install a control house for any required metering, protection and/or communication equipment. This area will share a fence and ground grid with the collector substation and have separate, unencumbered access for the Transmission Provider. The Interconnection Customer

shall perform and provide a CDEGS grounding analysis. AC station service will be supplied by the Interconnection Customer. DC power for the control house will be supplied by the Transmission Provider. Six (6), 345 kV and twelve (12), 34.5 kV combined CT/VT metering instrument transformers will be installed. A disconnect switch shall be installed on each side of each metering instrument transformer such that a visible open can be achieved.

Limber Substation

The new 345 kV line will terminate in an existing bay. If the ground fault duty increases above the growth factor, a CDEGS grounding analysis will be required. The following equipment has been identified as being required and may change during detailed design.

- 1 – 345 kV, breaker
- 2 – 345 kV, switch, breaker disconnect
- 1 – 345 kV, switch, line disconnect
- 3 – 345 kV, arrester
- 3 – 345 kV, combined CT/VT metering instrument transformer

C1-54

C1-54 Collector Substation

The Interconnection Customer will provide a separate graded, grounded, and fenced area along the perimeter of the Interconnection Customer's collector substation for the Transmission Provider to install a control house for any required metering, protection and/or communication equipment. This area will share a fence and ground grid with the collector substation and have separate, unencumbered access for the Transmission Provider. The Interconnection Customer shall perform and provide a CDEGS grounding analysis. AC station service will be supplied by the Interconnection Customer. DC power for the control house will be supplied by the Transmission Provider. Six (6), 345 kV and twelve (12), 34.5 kV combined CT/VT metering instrument transformers will be installed. A disconnect switch shall be installed on each side of each metering instrument transformer such that a visible open can be achieved.

TCS-09 POI Substation

The substation will be expanded to support the installation of a new 345 kV bay and new 345 kV line termination. A CDEGS grounding analysis is required. The following equipment has been identified as being required and may change during detailed design.

- 4 – 345 kV, breaker
- 7 – 345 kV, switch, breaker disconnect
- 1 – 345 kV, switch, line disconnect
- 3 – 345 kV, combined CT/VT metering instrument transformer
- 3 – 345 kV, arrester

7.8 Communication Requirements

Standard Transmission Provider fiber optic cable will be installed on the new 63.5-mile Black Rock–Sigurd 230 kV line**.**

C1-08/C1-09

The Interconnection Customer will install two OPGW fibers cables maintaining Transmission Provider required separation at all times on the Interconnection Customer tie-line to support protective relaying requirements of a 345 kV transmission line. The Transmission Provider will own and maintain both runs of fiber. At the transmission dead end fiber will transition from ADSS in the substation to OPGW on the transmission line. At the POI substation the Interconnection Customer will leave sufficient fiber at the final structure outside the substation for the transmission provider to hang the last span and terminate with ADSS fiber. The Transmission Provider will install diverse ADSS fiber cables from the dead-end fiber splices into the control house and terminate on separate panels. If the Interconnection Customer cannot install two runs of fiber the Transmission Provider assumes that it will construct microwave communications sites at each location to provide the second path.

Communications channelization, network, and transport equipment will be installed at each end of the fiber to support all metering, and SCADA requirements. The protective relays will utilize appropriate on-board optical interfaces to match with the single mode fiber cables for direct fiber relaying.

C1-16

The Interconnection Customer will install two OPGW fibers cables maintaining Transmission Provider required separation at all times on the Interconnection Customer tie-line to support protective relaying requirements of a 230 kV transmission line. The Transmission Provider will own and maintain both runs of fiber. At the transmission dead end fiber will transition from ADSS in the substation to OPGW on the transmission line. At the POI substation the Interconnection Customer will leave sufficient fiber at the final structure outside the substation for the transmission provider to hang the last span and terminate with ADSS fiber. The Transmission Provider will install diverse ADSS fiber cables from the dead-end fiber splices into the control house and terminate on separate panels. If the Interconnection Customer cannot install two runs of fiber the Transmission Provider assumes that it will construct microwave communications sites at each location to provide the second path.

Communications channelization, network, and transport equipment will be installed at each end of the fiber to support all metering, and SCADA requirements. The Protective relays will utilize appropriate on-board optical interfaces to match with the single mode fiber cables for direct fiber relaying.

C1-34

The Interconnection Customer will install OPGW fiber from the collector substation to the last structure outside the POI substation. The Transmission Provider will own and maintain the fiber. The transmission provider will install the final span of OPGW and splice to ADSS fiber to be installed to a patch panel in the POI control house.

At the POI substation, the Transmission Provider will install fiber and appropriate communications equipment to connect the new 138 KV control house to the existing building. The existing communications infrastructure will be utilized for connectivity to the Transmission Provider control centers.

The protective relays will utilize appropriate on-board optical interfaces to match with the single mode fiber cables for direct fiber relaying for the transmission tie-line.

C1-50
The Interconnection Customer will install two OPGW fibers cables maintaining Transmission Provider required separation at all times on the Interconnection Customer tie-line to support protective relaying requirements of a 230 kV transmission line. The Transmission Provider will own and maintain both runs of fiber. At the transmission dead end fiber will transition from ADSS in the substation to OPGW on the transmission line. At the POI substation the Interconnection Customer will leave sufficient fiber at the final structure outside the substation for the transmission provider to hang the last span and terminate with ADSS fiber. The Transmission Provider will install diverse ADSS fiber cables from the dead-end fiber splices into the control house and terminate on separate panels. If the Interconnection Customer cannot install two runs of fiber the Transmission Provider assumes that it will construct microwave communications sites at each location to provide the second path.

Communications channelization, network, and transport equipment will be installed at each end of the fiber to support all metering, and SCADA requirements. The Protective relays will utilize appropriate on-board optical interfaces to match with the single mode fiber cables for direct fiber relaying.

C1-51
The Interconnection Customer will install two OPGW fibers cables maintaining Transmission Provider required separation at all times on the Interconnection Customer tie-line to support protective relaying requirements of a 230 kV transmission line. The Transmission Provider will own and maintain both runs of fiber. At the transmission dead end fiber will transition from ADSS in the substation to OPGW on the transmission line. At the POI substation the Interconnection Customer will leave sufficient fiber at the final structure outside the substation for the transmission provider to hang the last span and terminate with ADSS fiber. The Transmission Provider will install diverse ADSS fiber cables from the dead-end fiber splices into the control house and terminate on separate panels. If the Interconnection Customer cannot install two runs of fiber the Transmission Provider assumes that it will construct microwave communications sites at each location to provide the second path.

Communications channelization, network, and transport equipment will be installed at each end of the fiber to support all metering, and SCADA requirements. The Protective relays will utilize appropriate on-board optical interfaces to match with the single mode fiber cables for direct fiber relaying.

C1-52
The Interconnection Customer will install two OPGW fibers cables maintaining Transmission Provider required separation at all times on the Interconnection Customer tie-line to support protective relaying requirements of a 230 kV transmission line. The Transmission Provider will own and maintain both runs of fiber. At the transmission dead end fiber will transition from ADSS in the substation to OPGW on the transmission line. At the POI substation the Interconnection Customer will leave sufficient fiber at the final structure outside the substation

for the transmission provider to hang the last span and terminate with ADSS fiber. The Transmission Provider will install diverse ADSS fiber cables from the dead-end fiber splices into the control house and terminate on separate panels. If the Interconnection Customer cannot install two runs of fiber the Transmission Provider assumes that it will construct microwave communications sites at each location to provide the second path.

Communications channelization, network, and transport equipment will be installed at each end of the fiber to support all metering, and SCADA requirements. The Protective relays will utilize appropriate on-board optical interfaces to match with the single mode fiber cables for direct fiber relaying.

OPGW fiber will be tapped and spliced into the new POI substation from the existing transmission line. DWDM/OTN optical transport equipment along with channelization equipment will be added to the substation control house to maintain communications to and from Oquirrh and Clover substations. The comm equipment will require separate communications battery and charger system. As well, modifications may be required at Oquirrh and Clover to adjust for the change in optical span length.

C1-54
The Interconnection Customer will install two OPGW fibers cables maintaining Transmission Provider required separation at all times on the Interconnection Customer tie-line to support protective relaying requirements of a 230 kV transmission line. The Transmission Provider will own and maintain both runs of fiber. At the transmission dead end fiber will transition from ADSS in the substation to OPGW on the transmission line. At the POI substation the Interconnection Customer will leave sufficient fiber at the final structure outside the substation for the transmission provider to hang the last span and terminate with ADSS fiber. The Transmission Provider will install diverse ADSS fiber cables from the dead-end fiber splices into the control house and terminate on separate panels. If the Interconnection Customer cannot install two runs of fiber the Transmission Provider assumes that it will construct microwave communications sites at each location to provide the second path.

Communications channelization, network, and transport equipment will be installed at each end of the fiber to support all metering, and SCADA requirements. The Protective relays will utilize appropriate on-board optical interfaces to match with the single mode fiber cables for direct fiber relaying.

7.9 Metering Requirements

Black Rock – Osceola line 1272 ACSR
The Black Rock-Osceola line has existing metering that will need to be moved to accommodate for the new phase-shifting transformer. Expect the existing metering transformers to be relocated and expect to require new metering cable to handle a new location for the metering transformers.

C1-08
Interchange Metering

The overall project metering will be located at the Point of Interconnection at Clover substation and rated for the total net generation of the Project. The Transmission Provider will specify and order all interconnection revenue metering, including the instrument transformers, meters, meter panel, junction box, and secondary metering wire. The primary metering transformers will be combination 345kV CT/VT units with extended range CTs for high-accuracy metering.

The metering design package will include two revenue quality meters with DNP real time digital data terminated at a metering interposition block. One meter will be designated as primary SCADA meter with DNP data delivered to the primary control center. A second meter will be designated as backup SCADA meter with DNP data delivered to the alternate control center. The metering data will include bidirectional KWH and KVARH revenue quantities. The meter data will also include instantaneous PF, MW, MVAR, MVA, per-phase voltage, and per-phase amps data.

GSU Metering

The GSU metering will be located at the Interconnection Customer's collector substation and rated for each individual transformer size. This will require two metering points. The Transmission Provider will specify and order all interconnection revenue metering, including the instrument transformers, meters, meter panels, junction boxes, and secondary metering wire. The primary metering transformers will be combination 345kV CT/VT units with extended range CTs for high-accuracy metering.

The metering design package will include two revenue quality meters for each metering point. The meters will output DNP real time digital data terminated at a metering interposition block. One meter will be designated as primary SCADA meter with DNP data delivered to the primary control center. A second meter will be designated as backup SCADA meter with DNP data delivered to the alternate control center. The metering data will include bidirectional KWH and KVARH revenue quantities. The meter data will also include instantaneous PF, MW, MVAR, MVA, per-phase voltage, and per-phase amps data.

An Ethernet connection is required for retail sales and generation accounting via the MV-90 translation system.

Generation Metering

The metering to separate solar and battery activity will be located at the Interconnection Customer's collector substation and rated for the capacity of each string. This will require four metering points. The Transmission Provider will specify and order all interconnection revenue metering, including the instrument transformers, meters, meter panels, junction boxes, and secondary metering wire. The primary metering transformers will be combination 34.5kV CT/VT units with extended range CTs for high-accuracy metering.

The metering design package will include two revenue quality meters for each metering point. The meters will output DNP real time digital data terminated at a metering interposition block. One meter will be designated as primary SCADA meter with DNP data delivered to the primary control center. A second meter will be designated as backup SCADA meter with DNP data delivered to the alternate control center. The metering data will include bidirectional KWH and KVARH revenue quantities. The meter data will also include instantaneous PF, MW, MVAR, MVA, per-phase voltage, and per-phase amps data.

An Ethernet connection is required for retail sales and generation accounting via the MV-90 translation system.

Station Service/Construction Power

The Project is within the Transmission Provider's service territory. Please note that prior to back feed, Interconnection Customer must arrange transmission retail meter service for electricity consumed by the Project that will be drawn from the transmission system when the Project is not generating. Interconnection Customer must call the Help Desk at 1-800-625-6078 to arrange this service. Approval for back feed is contingent upon obtaining station service.

C1-09

Interchange Metering

The overall project metering will be located at the Point of Interconnection at Clover substation and rated for the total net generation of the Project. The Transmission Provider will specify and order all interconnection revenue metering, including the instrument transformers, meters, meter panel, junction box, and secondary metering wire. The primary metering transformers will be combination 345kV CT/VT units with extended range CTs for high-accuracy metering.

The metering design package will include two revenue quality meters with DNP real time digital data terminated at a metering interposition block. One meter will be designated as primary SCADA meter with DNP data delivered to the primary control center. A second meter will be designated as backup SCADA meter with DNP data delivered to the alternate control center. The metering data will include bidirectional KWH and KVARH revenue quantities. The meter data will also include instantaneous PF, MW, MVAR, MVA, per-phase voltage, and per-phase amps data.

GSU Metering

The GSU metering will be located at the Interconnection Customer's collector substation and rated for each individual transformer size. This will require two metering points. The Transmission Provider will specify and order all interconnection revenue metering, including the instrument transformers, meters, meter panels, junction boxes, and secondary metering wire. The primary metering transformers will be combination 345kV CT/VT units with extended range CTs for high-accuracy metering.

The metering design package will include two revenue quality meters for each metering point. The meters will output DNP real time digital data terminated at a metering interposition block. One meter will be designated as primary SCADA meter with DNP data delivered to the primary control center. A second meter will be designated as backup SCADA meter with DNP data delivered to the alternate control center. The metering data will include bidirectional KWH and KVARH revenue quantities. The meter data will also include instantaneous PF, MW, MVAR, MVA, per-phase voltage, and per-phase amps data.

An Ethernet connection is required for retail sales and generation accounting via the MV-90 translation system.

Generation Metering

The metering to separate solar and battery activity will be located at the Interconnection Customer's collector Substation and rated for the capacity of each string. This will require four metering points. The Transmission Provider will specify and order all interconnection revenue metering, including the instrument transformers, meters, meter panels, junction boxes, and secondary metering wire. The primary metering transformers will be combination 34.5kV CT/VT units with extended range CTs for high-accuracy metering.

The metering design package will include two revenue quality meters for each metering point. The meters will output DNP real time digital data terminated at a metering interposition block. One meter will be designated as primary SCADA meter with DNP data delivered to the primary control center. A second meter will be designated as backup SCADA meter with DNP data delivered to the alternate control center. The metering data will include bidirectional KWH and KVARH revenue quantities. The meter data will also include instantaneous PF, MW, MVAR, MVA, per-phase voltage, and per-phase amps data.

An Ethernet connection is required for retail sales and generation accounting via the MV-90 translation system.

Station Service/Construction Power

The Project is within the Transmission Provider's service territory. Please note that prior to back feed, Interconnection Customer must arrange transmission retail meter service for electricity consumed by the Project that will be drawn from the transmission system when the Project is not generating. Interconnection Customer must call the Help Desk at 1-800-625-6078 to arrange this service. Approval for back feed is contingent upon obtaining station service.

C1-16

Interchange Metering

The overall project metering will be located at the Point of Interconnection at Black Rock substation and rated for the total net generation of the Project. The Transmission Provider will specify and order all interconnection revenue metering, including the instrument transformers, meters, meter panel, junction box, and secondary metering wire. The primary metering transformers will be combination 230kV CT/VT units with extended range CTs for high-accuracy metering.

The metering design package will include two revenue quality meters with DNP real time digital data terminated at a metering interposition block. One meter will be designated as primary SCADA meter with DNP data delivered to the primary control center. A second meter will be designated as backup SCADA meter with DNP data delivered to the alternate control center. The metering data will include bidirectional KWH and KVARH revenue quantities. The meter data will also include instantaneous PF, MW, MVAR, MVA, per-phase voltage, and per-phase amps data.

An Ethernet connection is required for retail sales and generation accounting via the MV-90 translation system.

Generation Metering

The metering to separate solar and battery activity will be located at the Interconnection Customer's collector Substation and rated for the capacity of each set of strings. This will require two metering points. The Transmission Provider will specify and order all interconnection revenue metering, including the instrument transformers, meters, meter panels, junction boxes, and secondary metering wire. The primary metering transformers will be combination 34.5kV CT/VT units with extended range CTs for high-accuracy metering.

The metering design package will include two revenue quality meters for each metering point. The meters will output DNP real time digital data terminated at a metering interposition block. One meter will be designated as primary SCADA meter with DNP data delivered to the primary control center. A second meter will be designated as backup SCADA meter with DNP data delivered to the alternate control center. The metering data will include bidirectional KWH and KVARH revenue quantities. The meter data will also include instantaneous PF, MW, MVAR, MVA, per-phase voltage, and per-phase amps data.

Station Service/Construction Power

The Project is within the Transmission Provider's service territory. Please note that prior to back feed, Interconnection Customer must arrange transmission retail meter service for electricity consumed by the Project that will be drawn from the transmission system when the Project is not generating. Interconnection Customer must call the Help Desk at 1-800-625-6078 to arrange this service. Approval for back feed is contingent upon obtaining station service.

C1-34

Interchange Metering

The overall project metering will be located at the Point of Interconnection at Mercer substation and rated for the total net generation of the Project. The Transmission Provider will specify and order all interconnection revenue metering, including the instrument transformers, meters, meter panel, junction box, and secondary metering wire. The primary metering transformers will be combination 138kV CT/VT units with extended range CTs for high-accuracy metering.

The metering design package will include two revenue quality meters with DNP real time digital data terminated at a metering interposition block. One meter will be designated as primary SCADA meter with DNP data delivered to the primary control center. A second meter will be designated as backup SCADA meter with DNP data delivered to the alternate control center. The metering data will include bidirectional KWH and KVARH revenue quantities. The meter data will also include instantaneous PF, MW, MVAR, MVA, per-phase voltage, and per-phase amps data.

A Direct Serial connection is required for retail sales and generation accounting via the MV-90 translation system.

Station Service/Construction Power

The Project is within the Transmission Provider's service territory. Please note that prior to back feed, Interconnection Customer must arrange transmission retail meter service for electricity consumed by the Project that will be drawn from the transmission system when the Project is not generating. Interconnection Customer must call the Help Desk at 1-800-625-6078 to arrange this service. Approval for back feed is contingent upon obtaining station service.

C1-50

Interchange Metering

The overall project metering will be located at the Point of Interconnection at TCS-09 POI substation and rated for the total net generation of the Project. The Transmission Provider will specify and order all interconnection revenue metering, including the instrument transformers, meters, meter panel, junction box, and secondary metering wire. The primary metering transformers will be combination 345kV CT/VT units with extended range CTs for high-accuracy metering.

The metering design package will include two revenue quality meters with DNP real time digital data terminated at a metering interposition block. One meter will be designated as primary SCADA meter with DNP data delivered to the primary control center. A second meter will be designated as backup SCADA meter with DNP data delivered to the alternate control center. The metering data will include bidirectional KWH and KVARH revenue quantities. The meter data will also include instantaneous PF, MW, MVAR, MVA, per-phase voltage, and per-phase amps data.

GSU Metering

The GSU metering will be located at the Interconnection Customer's collector substation and rated for each individual transformer size. This will require two metering points. The Transmission Provider will specify and order all interconnection revenue metering, including the instrument transformers, meters, meter panels, junction boxes, and secondary metering wire. The primary metering transformers will be combination 345kV CT/VT units with extended range CTs for high-accuracy metering.

The metering design package will include two revenue quality meters for each metering point. The meters will output DNP real time digital data terminated at a metering interposition block. One meter will be designated as primary SCADA meter with DNP data delivered to the primary control center. A second meter will be designated as backup SCADA meter with DNP data delivered to the alternate control center. The metering data will include bidirectional KWH and KVARH revenue quantities. The meter data will also include instantaneous PF, MW, MVAR, MVA, per-phase voltage, and per-phase amps data.

An Ethernet connection is required for retail sales and generation accounting via the MV-90 translation system.

Generation Metering

The metering to separate solar and battery activity will be located at the Interconnection Customer's collector substation and rated for the capacity of each string. This will require four metering points. The Transmission Provider will specify and order all interconnection revenue metering, including the instrument transformers, meters, meter panels, junction boxes, and secondary metering wire. The primary metering transformers will be combination 34.5kV CT/VT units with extended range CTs for high-accuracy metering.

The metering design package will include two revenue quality meters for each metering point. The meters will output DNP real time digital data terminated at a metering interposition block. One meter will be designated as primary SCADA meter with DNP data delivered to the primary control center. A second meter will be designated as backup SCADA meter with DNP data delivered to the alternate control center. The metering data will include bidirectional KWH and KVARH revenue quantities. The meter data will also include instantaneous PF, MW, MVAR, MVA, per-phase voltage, and per-phase amps data.

An Ethernet connection is required for retail sales and generation accounting via the MV-90 translation system.

Station Service/Construction Power

The Project is within the Transmission Provider's service territory. Please note that prior to back feed, Interconnection Customer must arrange transmission retail meter service for electricity consumed by the Project that will be drawn from the transmission system when the Project is not generating. Interconnection Customer must call the Help Desk at 1-800-625-6078 to arrange this service. Approval for back feed is contingent upon obtaining station service.

C1-51

Interchange Metering

The overall project metering will be located at the Point of Interconnection at Clover substation and rated for the total net generation of the Project. The Transmission Provider will specify and order all interconnection revenue metering, including the instrument transformers, meters, meter panel, junction box, and secondary metering wire. The primary metering transformers will be combination 345kV CT/VT units with extended range CTs for high-accuracy metering.

The metering design package will include two revenue quality meters with DNP real time digital data terminated at a metering interposition block. One meter will be designated as primary SCADA meter with DNP data delivered to the primary control center. A second meter will be designated as backup SCADA meter with DNP data delivered to the alternate control center. The metering data will include bidirectional KWH and KVARH revenue quantities. The meter data will also include instantaneous PF, MW, MVAR, MVA, per-phase voltage, and per-phase amps data.

GSU Metering

The GSU metering will be located at the Interconnection Customer's collector substation and rated for each individual transformer size. This will require two metering points. The Transmission Provider will specify and order all interconnection revenue metering, including the instrument transformers, meters, meter panels, junction boxes, and secondary metering wire. The primary metering transformers will be combination 345kV CT/VT units with extended range CTs for high-accuracy metering.

The metering design package will include two revenue quality meters for each metering point. The meters will output DNP real time digital data terminated at a metering interposition block. One meter will be designated as primary SCADA meter with DNP data delivered to the primary control center. A second meter will be designated as backup SCADA meter with DNP data delivered to the alternate control center. The metering data will include bidirectional KWH and KVARH revenue quantities. The meter data will also include instantaneous PF, MW, MVAR, MVA, per-phase voltage, and per-phase amps data.

An Ethernet connection is required for retail sales and generation accounting via the MV-90 translation system.

Generation Metering

The metering to separate solar and battery activity will be located at the Interconnection Customer's collector substation and rated for the capacity of each string. This will require four metering points. The Transmission Provider will specify and order all interconnection revenue metering, including the instrument transformers, meters, meter panels, junction boxes, and secondary metering wire. The primary metering transformers will be combination 34.5kV CT/VT units with extended range CTs for high-accuracy metering.

The metering design package will include two revenue quality meters for each metering point. The meters will output DNP real time digital data terminated at a metering interposition block. One meter will be designated as primary SCADA meter with DNP data delivered to the primary control center. A second meter will be designated as backup SCADA meter with DNP data delivered to the alternate control center. The metering data will include bidirectional KWH and KVARH revenue quantities. The meter data will also include instantaneous PF, MW, MVAR, MVA, per-phase voltage, and per-phase amps data.

An Ethernet connection is required for retail sales and generation accounting via the MV-90 translation system.

Station Service/Construction Power

The Project is within the Transmission Provider's service territory. Please note that prior to back feed, Interconnection Customer must arrange transmission retail meter service for electricity consumed by the Project that will be drawn from the transmission system when the Project is not generating. Interconnection Customer must call the Help Desk at 1-800-625-6078 to arrange this service. Approval for back feed is contingent upon obtaining station service.

C1-52

Interchange Metering

The overall project metering will be located at the Point of Interconnection at Limber substation and rated for the total net generation of the Project. The Transmission Provider will specify and order all interconnection revenue metering, including the instrument transformers, meters, meter panel, junction box, and secondary metering wire. The primary metering transformers will be combination 345kV CT/VT units with extended range CTs for high-accuracy metering.

The metering design package will include two revenue quality meters with DNP real time digital data terminated at a metering interposition block. One meter will be designated as primary SCADA meter with DNP data delivered to the primary control center. A second meter will be designated as backup SCADA meter with DNP data delivered to the alternate control center. The metering data will include bidirectional KWH and KVARH revenue quantities. The meter data will also include instantaneous PF, MW, MVAR, MVA, per-phase voltage, and per-phase amps data.

GSU Metering

The GSU metering will be located at the Interconnection Customer's collector substation and rated for each individual transformer size. This will require two metering points. The Transmission Provider will specify and order all interconnection revenue metering, including the instrument transformers, meters, meter panels, junction boxes, and secondary metering wire. The primary metering transformers will be combination 345kV CT/VT units with extended range CTs for high-accuracy metering.

The metering design package will include two revenue quality meters for each metering point. The meters will output DNP real time digital data terminated at a metering interposition block. One meter will be designated as primary SCADA meter with DNP data delivered to the primary control center. A second meter will be designated as backup SCADA meter with DNP data delivered to the alternate control center. The metering data will include bidirectional KWH and KVARH revenue quantities. The meter data will also include instantaneous PF, MW, MVAR, MVA, per-phase voltage, and per-phase amps data.

An Ethernet connection is required for retail sales and generation accounting via the MV-90 translation system.

Generation Metering

The metering to separate solar and battery activity will be located at the Interconnection Customer's collector substation and rated for the capacity of each string. This will require four metering points. The Transmission Provider will specify and order all interconnection revenue metering, including the instrument transformers, meters, meter panels, junction boxes, and secondary metering wire. The primary metering transformers will be combination 34.5kV CT/VT units with extended range CTs for high-accuracy metering.

The metering design package will include two revenue quality meters for each metering point. The meters will output DNP real time digital data terminated at a metering interposition block. One meter will be designated as primary SCADA meter with DNP data delivered to the primary control center. A second meter will be designated as backup SCADA meter with DNP data delivered to the alternate control center. The metering data will include bidirectional KWH and KVARH revenue quantities. The meter data will also include instantaneous PF, MW, MVAR, MVA, per-phase voltage, and per-phase amps data.

An Ethernet connection is required for retail sales and generation accounting via the MV-90 translation system.

Station Service/Construction Power

The Project is within the Transmission Provider's service territory. Please note that prior to back feed, Interconnection Customer must arrange transmission retail meter service for electricity consumed by the Project that will be drawn from the transmission system when the Project is not generating. Interconnection Customer must call the Help Desk at 1-800-625-6078 to arrange this service. Approval for back feed is contingent upon obtaining station service.

C1-54

Interchange Metering

The overall project metering will be located at the Point of Interconnection at TCS-09 POI substation and rated for the total net generation of the Project. The Transmission Provider will specify and order all interconnection revenue metering, including the instrument transformers, meters, meter panel, junction box, and secondary metering wire. The primary metering transformers will be combination 345kV CT/VT units with extended range CTs for high-accuracy metering.

The metering design package will include two revenue quality meters with DNP real time digital data terminated at a metering interposition block. One meter will be designated as primary SCADA meter with DNP data delivered to the primary control center. A second meter will be designated as backup SCADA meter with DNP data delivered to the alternate control center. The metering data will include bidirectional KWH and KVARH revenue quantities. The meter data will also include instantaneous PF, MW, MVAR, MVA, per-phase voltage, and per-phase amps data.

A direct serial connection is required for retail sales and generation accounting via the MV-90 translation system.

Generation Metering

The metering to separate solar and battery activity will be located at the Interconnection Customer's collector substation and rated for the capacity of each string. This will require two metering points. The Transmission Provider will specify and order all interconnection revenue metering, including the instrument transformers, meters, meter panels, junction boxes, and secondary metering wire. The primary metering transformers will be combination 34.5kV CT/VT units with extended range CTs for high-accuracy metering.

The metering design package will include two revenue quality meters for each metering point. The meters will output DNP real time digital data terminated at a metering interposition block. One meter will be designated as primary SCADA meter with DNP data delivered to the primary control center. A second meter will be designated as backup SCADA meter with DNP data delivered to the alternate control center. The metering data will include bidirectional KWH and KVARH revenue quantities. The meter data will also include instantaneous PF, MW, MVAR, MVA, per-phase voltage, and per-phase amps data.

An Ethernet connection is required for retail sales and generation accounting via the MV-90 translation system.

Station Service/Construction Power

The Project is within the Transmission Provider's service territory. Please note that prior to back feed, Interconnection Customer must arrange transmission retail meter service for electricity consumed by the Project that will be drawn from the transmission system when the Project is not generating. Interconnection Customer must call the Help Desk at 1-800-625-6078 to arrange this service. Approval for back feed is contingent upon obtaining station service.

8.0 CONTINGENT FACILITIES (ERIS)

Contingent Facilities Table

Potential Contingent Facility Description	Outage(s)	Pre-CA5 Overload/ Violation Level	Post-CA5 Overload/ Violation Level	% Change	Contingent Facility (Yes/No)	Responsible Entity	Planned ISD
Four breaker ring bus adjacent to South Milford substation	N/A	N/A	N/A	N/A	No	Q0820	TBD
Milford 138-46 kV XFMR upgrade to 112 MVA	N-0	150%	150%	0%	No	Transition Cluster	TBD

Contingent Facilities Table

Potential Contingent Facility Description	Outage(s)	Pre-CA5 Overload/ Violation Level	Post-CA5 Overload/ Violation Level	% Change	Contingent Facility (Yes/No)	Responsible Entity	Planned ISD
Milford-South Milford 2nd 397 ACSR 46 kV line and additional 46 kV breaker at Milford	Milford-South Milford 46 kV	185% (estimated. Case won't solve)	185% (estimated. Case won't solve)	0%	No	Q0820	TBD
Cameron - Tushar - Sevier Tap- Sigurd 138 kV line	CB 112 @ Parowan	100.6%	100.7%	0.1%	No	Q0820	TBD
Gateway South	N-2 Mona - Mercer #2 & #4	TCS 09 POI – Camp Williams 90% Camp Williams – Clover 78%	TCS 09 POI – Camp Williams 127.9% Camp Williams – Clover 99.1%	6%	Yes	PAC	4Q24
Lakeside I RAS Modification	Dynamo - Tri City and Dynamo - Shoreline - Tri City 345 kV lines (credible n-2)	Highland - Hale 124.5% Timp - Cherrywood 126.5%	Highland - Hale 129.4% Timp - Cherrywood 130.9%	4.9% 4.4%	Yes	PAC	1Q22
Magna Cap Bank		0.873 pu @ Magna 138kV	0.860 pu @ Magna 138kV	0.013		PAC	2Q22
Spanish Fork – Mercer	Camp Williams – Steel Mill & Camp Williams – Spanish Fork	Spanish Fork – Hale 123.4% Tanner – Hale 115.8%	Spanish Fork – Hale 129.6% Tanner – Hale 121.6%	6.2% 5.8%	Yes	Transition Cluster	TBD
Huntington Jumpers	Emery – Spanish Fork	Emery – Huntington 100.5%	Emery – Huntington 103.5%	3%	Yes	Transition Cluster	TBD
Camp Williams transformer and yard project	N-1-1				No	PAC	

9.0 COST ESTIMATE (ERIS)

The following estimate represents only scopes of work that will be performed by the Transmission Provider. Costs for any work being performed by the Interconnection Customer and/or Affected Systems are not included.

9.1 Interconnection Facilities

The following facilities are directly assigned to Interconnection Customer(s) using such facilities. If multiple Interconnection Requests are utilizing the same Transmission Provider

Interconnection Facilities the costs shall be shared pursuant to Section 39.2.3 of Transmission Provider's OATT.

C1-08/C1-09

Collector Substation $1,120,000
Control house, relaying, communications, and metering

Clover Substation $1,260,000
Line position and metering

Total: $2,380,000

C1-16

Collector Substation $620,000
Control house, relaying, communications, and metering

Black Rock Substation $750,000
Line termination and metering

Total: $1.370,000

C1-34

Collector Substation $420,000
Control house, relaying, communications, and metering

Mercer Substation $700,000
Line termination and metering

Total: $1,120,000

C1-50

Collector Substation $1,260,000
Control house, relaying, communications, and metering

Mona Substation $1,120,000
Line termination and metering

Total: $2,380,000

C1-51

Collector Substation $1,260,000
Control building, communications, and metering equipment

Clover Substation $1,120,000
Line termination and metering

Total: $2,380,000

C1-52

Collector Substation $1,260,000
Control house, relaying, communications, and metering

Limber Substation $1,120,000
Line termination and metering

Total: $2,380,000

C1-54
Collector Substation $700,000
Control house, relaying, communications, and metering

TCS-09 POI Substation $1,120,000
Line termination and metering

Total: $1,820,000

9.2 Station Equipment

The following are Network Upgrades which are allocated based on the number of Generating Facilities interconnecting at an individual station on a per Interconnection Request basis. Interconnection Requests utilizing the same Interconnection Facilities shall be consider one request for this allocation.

C1-08/C1-09
Clover Substation $2465,000
New 345kV bay and line position

C1-16
Black Rock Substation $840,000
Install new 230kV circuit breaker

C1-34
Mercer Substation $16,400,000
New 138kV line position, 345kV breaker bays and 345-138kV transformer

C1-50
TCS09 Substation $1,400,000
New line position

C1-51
Clover Substation $2,465,000
New line position

C1-52
Limber Substation $1,790,000
New line position

C1-54
TCS-09 POI Substation $6,950,000
Expand yard, install new 345kV bay and line position

9.3 Network Upgrades

The funding responsibility for Network Upgrades other than those identified in the previous section shall be allocated based on the proportional capacity of each individual Generating Facility.

Black Rock Substation *New 230kV bay and new 230kV phase shifting transformer*	$20,100,000
Sigurd Substation *New 230kV bay and replace two 345-230kV transformers*	$15,000,000
Oquirrh Substation *New 345kV bay and 345-138kV transformer*	$10,750,000
McClelland Substation *Relay settings*	$20,000
Midvalley Substation *Relay settings*	$20,000
Emery Substation *Relay settings*	$20,000
Spanish Fork Substation *Relay settings*	$20,000
Camp Williams Substation *Relay settings*	$20,000
Mona Substation *Relay settings*	$20,000
Clover Substation *Relay settings*	$20,000
Nebo Substation *Relay settings*	$20,000
Mona Substation *Replace circuit breaker*	$820,000
Emery–Spanish Fork Transmission Line *Increase clearance of five spans*	$1,860,000

Camp Williams–Mona #1 Transmission Line *Increase clearance of twenty spans*	$4,700,000
McClelland–Mid Valley Transmission Line *Rebuild 6.5 miles of transmission line*	$12,800,000
Black Rock–Sigurd Transmission Line *Build new ~63.5-mile 230kV transmission line*	$42,000,000
Clover–Nebo Transmission Line *Rebuild 5.6 miles of 138kV transmission line*	$4,700,000
Black Rock–Sigurd Transmission Line *Install 64 miles of OPGW*	$3,160,000
Black Rock–Osceola Transmission Line *Reroute line into new bay at Black Rock substation*	$220,000
Mercer Substation *New transmission line between 138kV and 345kV yards*	$480,000
Clover Substation *Add communications*	$110,000
Clover–Limber Transmission Line *Install approximately 1,700 feet of new 500kV line*	$3,900,000
Limber–Oquirrh Transmission Line *Install approximately 1,500 feet of new 345kV line*	$2,500,000
Network Upgrade Total:	**$123,260,000**

9.4 Total Estimated Project Costs

C1-08

Interconnection Facilities	$1,190,000
Station Equipment	$1,233,000
Network Upgrades	$12,650,000
Total:	**$15,073,000**

C1-09

Interconnection Facilities	$1,190,000
Station Equipment	$1,233,000
Network Upgrades	$12,650,000
Total:	**$15,073,000**

C1-16

Interconnection Facilities	$1,370,000
Station Equipment	$840,000
Network Upgrades	$12.593,000
	Total: $14,803,000

C1-34

Interconnection Facilities	$1,120,000
Station Equipment	$16,400,000
Network Upgrades	$12,656,000
	Total: $30,176,000

C1-50

Interconnection Facilities	$2,380,000
Station Equipment	$1,400,000
Network Upgrades	$25,313,000
	Total: $29,093,000

C1-51

Interconnection Facilities	$2,380,000
Station Equipment	$2,465,000
Network Upgrades	$18,984,000
	Total: $23,829,000

C1-52

Interconnection Facilities	$2,380,000
Station Equipment	$1,790,000
Network Upgrades	$15,820,000
	Total: $19,990,000

C1-54

Interconnection Facilities	$1,820,000
Station Equipment	$6,950,000
Network Upgrades	$12,593,000
	Total: $21,363,000

10.0 SCHEDULE (ERIS)

The Transmission Provider estimates it will require approximately 72 months to design, procure and construct the facilities described in this report following the execution of Interconnection Agreements. The schedule will be further developed and optimized during the Facilities Studies.

11.0 TRANSMISSION PROVIDER SYSTEM REQUIREMENTS – NRIS

All requirements in this NRIS section are in addition to all ERIS requirements.

11.1 Transmission System Requirements

- Separate 2.5 miles of the Camp Williams – Mona #1 & Camp Williams – Clover #3 345 kV lines

11.2 Transmission Line Requirements

The following existing transmission lines are required to be reconfigured. The Transmission Provider assumes a new transmission line will need to be constructed requiring new rights-of-way and possible federal environmental permitting.

- Separate 2.5 miles of the Camp Williams – Mona #1 & Camp Williams – Clover #3 345 kV lines

12.0 CONTINGENT FACILITIES (NRIS)

None

13.0 COST ESTIMATE (NRIS)

The following estimate represents only scopes of work that will be performed by the Transmission Provider. Costs for any work being performed by the Interconnection Customer and/or Affected Systems are not included.

13.1 Network Upgrades

The funding responsibility for Network Upgrades other than those identified in the previous section shall be allocated based on the proportional capacity of each individual Generating Facility.

Camp Williams-Mona/Camp Williams-Clover Double Circuit Transmission Lines
Separate Lines $5,100,000

13.2 Total Estimated Project Costs

C1-08 $583,000

C1-09 $583,000

C1-16 $580,000

C1-34 $583,000

C1-50 $1,167,000

C1-51 $875,000

C1-52 $729,000

14.0 AFFECTED SYSTEMS

Transmission Provider has identified the following affected systems: NV Energy, Deseret Generation And Transmission Cooperative

A copy of this report will be shared with each Affected System.

15.0 APPENDICES

Appendix 1: Cluster Area Power Flow and Stability Study Results
Appendix 2: Higher Priority Requests
Appendix 3: Property Requirements

15.1 Appendix 1: Cluster Area Power Flow and Stability Study Results

The Western Electricity Coordinating Council (WECC) approved 2020 Heavy Summer case was used to perform the power flow studies using PSS/E version 34.6. The 2020 Heavy Summer case was modified for the study year, 2025. The local 345 kV, 230 kV and 138 kV transmission system outages were considered during the study. The following table describes the outage, the issue(s) that arises from each outage and the proposed mitigation.

-

Category	Outage(s)	Issue(s)	Mitigation
N-0		Gonder - Osceola 230 kV line OL 106%	Phase Shifting transformer at Black Rock
P1-2	Pavant – Sigurd 230 kV line	Gonder – Osceola 230 kV line 113%	Phase Shifting transformer at Black Rock & Build a new 63.5-mile Black Rock – Sigurd 230 kV line
	Osceola – Gonder 230 kV line	Pavant – Sigurd 230 kV line 112%	Phase Shifting transformer at Black Rock & Build a new 63.5-mile Black Rock – Sigurd 230 kV line
P1-3	Oquirrh 345/138 transformer	Camp Williams – Sunrise 138 kV line 113.99% Sunrise – Oquirrh 138 kV line 101.21%	Install an additional 700 MVA 345/138 transformer at Oquirrh
P2-2	Cottonwood 138 kV bus fault	Midvalley – McClelland 138 kV line 106.47%	Rebuild 6.5 miles Midvalley - McClelland 138 kV line with 1272 ACSR
P2-3	Midvalley CB 344	90th South – South Park Tap 102%	Loop 90th South – Terminal 345 kV line into Midvalley
	Cottonwood CBs 130,125,123	McClelland – Midvalley 138 kV line 106.47%	Rebuild 6.5 miles Midvalley – McClelland 138 kV line with 1272 ACSR

Category	Outage(s)	Issue(s)	Mitigation
	Cottonwood CB 128	McClelland – Midvalley 138 kV line 107%	Rebuild 6.5 miles Midvalley – McClelland 138 kV line with 1272 ACSR
	Oquirrh CB 348	Camp Williams – Sunrise 138 kV line 119.5% Sunrise – Oquirrh 138 kV line 107.61%	Install an additional 700 MVA 345/138 transformer at Oquirrh
	Oquirrh CB 133	Camp Williams – Sunrise 138 kV line 113.63% Sunrise – Oquirrh 138 kV line 100.79%	Install an additional 700 MVA 345/138 transformer at Oquirrh
	Huntington CB 321	Emery – Spanish Fork 345 kV line 101.5%	Improve Clearance on 5 spans of the Emery – Spanish Fork 345 kV line
P7	Oquirrh – Terminal #1 & #2 345 kV lines	90th South – Midvalley 345 kV line 115.11% Oquirrh 345/138 transformer 103.13%	Build a new 16-mile Oquirrh – Terminal #3 345 kV line Install an additional 700 MVA 345/138 transformer at Oquirrh
	Camp Williams – Mercer #1 & #2 345 kV lines	Camp Williams – Mona #1 POI substation – Camp Williams line 113.93%	Improve Clearance on 20 spans of the Camp Williams – Mona #1 345 kV line
	Mercer – Mona #1 & #2 345 kV lines	Camp Williams – Mona #1 POI substation – Camp Williams line 113.78%	Improve Clearance on 20 spans of the Camp Williams – Mona #1 345 kV line

Category	NR Outage(s)	Issue(s)	Mitigation
P7	Camp Williams – Mona #1 & Camp Williams – Clover #3 345 kV lines	Clover – Nebo 138 kV line 105.13%	Separate Camp Williams – Mona #1 & Camp Williams –

Category	NR Outage(s)	Issue(s)	Mitigation
			Clover #3 345 kV lines for 2.5 miles

The following provides a summary list of required mitigation:

- Construct a new 63.5-mile Black Rock – Sigurd 230 kV line 1272 ACSR. Requires a single new breaker at Black Rock. A new bay and 230 kV two breakers at Sigurd are required.
- Install a 230 kV phase shifting transformer on the Black Rock – Osceola line at Black Rock substation. Requires new bus and two 230 kV breakers at Black Rock.
- Install a new 345/138 kV 700 MVA transformer at Oquirrh substation in an existing 345 kV bay. Add a 138 kV breaker to the 138 kV existing bay.
- Rebuild McClelland – Midvalley 138 kV line (6.5 miles) to 1272 ACSR
- Loop the 90th South -Terminal 345 kV line into the Midvalley substation (planned Energy Gateway project)
- Fix clearance issue on 5 spans of the Emery – Spanish Fork 345 kV line
- Fix clearance issue on 20 spans of the Camp Williams – Mona #1 345 kV line
- Construct a new 14-mile Oquirrh – Terminal #3 345 kV line. Use existing bays at Oquirrh and Terminal (planned Energy Gateway project)
- Rebuild 5.6 miles of the Clover – Nebo 138 kV line to 1272 ACSR
- Replace 345/230 kV transformers #5 and #6 at Sigurd with 700 MVA units each

15.2 Appendix 2: Higher Priority Requests

All active higher priority Transmission Provider projects, and transmission service and/or generator interconnection requests will be considered in this cluster area study and are identified below. If any of these requests are withdrawn, the Transmission Provider reserves the right to restudy this request, as the results and conclusions contained within this study could significantly change.

Transmission/Generation Interconnection Queue Requests considered:

LGI Q#	MW	TSR Q#
632	2.99	
634	99	
636	99	
642	58	
752	40	2867/2939
763	200	2872/2873
764	80	2782
777	100	
778	200	2879
787	200	
788	200	
792	80	
799	67	
804	80	2602
805	95	
815	20	
823	178	
838	525	
TCS-07	20	
TCS-09	300	
TCS-25	30	
TCS-41	31.1	

15.3 Appendix 3: Property Requirements

Property Requirements for Point of Interconnection Substation

Requirements for rights of way easements
Rights of way easements will be acquired by the Interconnection Customer in the Transmission Provider's name for the construction, reconstruction, operation, maintenance, repair, replacement and removal of Transmission Provider's Interconnection Facilities that will be owned and operated by Transmission Provider. Interconnection Customer will acquire all necessary permits for the Project and will obtain rights of way easements for the Project on Transmission Provider's easement form.

Real Property Requirements for Point of Interconnection Substation
Real property for a POI substation will be acquired by an Interconnection Customer to accommodate the Interconnection Customer's Project. The real property must be acceptable to Transmission Provider. Interconnection Customer will acquire fee ownership for interconnection substation unless Transmission Provider determines that other than fee ownership is acceptable; however, the form and instrument of such rights will be at Transmission Provider's sole discretion. Any land rights that Interconnection Customer is planning to retain as part of a fee property conveyance will be identified in advance to Transmission Provider and are subject to the Transmission Provider's approval.

The Interconnection Customer must obtain all permits required by all relevant jurisdictions for the planned use including but not limited to conditional use permits, Certificates of Public Convenience and Necessity, California Environmental Quality Act, as well as all construction permits for the Project.

If eligible, Interconnection Customer will not be reimbursed through network upgrades for more than the market value of the property.

As a minimum, real property must be environmentally, physically, and operationally acceptable to Transmission Provider. The real property shall be a permitted or able to be permitted use in all zoning districts. The Interconnection Customer shall provide Transmission Provider with a title report and shall transfer property without any material defects of title or other encumbrances that are not acceptable to Transmission Provider. Property lines shall be surveyed and show all encumbrances, encroachments, and roads.

Examples of potentially unacceptable environmental, physical, or operational conditions could include but are not limited to:

1. Environmental: known contamination of site; evidence of environmental contamination by any dangerous, hazardous or toxic materials as defined by any governmental agency; violation of building, health, safety, environmental, fire, land use, zoning or other such regulation; violation of ordinances or statutes of any governmental entities having jurisdiction over the property; underground or above ground storage tanks in area; known remediation sites on property; ongoing mitigation activities or monitoring activities; asbestos; lead-based paint, etc. A

phase I environmental study is required for land being acquired in fee by the Transmission Provider unless waived by Transmission Provider.

2. Physical: inadequate site drainage; proximity to flood zone; erosion issues; wetland overlays; threatened and endangered species; archeological or culturally sensitive areas; inadequate sub-surface elements, etc. Transmission Provider may require Interconnection Customer to procure various studies and surveys as determined necessary by Transmission Provider.

Operational: inadequate access for Transmission Provider's equipment and vehicles; existing structures on land that require removal prior to building of substation; ongoing maintenance for landscaping or extensive landscape requirements; ongoing homeowner's or other requirements or restrictions (e.g., Covenants, Codes and Restrictions, deed restrictions, etc.) on property which are not acceptable to the Transmission Provider.

Exhibit C

Aurora Energy Research Eolian - Storage Solutions to Unlock Greater Transmission Capacity for Large Load

AURORA ENERGY RESEARCH

Eolian – Storage Solutions to Unlock Greater Transmission Capacity for Large Load

Prepared by Aurora Energy Research for Eolian

April 2026

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Executive Summary

Key Takeaways:

Data center development in New Albany, OH is constrained by the local transmission availability under the N-1 contingency study.

The addition of the 200MW Flint BESS will improve local transmission reliability and allow more data center load to interconnect.

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A 2026 power flow study executed by PowerGEM (the Study)[1], analyzed the impact on local transmission available of adding the 200MW 4-hour duration Flint BESS at Jug St node in New Albany, OH, using the Peak Load and Light Load conditions under 2028 RTEP assumptions.

This report provides a summary of the Study, which demonstrates that Flint BESS can allow additional load to connect in the surrounding area.

Node Name	Additional transmission availability[2] with BESS Discharge (MW)[2]	Limiting scenario without BESS	Limiting scenario with BESS dispatch	Additional Transmission Availability with BESS discharge if only assuming Peak Load conditions (MW)
Jug St	167	Light Load	Light Load	163
Innovation	110	Light Load	Light Load	221
Badger	174	Light Load	Light Load	168

In addition, if AEP chooses to upgrade the shared limiting transmission element of all three nodes under N-1 contingency study, the *Babbit1-2 to Babbit5-6 tap line*, the battery could be even more effective in unlocking additional grid capacity for data center load.

Node Name	Additional transmission availability[1] with BESS Discharge (MW)[2]	Limiting scenario without BESS	Limiting scenario with BESS dispatch	Additional transmission availability with BESS discharge if only assuming Peak Load conditions (MW)
Jug St	711	Light Load	Peak Load	200
Innovation	38	Light Load	Light Load	-99
Badger	511	Light Load	Peak Load	55

The Study demonstrates that battery resources at strategic grid locations can increase transmission capacity, which can bring more nodal transmission availability than the battery's own capacity.

1) PowerGEM report titled "200MW Energy Storage Effect on Additional Load Hosting Capability". 2) Using the lowest transmission availability under Peak Load and Light Load conditions. 2) Transmission availability unlocked by BESS for each node is exclusive. Adding the maximum load to one node reduces the maximum load can connect to other nodes.
Sources: Aurora Energy Research, PowerGEM

Agenda

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I. Modeling assumptions and methodology

II. Forecast results deep dive

III. Conclusion

IV. Appendix

Area Transmission Map| The Study utilized the 2028 summer case from PJM, including all the TC2 related transmission upgrades near New Albany, Ohio

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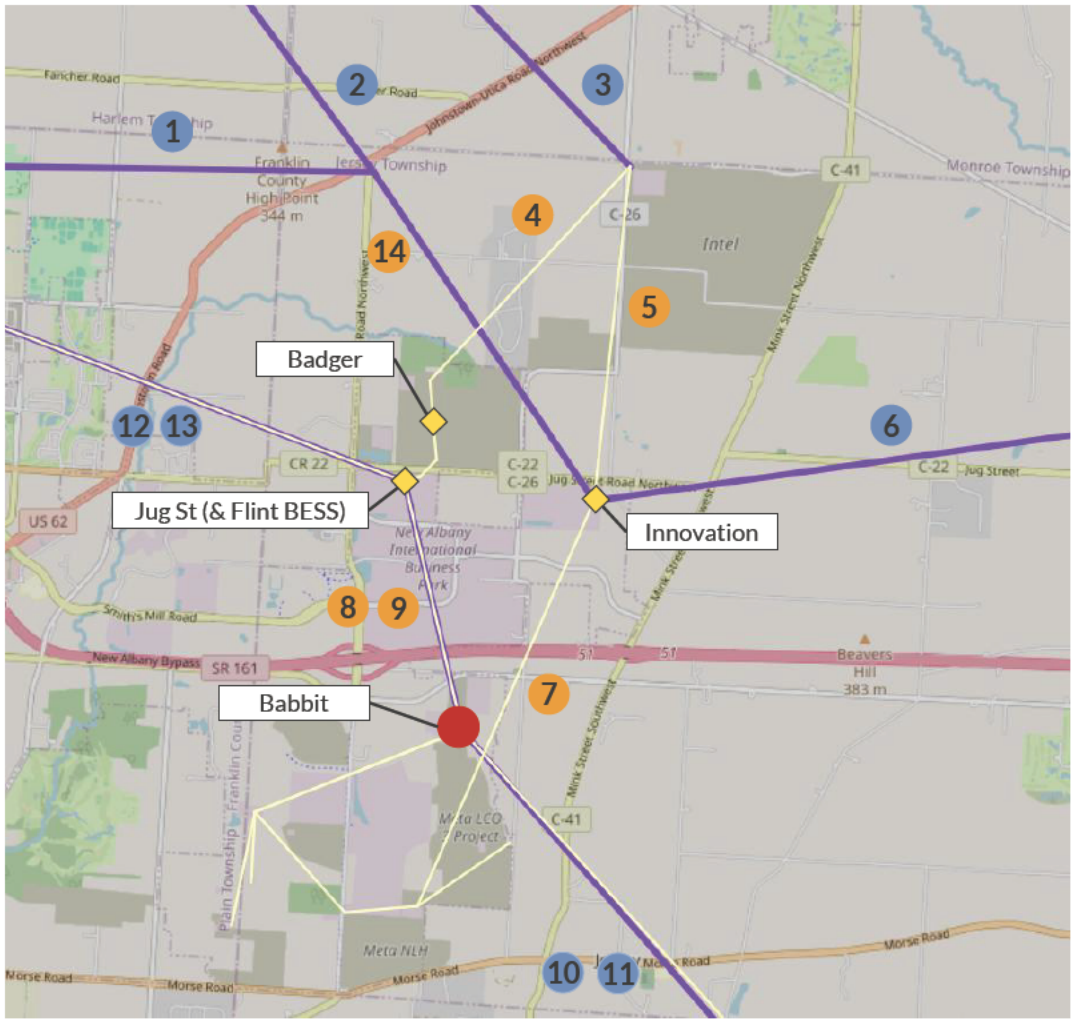


	Key local transmission lines	Voltage (kV)
1	Corridor – Curleys	345
2	Vassel – Curleys	345
3	Vassel – Green Chapel	345
4	Green Chapel – Horizon	138
5	Green Chapel - Innovation	138
6	Innovation – Conesville	345
7	Innovation - Brie	138
8	Jug St – Babbit	345
9	Jug St – Babbit	138
10	Babbit – Kirk	345
11	Babbit – Hazelton	138
12	Corridor – Jug St	345
13	Corridor – Jug St	138
14	Curleys - Innovation	345

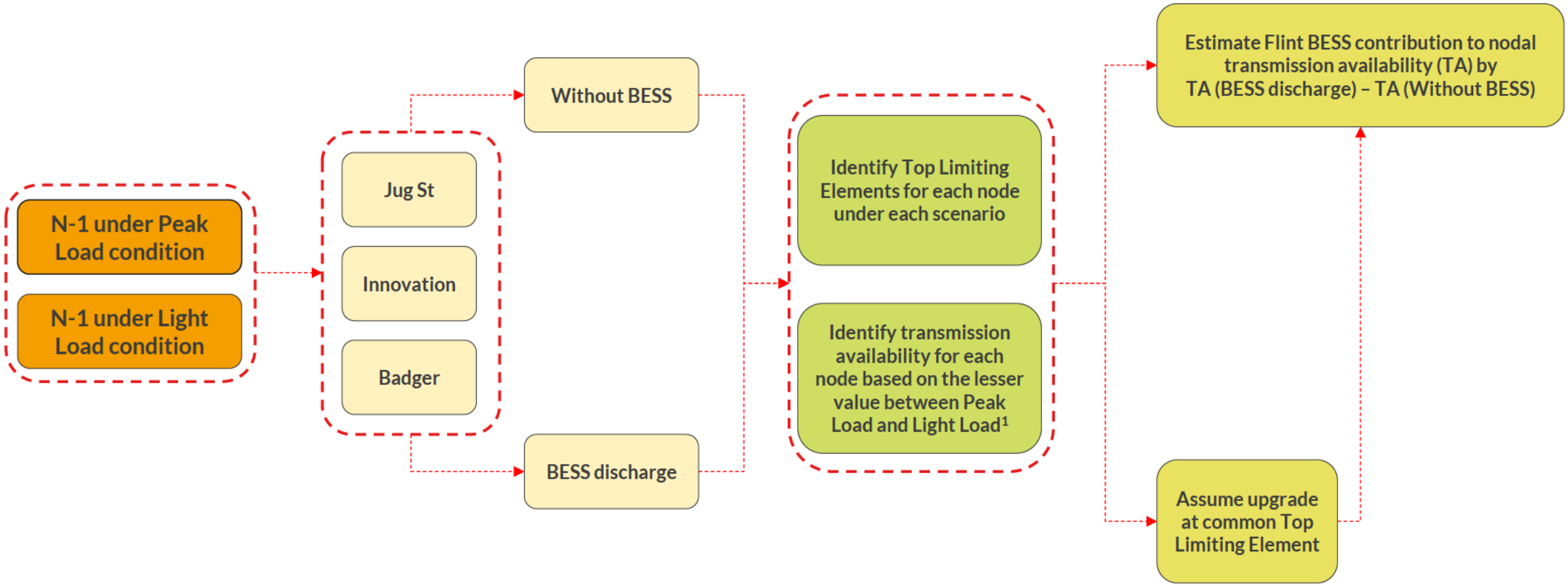
345kV | 138kV | External connections | Local transmission loop | Key limiting element | Studied substations

Sources: PJM, Aurora Energy Research, PowerGEM

Study methodology| The Study estimated benefits of Flint BESS by analyzing the changes in nodal transmission availability under N-1 contingency

AURORA

Flow chart showing methodology used in the Study



1) For example, when BESS discharge, transmission availability at Jug St is 200MW under Peak Load and 100MW under Light Load, the transmission availability would be 100MW.

Sources: Aurora Energy Research, PowerGEM

Agenda

AURORA

I. Modeling assumptions and methodology

II. Forecast results deep dive

III. Conclusion

IV. Appendix

Transmission Availability| The addition of a 200MW BESS at Jug St node increases transmission availability for all three nodes by at least 110MW

AURORA

A node's transmission availability is determined by the lesser value between the Peak Load Conditions and the Light Load Conditions under N-1 contingency reliability study.

Nodal transmission availability with and without Flint BESS (MW)

Node name	Without BESS	With BESS	BESS impact[1]	BESS impact under Peak Load condition[2]
Jug St	242	409	167	163
Innovation	346	456	110	221
Badger	251	425	174	168

I All three nodes have the same top limiting element *(Babbit 5-6 TO 1-2 138kV)* in both Peak Load and Light Load conditions, with or without Flint BESS discharging, with only one exception for Innovation node in Light Load conditions with battery discharging.

II The only different top limiting element (*ANGUIN TO EMPEROR 138kV)* happens when Flint BESS is discharging under the Light Load Conditions for Innovation node.

1) Additional transmission availability unlocked by BESS for each node is exclusive. Adding the maximum load to one node would reduce the maximum load the other load can interconnect. 2) BESS impact considering only the limits under Peak Load Conditions.

Sources: Aurora Energy Research, PJM, PowerGEM

Babbit 5-6 TO 1-2 tap line| Upgrading this short top limiting tap line will be affordable and enable additional load additions

AURORA

Map showing the Babbit tap line identified

Identified Babbit 5-6 TO 1-2 tap line

- Babbit 5-6 TO 1-2 tap line is identified as the top limiting element for all nodes under all scenarios, except one. The line is only ~600ft long at 138kV, which would be relatively low-cost to upgrade.
- Given the high potential upside of upgrading the Babbit 5-6 TO 1-2 tap line, the Study provides a sensitivity assuming the tap line is upgraded.

Sources: Aurora Energy Research, PJM, Eolian

Transmission Availability with Babbit 5-6 TO 1-2 upgrade| Flint BESS increases transmission availability for Jug St and Badger by more than 500MW

A node's transmission availability is determined by the lesser value between the Peak Load Conditions and the Light Load Conditions under N-1 contingency reliability study.

Nodal transmission availability with and without Flint BESS (MW)

Node name	Without BESS	With BESS	BESS impact[1]	BESS impact under Peak Load condition[2]
Jug St	250	961	711	200
Innovation	418	456	38	-99
Badger	305	816	511	55

I With the *Babbit 5-6 TO 1-2 138kV* Tap line upgraded, both Jug St and Badger nodes see significantly greater benefits from the addition of Flint BESS. Battery dispatch can have up to 700MW impact on transmission availability because it allows Jug St and Badger nodes' most limiting scenario switches from Light Load to Peak Load and seeing different top limiting elements.

II Impact on Innovation node is limited as it has a top limiting element other than the *Babbit 5-6 TO 1-2 138kV* Tap line.

1) Additional transmission availability unlocked by BESS for each node is exclusive. Adding the maximum load to one node would reduce the maximum load the other load can interconnect. 2) BESS impact considering only the limits under Peak Load Conditions.

Sources: Aurora Energy Research, PJM, PowerGEM

Agenda

AURORA

I. Modeling assumptions and methodology

II. Forecast results deep dive

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IV. Appendix

Conclusion

- In locating Flint BESS adjacent to load at a strategic location on the grid, the BESS unlocks more transmission availability for the nearby nodes than its own nameplate capacity of 200MW.
- If generation, including BESS, is developed in response to anticipated large load and strategically co-located, it can reduce or defer the transmission upgrades required to interconnect that load. Absent such generation, those upgrade costs would be recovered by PJM through interconnection charges to the load.
- The Study demonstrates that appropriately sited BESS can enable large load interconnection. The avoided transmission upgrade costs can be passed on to the BESS developer and promote additional BESS deployment.

Sources: Aurora Energy Research, PJM, PowerGEM

Agenda

AURORA

I. Modeling assumptions and methodology

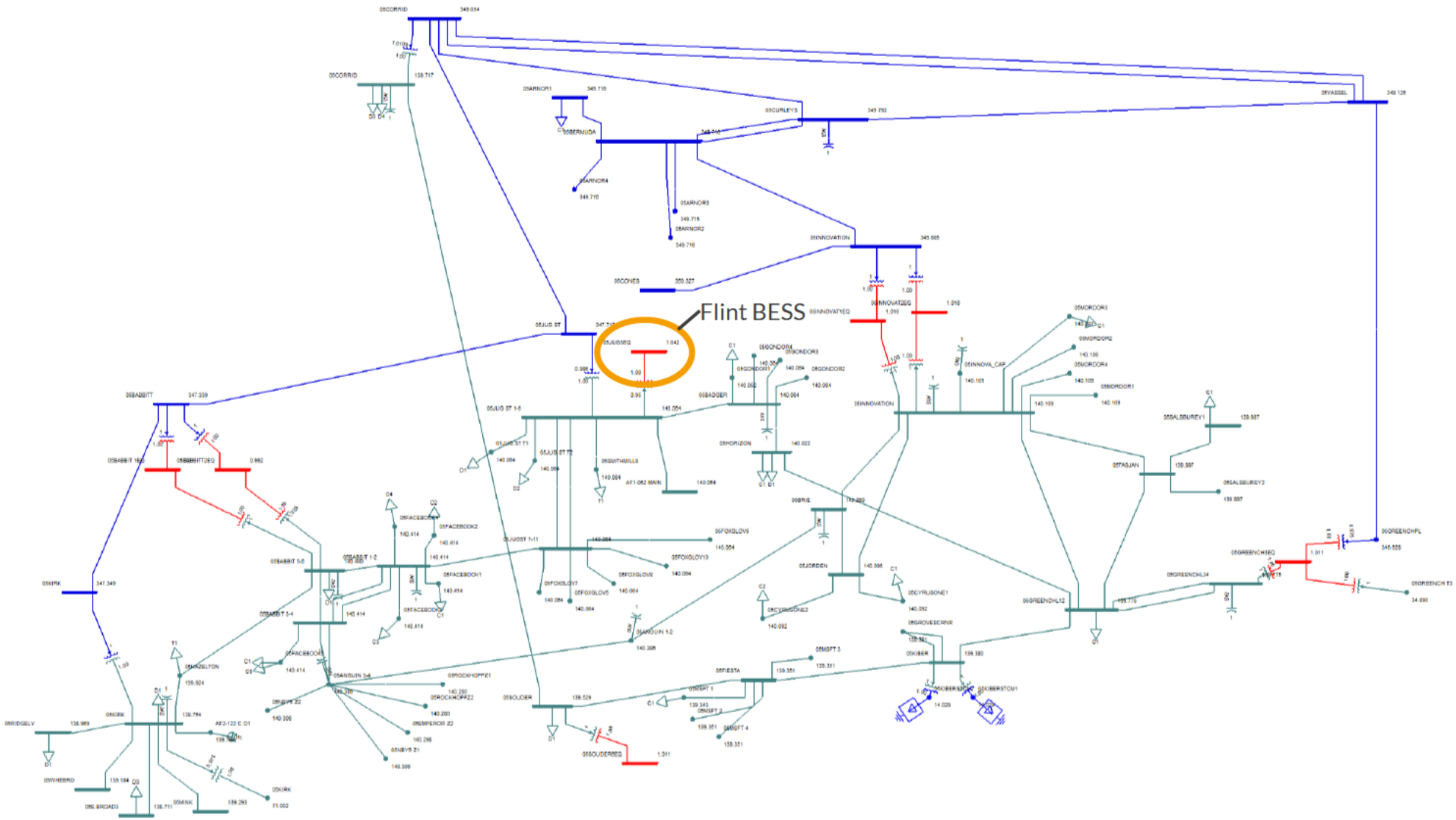
II. Forecast results deep dive

III. Conclusion

IV. Appendix

Transmission topology used in the Study

AURORA



Sources: PJM, PowerGEM

Details and disclaimer

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Exhibit D

PowerGEM Report 200 MW Energy Storage Effect on Additional Load Hosting Capability

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Final Report

200 MW Energy Storage Effect on Additional Load Hosting Capability

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Introduction

This study examined the impacts on system reliability at the Innovation Lab 138 kV, Badger 138 kV, and Jug St 138 kV substations resulting from a 200 MW energy storage project at Jug St 138 kV ("the Project" or "Flint Storage"). Specifically, the study evaluated the load hosting capability at these locations in order to quantify the impact of the Project on transmission capacity in the immediate area. The Transition Cluster #2 Generation Interconnection Study (TC2) model was used to assess potential transmission limits under both charging and discharging scenarios.

Project description

The PJM Generation Deliverability methodology (i.e., for interconnection of new generators including storage) considers the storage as discharging (injection) at full under peak load conditions, and both discharging (injection) and charging (withdraw) at full under light load conditions. The Developer's Project, AF1-062, is a 200 MW storage project with a point-of-interconnection (POI) at the Jug St 138 kV bus. The project has a projected interconnection in-service date of November 2026 and a project COD date of May 2027 (approximately). Multiple new data centers are planned to be connected in the vicinity of the AF1-062 storage project. AF1-062 has a signed ISA and is currently in the Procurement and Construction stages.

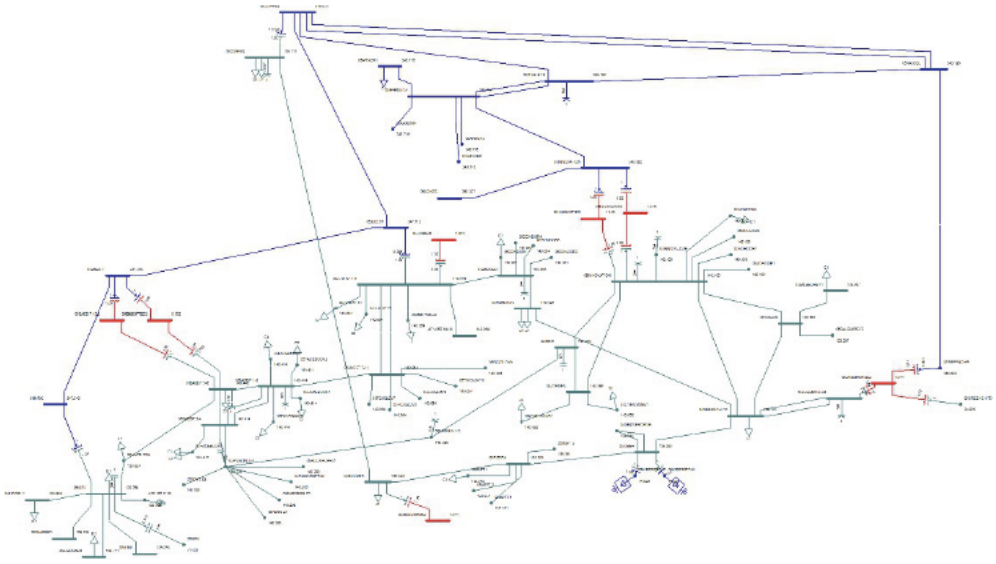

Figure 1. The TC2 area diagram near project location

The Flint Storage project is modeled using the parameters from Table 1.

Table 1: Flint Storage Model Parameters

Parameter	Maximum	Minimum
P (Active Power)	200.0 MW	-200.0 MW
Q (Reactive Power)	66.0 Mvar	-66.0 Mvar

Methodology

Reliability Analysis

1. Innovation Lab 138 kV – Peak Load Discharging

The maximum additional load at Innovation Lab 138 kV is determined (in MW) under full discharging (i.e., injection) during peak load conditions. The additional load is obtained by gradually increasing the injection at the study bus until new system reliability violations are observed in the model.

2. Innovation Lab 138 kV – Light Load Charging

Full charging (i.e., withdrawal) is simulated under light load conditions using a process similar to the peak load scenario.

3. Innovation Lab 138 kV – Light Load Discharging

The same approach is applied to evaluate the maximum additional load at Innovation Lab 138 kV under discharging conditions during light load.

4. Jug St 138 kV and Badger 138 kV

The procedure described above is repeated to determine the maximum additional load that can be accommodated at Jug St 138 kV and Badger 138 kV.

Analysis included determining potential thermal limitations for the Project for both charging and discharging of N-1 and N-1-1 analysis for both peak load and light load with and without any additional load.

Study Database

To determine the amount of additional load that can be accommodated at the Jug St 138 kV, Badger 138 kV, and Innovation Lab 138 kV buses with the addition of the Project, five scenarios were evaluated under N-1 conditions. The analysis is based on the TC2 base cases ("Base Cases"), which represent peak load and light load system conditions. The study models correspond to the following scenarios:

1. Case-SP: Base Case with peak load conditions and without Flint Storage,
2. Case-SP-Discharging: Same as Case-SP, but with Flint Storage discharging at full capacity,
3. Case-LL: Base Case with light load conditions and without Flint Storage,
4. Case-LL-Discharging: Same as Case-LL, but with Flint Storage discharging at full capacity,
5. Case-LL-Charging: Same as Case-LL, but with Flint Storage charging at full capacity.

N-1 Results: Peak Load Conditions

The N-1 analysis was performed considering contingency events including breaker failure, single element, bus, and tower contingencies. The results of the N-1 analysis were then applied to determine the load hosting capability at the three study buses with Flint Storage charging under peak load conditions, in accordance with the applicable N-1 analysis criteria.

The load hosting capability analysis indicates that Flint Storage discharging at 200 MW increases hosting capability at all three study buses, as shown in Table 2. Under N-1 conditions, Innovation Lab 138 kV can host up to 674 MW without thermal issues, while Jug St 138 kV and Badger 138 kV can host up to 497 MW and 513 MW, respectively. For load hosting capacity calculations, the first limiting element is treated as the binding constraint.

Table 2: Load Hosting Capability under Peak Load Conditions

Scenario	Flint Storage status	Bus Name	Load Hosting Capability (MW)	Incremental Load Hosting Capability (MW)
1	Without Flint Storage	Jug St	333.5	NA
		Innovation Lab	453.6	NA
		Badger	345.6	NA
2	Discharging (200 MW)	Jug St	496.7	163.2
		Innovation Lab	674.1	220.5
		Badger	513.2	167.6

The top limiting elements for load hosting capability under peak load are presented in Table 3.

Table 3: Top Limiting Elements for Load Hosting Capability under Peak Load Conditions

Scenario	Flint Storage status	Bus Name	Top Limiting Element	Contingency Name
1	Without Flint Storage	Jug St	05BABBIT 5-6 138 kV to 05BABBIT 1-2 138 kV	AEP_P4_#15956_05BABBIT 5-6 138_L_SRT-A
		Innovation Lab	05BABBIT 5-6 138 kV to 05BABBIT 1-2 138 kV	AEP_P4_#15956_05BABBIT 5-6 138_L_SRT-A
		Badger	05BABBIT 5-6 138 kV to 05BABBIT 1-2 138 kV	AEP_P4_#15956_05BABBIT 5-6 138_L_SRT-A
2	Discharging (200 MW)	Jug St	05BABBIT 5-6 138 kV to 05BABBIT 1-2 138 kV	AEP_P4_#15956_05BABBIT 5-6 138_L_SRT-A
		Innovation Lab	05BABBIT 5-6 138 kV to 05BABBIT 1-2 138 kV	AEP_P4_#15956_05BABBIT 5-6 138_L_SRT-A
		Badger	05BABBIT 5-6 138 kV to 05BABBIT 1-2 138 kV	AEP_P4_#15956_05BABBIT 5-6 138_L_SRT-A

However, since the 05BABBIT 5–6 to 05BABBIT 1–2 138 kV line is a short tap line and relatively easy to upgrade, we also evaluated additional constraints. The second and third limiting elements for load hosting capability under peak load conditions with battery discharging are also listed in Table 4, along with their DC DFAX values and the available branch headroom

(MW). Using these two values, we can approximate the load hosting capacity limited by the second and third limiting elements under the peak load conditions with battery discharging. Table 5 shows the second and third limiting elements without battery storage.

Table 4: Second and Third Limiting Elements for Load Hosting Capability under Peak Load Conditions with Battery Discharging

Limiting element	Bus Name	Top Limiting Element	Contingency Name	Dfax	Branch Headroom	Implied Capability
Second	Jug St	05JUG ST 345 kV to 05JUG ST 1-6 138 kV	AEP_P7-1_#17720___SRT-A	0.3035	291.6	961
	Innovation Lab	05ANGUIN 3-4 138 kV to 05EMPEROR Z2 138 kV	AEP_P4_#17886_05ANGUIN 3-4 138_H2_SRT-A	0.1852	153.4	828
	Badger	05GREENCHL12 138 kV to 05HORIZON 138 kV	AEP_P1-2_#17688_SRT-A	1.0000	816.2	816
Third	Jug St	05BABBIT 1-2 138 kV to 05JUGST 7-11 138 kV	AEP_P4_#17926_05HORIZON 138_CB4_SRT-A	0.6977	810.2	1161
	Innovation Lab	05ROBERT 138 kV to 05KENNY 138 kV	AEP_P4_#7246_05VASSEL 345_J_SRT-A	0.0301	46.3	1538
	Badger	05HORIZON 138 kV to 05BADGER 138 kV	AEP_P4_#17907_05BADGER 138_CB7_SRT-A	1.0000	871.0	871

Table 5: Second and Third Limiting Elements for Load Hosting Capability under Peak Load Conditions without Flint Battery

Limiting element	Bus Name	Top Limiting Element	Contingency Name	Dfax	Branch Headroom	Implied Capability
Second	Jug St	05JUG ST 345 kV to 05JUG ST 1-6 138 kV	AEP_P7-1_#17720___SRT-A	0.3032	230.8	761
	Innovation Lab	05ANGUIN 3-4 138 kV to 05EMPEROR Z2 138 kV	AEP_P4_#17884_05ANGUIN 3-4 138_G2_SRT-A	0.1853	171.7	927
	Badger	05GREENCHL12 138 kV to 05HORIZON 138 kV	AEP_P1-2_#17688_SRT-A	1	816.1	816
Third	Jug St	05BABBIT 1-2 138 kV to 05JUGST 7-11 138 kV	AEP_P1-2_#17919_SRT-A	0.6969	672.8	965
	Innovation Lab	05ROBERT 138 kV to 05KENNY 138 kV	AEP_P4_#7246_05VASSEL 345_J_SRT-A	0.030	40.5	1350
	Badger	05JUG ST 345 kV to 05JUG ST 1-6 138 kV	AEP_P7-1_#17720___SRT-A	0.3032	230.8	761

N-1 Results: Light Load Conditions

The load hosting capability analysis indicates that Flint Storage discharging increases hosting capability at all three study buses, as shown in Table 6. However, when the Project is charging, the load hosting capability at the three study buses decreases due to increased thermal loading on limiting elements. It is important to note that it is very unlikely the Project would charge during an N-1 condition; in such a scenario market signals would disincentivize charging during these times and PJM may not permit the battery to charge.

Table 6: Load Hosting Capability under Light Load Conditions

Scenario	Flint Storage status	Bus Name	Load Hosting Capability (MW)	Incremental Load Hosting Capability (MW)
3	Without Flint Storage	Jug St	241.9	NA
		Innovation Lab	345.8	NA
		Badger	251.3	NA
4	Discharging (200 MW)	Jug St	408.9	167.0
		Innovation Lab	456.3	110.5
		Badger	424.8	173.5
5	Charging (200 MW)	Jug St	71.9	-170.0
		Innovation Lab	98.4	-247.4
		Badger	74.4	-176.9

The top primary limiting elements for load hosting capability under light load conditions are presented in Table 7.

Table 7: Limiting Elements for Load Hosting Capability under Light Load Conditions

Scenario	Flint Storage status	Bus Name	Top Limiting Element	Contingency Name
3	Without Flint Storage	Jug St	05BABBIT 5-6 138 kV to 05BABBIT 1-2 138 kV	AEP_P4_#15956_05BABBIT 5-6 138_L_SRT-A
		Innovation Lab	05BABBIT 5-6 138 kV to 05BABBIT 1-2 138 kV	AEP_P4_#15956_05BABBIT 5-6 138_L_SRT-A
		Badger	05BABBIT 5-6 138 kV to 05BABBIT 1-2 138 kV	AEP_P4_#15956_05BABBIT 5-6 138_L_SRT-A
4	Discharging (200 MW)	Jug St	05BABBIT 5-6 138 kV to 05BABBIT 1-2 138 kV	AEP_P4_#15956_05BABBIT 5-6 138_L_SRT-A
		Innovation Lab	05ANGUIN 3-4 138 kV 05EMPEROR Z2 138 kV	AEP_P4_#17935_05ANGUIN 3-4 138_F2_SRT-A
		Badger	05BABBIT 5-6 138 kV to 05BABBIT 1-2 138 kV	AEP_P4_#15957_05BABBIT 5-6 138_L2_SRT-A
5	Charging (200 MW)	Jug St	05BABBIT 5-6 138 kV to 05BABBIT 1-2 138 kV	AEP_P4_#15957_05BABBIT 5-6 138_L2_SRT-A
		Innovation Lab	05BABBIT 5-6 138 kV to 05BABBIT 1-2 138 kV	AEP_P4_#15957_05BABBIT 5-6 138_L2_SRT-A
		Badger	05BABBIT 5-6 138 kV to 05BABBIT 1-2 138 kV	AEP_P4_#15957_05BABBIT 5-6 138_L2_SRT-A

From Tables 3 or 7, the 05BABBIT 5-6 to 05BABBIT 1-2 138 kV is identified as the binding element for load hosting capability across all three buses. As additional load is withdrawn, this line reaches its contingency rating under the N-1 contingency corresponding to the outage of the 05BABBIT 5-6 to 05BABBIT 3-4 138 kV line

Taking Jug St 138 kV as an example: in the Base Case with light load conditions (Case-LL, Figure 2), the 05BABBIT 5-6 to 05BABBIT 1-2 138 kV line is loaded at 58%. When Flint Storage is charging at full capacity (Case-LL-Charging, Figure 3), the loading on this line increases to 62%, and the flow direction on the 05JUG ST 7-11 to 05BABBIT 1-2 138 kV line is reversed. After applying the N-1 contingency that opens the 05BABBIT 5-6 to 05BABBIT 3-4 138 kV line, the 05BABBIT 5-6 to 05BABBIT 1-2 138 kV line is loaded to 98%, indicating a highly stressed condition (Figure 4). When the load at Jug St Bus is further increased to its maximum hosting capability, the 05BABBIT 5-6 to 05BABBIT 1-2 138 kV line reaches its contingency rating (Figure 5).

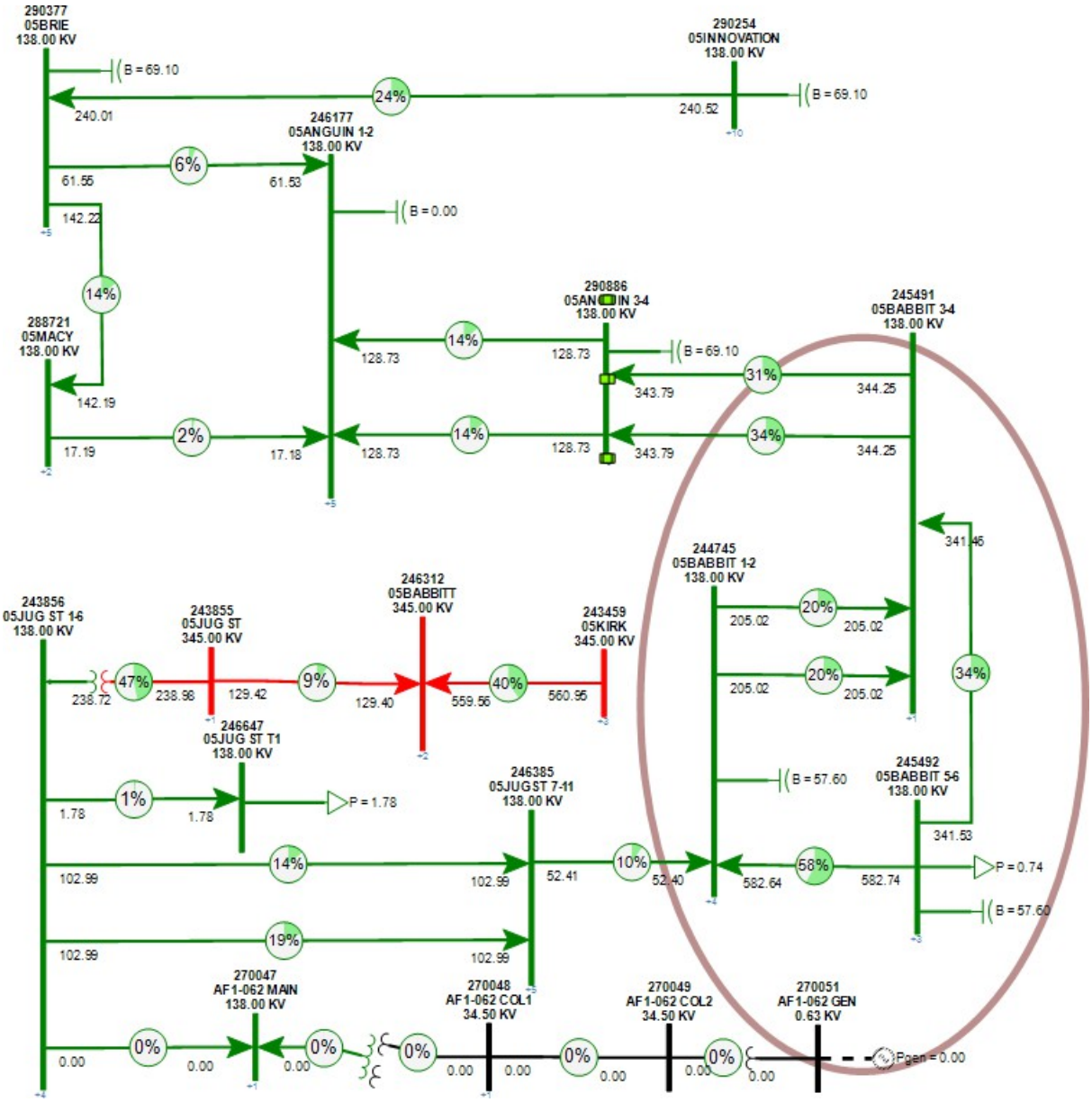

Figure 2. Power flow in Case-LL

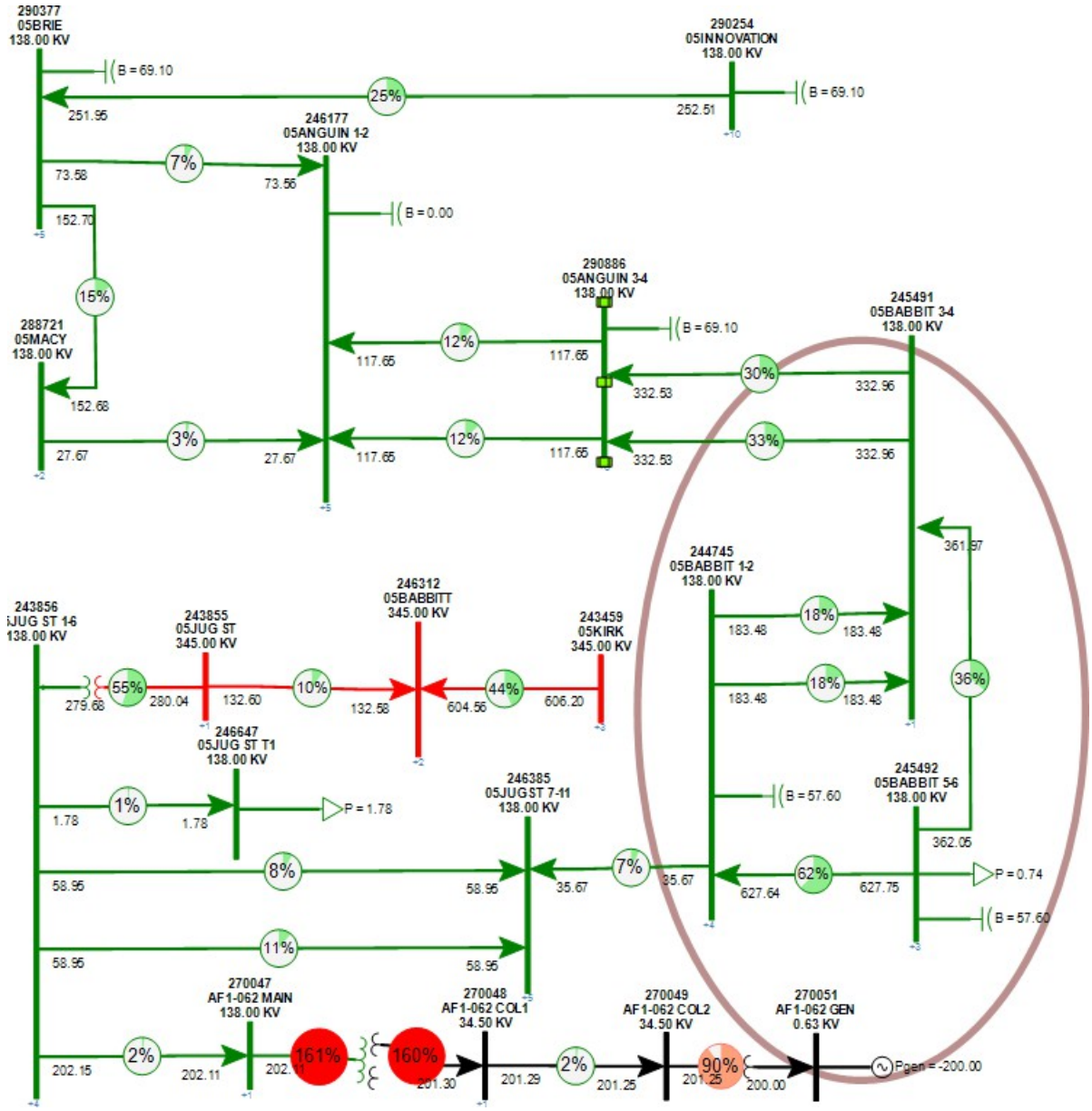

Figure 3. Power flow in Case-LL-Charging

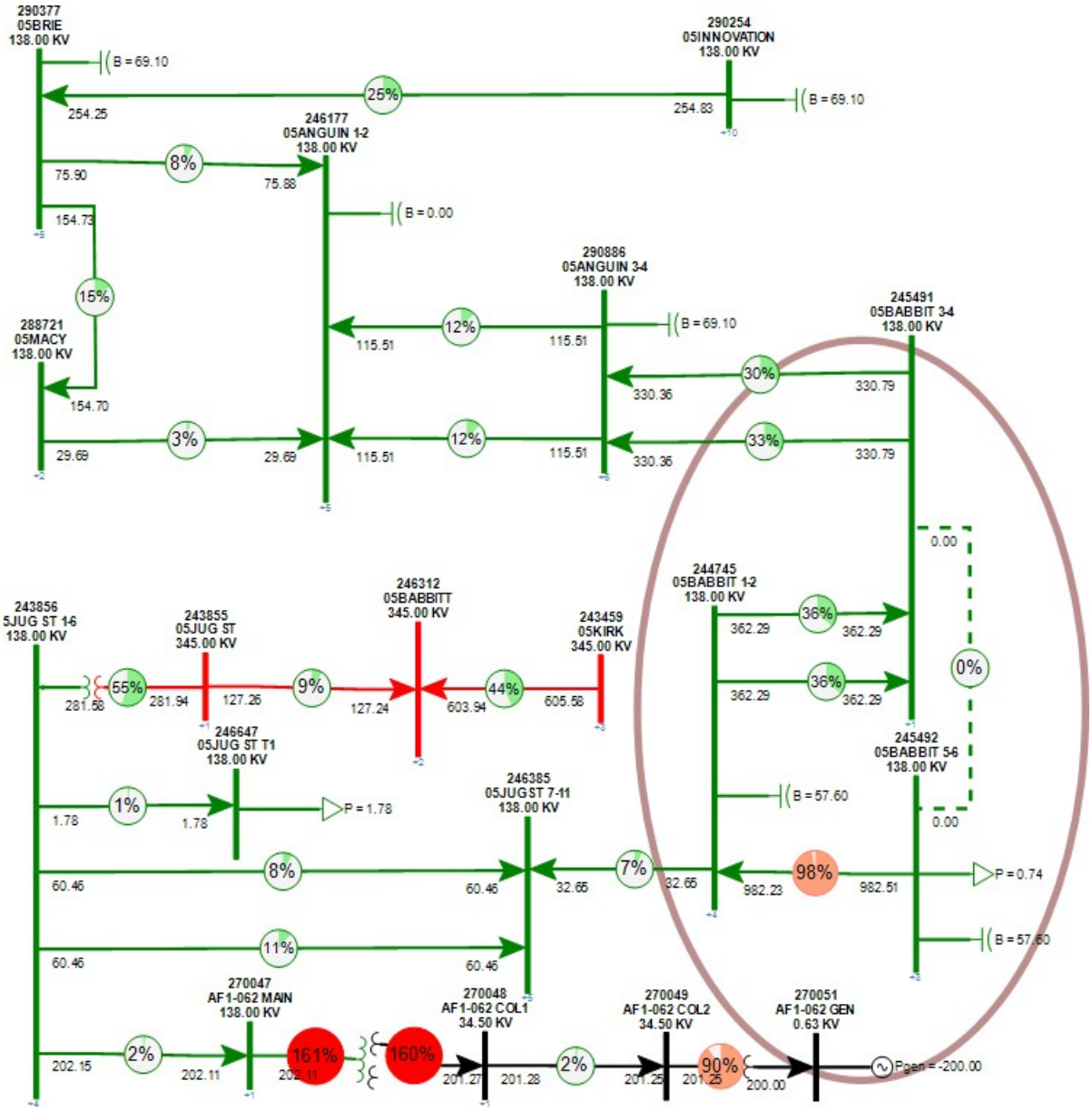

Figure 4. Power flow in Case-LL-Charging under the N-1 contingency

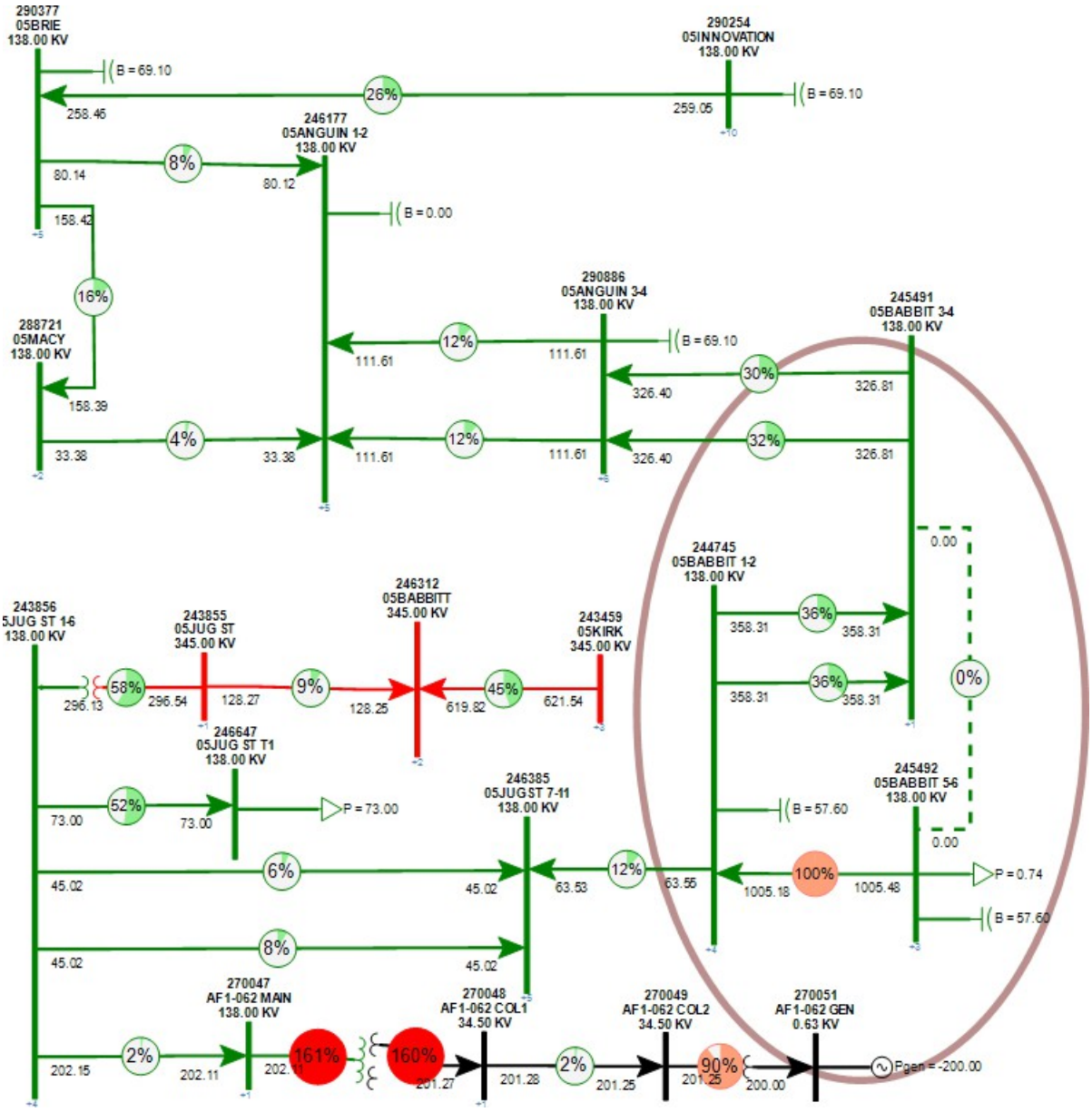

Figure 5. Power flow in Case-LL-Charging under the N-1 contingency with max load hosting at Jug St 138 kV

However, since the 05BABBIT 5–6 to 05BABBIT 1–2 138 kV line is a short tap line and relatively easy to upgrade, we also evaluated additional constraints. The second and third limiting elements for load hosting capability under light load conditions with battery discharging are listed in Table 8, along with their DC DFAX values and available branch headroom (MW). Using these two values, we can approximate the load hosting capacity limited by the second and third limiting elements. The second and third limiting elements for load hosting capability under light load conditions with battery charging are listed in Table 9. The second and third limiting elements for load hosting capability under light load conditions without the battery are listed in Table 10. These tables demonstrate that if this simple element is upgraded, there are hundreds of MWs of new load serving capacity available even in light load contingency charging conditions, and that in the cases with Flint discharging there are up to 800 MW of new capacity created.

Table 8: Second and Third Limiting Elements for Load Hosting Capability under Light Load Conditions with Battery Discharging

Limiting element	Bus Name	Top Limiting Element	Contingency Name	Dfax	Branch Headroom	Implied Capability
Second	Jug St	05JUG ST 345 kV to 05JUG ST 1-6 138 kV	AEP_P7-1_#17720___SRT-A	0.3035	320.1	1055
	Innovation Lab	05BABBIT 5-6 138 kV to 05BABBIT 3-4 138 kV	AEP_P4_#15960_05BABBIT 5-6 138_N2_SRT-A	0.2238	156.1	697
	Badger	05GREENCHL12 138 kV to 05HORIZON 138 kV	AEP_P1-2_#17688_SRT-A	1.0000	819.6	820
Third	Jug St	05BABBIT 1-2 138 kV to 05JUGST 7-11 138 kV	AEP_P4_#17926_05HORIZON 138_CB4_SRT-A	0.6977	828.1	1187
	Innovation Lab	05GREENCHL12 138 kV to 05INNOVATION 138 kV	AEP_P7-1_#14452___SRT-A	0.3389	604.8	1785
	Badger	05HORIZON 138 kV to 05BADGER 138 kV	AEP_P4_#13998_05JUG ST 1-6 138_D_SRT-A	1.0000	871.0	871

Table 9: Second and Third Limiting Elements for Load Hosting Capability under Light Load Conditions with Battery Charging

Limiting element	Bus Name	Top Limiting Element	Contingency Name	Dfax	Branch Headroom	Implied Capability
Second	Jug St	05JUG ST 345 kV to 05JUG ST 1-6 138 kV	AEP_P7-1_#17720___SRT-A	0.3035	197.9	652
	Innovation Lab	05ANGUIN 3-4 138 kV to 05EMPEROR Z2 138 kV	AEP_P4_#17884_05ANGUIN 3-4 138_G2_SRT-A	0.1852	146.4	790
	Badger	05JUG ST 345 kV to 05JUG ST 1-6 138 kV	AEP_P7-1_#17720___SRT-A	0.3035	197.9	652
Third	Jug St	05BABBIT 1-2 138 kV to 05JUGST 7-11 138 kV	AEP_P4_#17926_05HORIZON 138_CB4_SRT-A	0.6977	549.4	787
	Innovation Lab	05GREENCHL12 138 kV to 05INNOVATION 138 kV	AEP_P7-1_#14452___SRT-A	0.3389	641.6	1893
	Badger	05HORIZON 138 kV to 05BADGER 138 kV	AEP_P4_#17927_05HORIZON 138_CB5_SRT-A	0.6977	549.4	787

Table 10: Second and Third Limiting Elements for Load Hosting Capability under Light Load Conditions without Flint Battery

Limiting element	Bus Name	Top Limiting Element	Contingency Name	Dfax	Branch Headroom	Implied Capability
Second	Jug St	05BABBIT 5-6 138 kV to 05BABBIT 3-4 138 kV	AEP_P4_#15960_05BABBIT 5-6 138_N2_SRT-A	0.3735	93.4	250
	Innovation Lab	05BABBIT 5-6 138 kV to 05BABBIT 3-4 138 kV	AEP_P4_#15960_05BABBIT 5-6 138_N2_SRT-A	0.2233	93.4	418
	Badger	05BABBIT 5-6 138 kV to 05BABBIT 3-4 138 kV	AEP_P4_#15960_05BABBIT 5-6 138_N2_SRT-A	0.30632	93.4	305
Third	Jug St	05JUG ST 345 kV to 05JUG ST 1-6 138 kV	AEP_P7-1_#17720___SRT-A	0.3953	259.3	656
	Innovation Lab	05ANGUIN 3-4 138 kV to 05EMPEROR Z2 138 kV	AEP_P4_#17935_05ANGUIN 3-4 138_F2_SRT-A	0.18535	127.1	686
	Badger	05GREENCHL12 138 kV to 05HORIZON 138 kV	AEP_P1-2_#17688_SRT-A	1	819.5	820

N-1-1 Results: Peak Load Conditions

N-1-1 analysis is based on the first and the second contingency being single design contingencies as consistent with the PJM study processes.

With storage = 0 MW and no additional load at Jug / Badger / Innovation Lab 138 kV, there are thermal overloads for Maliszewski bus tie, Maliszewski to Polaris 138 kV circuit and Robert to Kenny 138 kV circuit as shown in Table 11.

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Table 11: Thermal Results for N-1-1 without storage and no additional load

First Level Contingency	Second Level Contingency	Monitored Facility	Contingency Rating (MVA)	% Loading
AEP P1-2 #7216 406223 SRT-A	AEP P1-3 #8072 05MARYSV 765 2 SRT-A	05MALIS 138 kV to 05MALISX 138 kV	1054	122
AEP P1-3 #8072 05MARYSV 765 2 SRT-A	AEP P1-2 #7216 406223 SRT-A	05MALIS 138 kV to 05MALISX 138 kV	1054	122
AEP P1-3 #7222 05MALIS 765 1 SRT-A	AEP P1-2 #7207 9700575 SRT-A	05ROBERT 138 kV to 05KENNY 138 kV	184	116
AEP P1-2 #7207 9700575 SRT-A	AEP P1-3 #7222 05MALIS 765 1 SRT-A	05ROBERT 138 kV to 05KENNY 138 kV	184	115
AEP P1-2 #7216 406223 SRT-A	Base Case	05MALIS 138 kV to 05MALISX 138 kV	1054	109
AEP P1-2 #7216 406223 SRT-A	AEP P1-2 #5832 6228 SRT-A	05MALIS 138 kV to 05POLARS 138 kV	310	109
AEP P1-2 #5832 6228 SRT-A	AEP P1-2 #7216 406223 SRT-A	05MALIS 138 kV to 05POLARS 138 kV	310	109

With storage = -200 MW in charging mode and no additional load at Jug / Badger / Innovation Lab 138 kV, the observed overloads are worsened as shown in Table 12. Therefore, during some first contingency conditions, the Flint storage may be constrained to charge during peak load conditions (based on decisions from PJM and/or pricing signals that would disincentivize charging).

Table 12: Thermal Results for N-1-1 with storage charging and no additional load

First Level Contingency	Second Level Contingency	Monitored Facility	Contingency Rating (MVA)	% Loading
AEP P1-2 #7216 406223 SRT-A	AEP P1-3 #8072 05MARYSV 765 2 SRT-A	05MALIS 138 kV to 05MALISX 138 kV	1054	124
AEP P1-3 #8072 05MARYSV 765 2 SRT-A	AEP P1-2 #7216 406223 SRT-A	05MALIS 138 kV to 05MALISX 138 kV	1054	124
AEP P1-3 #7222 05MALIS 765 1 SRT-A	AEP P1-2 #7207 9700575 SRT-A	05ROBERT 138 kV to 05KENNY 138 kV	184	121
AEP P1-2 #7207 9700575 SRT-A	AEP P1-3 #7222 05MALIS 765 1 SRT-A	05ROBERT 138 kV to 05KENNY 138 kV	184	120
AEP P1-2 #7216 406223 SRT-A	Base Case	05MALIS 138 kV to 05MALISX 138 kV	1054	111
AEP P1-2 #7216 406223 SRT-A	AEP P1-2 #5832 6228 SRT-A	05MALIS 138 kV to 05POLARS 138 kV	310	111
AEP P1-2 #5832 6228 SRT-A	AEP P1-2 #7216 406223 SRT-A	05MALIS 138 kV to 05POLARS 138 kV	310	110

With storage = 200 MW in discharging mode and no additional load at Jug / Badger / Innovation Lab 138 kV, the observed overloads are less severe by a small percentage as shown in Table 13. Therefore, when the storage is discharging during peak load, the storage at discharging mode slightly reduces thermal loading on the overloaded limiting elements.

Table 13: Thermal Results for N-1-1 with storage discharging and no additional load

First Level Contingency	Second Level Contingency	Monitored Facility	Contingency Rating (MVA)	% Loading
AEP P1-2 #7216 406223 SRT-A	AEP P1-3 #8072 05MARYSV 765 2 SRT-A	05MALIS 138 kV to 05MALISX 138 kV	1054	120
AEP P1-3 #8072 05MARYSV 765 2 SRT-A	AEP P1-2 #7216 406223 SRT-A	05MALIS 138 kV to 05MALISX 138 kV	1054	120
AEP P1-3 #7222 05MALIS 765 1 SRT-A	AEP P1-2 #7207 9700575 SRT-A	05ROBERT 138 kV to 05KENNY 138 kV	184	112
AEP P1-2 #7207 9700575 SRT-A	AEP P1-3 #7222 05MALIS 765 1 SRT-A	05ROBERT 138 kV to 05KENNY 138 kV	184	111
AEP P1-2 #7216 406223 SRT-A	Base Case	05MALIS 138 kV to 05MALISX 138 kV	1054	107
AEP P1-2 #7216 406223 SRT-A	AEP P1-2 #5832 6228 SRT-A	05MALIS 138 kV to 05POLARS 138 kV	310	107
AEP P1-2 #5832 6228 SRT-A	AEP P1-2 #7216 406223 SRT-A	05MALIS 138 kV to 05POLARS 138 kV	310	106

To summarize, when Flint Storage is discharging for peak load condition, it alleviates N-1-1 overloads and improves load-hosting capacity at Jug / Badger / Innovation Lab 138 kV. When charging at maximum power for peak load condition, it aggravates N-1-1 overloads, so PJM operators may limit Flint Storage charging at peak load under certain first level N-1 conditions.

The result can be further illustrated in Table 14 as we can see the incremental load hosting capability for all three buses under peak load N-1-1 condition.

Table 14: Load Hosting Capability under Peak Load Conditions (N-1-1)

Scenario	Flint Storage status	Bus Name	Top Limiting Element	Incremental Load Hosting Capability (MW)
1	Without Flint Storage	Jug St	05MALIS 138 kV to 05MALISX 138 kV	NA
		Innovation Lab	05MALIS 138 kV to 05MALISX 138 kV	NA
		Badger	05MALIS 138 kV to 05MALISX 138 kV	NA
2	Discharging (200 MW)	Jug St	05MALIS 138 kV to 05MALISX 138 kV	246
		Innovation Lab	05MALIS 138 kV to 05MALISX 138 kV	193
		Badger	05MALIS 138 kV to 05MALISX 138 kV	192

Conclusions

The N-1 reliability analysis demonstrates that Flint Storage discharging during peak load conditions significantly increases the load hosting capability at the three study buses. **Specifically, incremental hosting capability of approximately 221 MW at Innovation Lab 138 kV, 168 MW at Badger 138 kV, and 163 MW at Jug St 138 kV was observed.**

This benefit is increased at Jug Street and Badger when the top limiting element is fixed/eliminated (a line of ~ 600 linear feet) and the incremental transmission capacity changes to 711MW and 511MW, respectively. Innovation experiences a less significant increase from this upgrade of just 38 additional MW.

The N-1-1 reliability analysis demonstrates that Flint Storage discharging during peak load conditions significantly increases the load hosting capability at the three study buses. Specifically, incremental hosting capability of approximately 246 MW at Innovation Lab 138 kV, 193 MW at Badger 138 kV, and 192 MW at Jug St 138 kV was observed.

In contrast and not suprisingly, when Flint Storage is in charging mode during N-1 conditions, no additional load hosting capability is available. Instead, the Project's charging operation reduces the load hosting capability at all three buses due to increased stress on the transmission system. As discussed above, it is highly unlikely that the Project would charge during an N-1 scenario.

Overall, the study results indicate that Flint Storage increases transmission capacity during peak load by supporting additional demand while discharging in both N-1 and N-1-1 scenarios.